An entry–exit law for the resurrection of dead neurons under smoothed ReLU activation

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Abstract

A neuron dies when its pre-activation margin is negative on every datum, and under hard ReLU its gradient is exactly zero, so it never returns. In the sharp-activation limit of a shallow softplus network under gradient flow the dynamics is instead hybrid with memory: we prove that the dead unit is released exactly when the signed time integral of the residual excluding it returns to zero, at rate $O(\log\beta/\beta)$ and with a first-order release coefficient in which the arrival speed has cancelled. A sweep of 600 instances isolates 57 satisfying the hypotheses; the law holds in all 57, median relative error 2.7%, coefficients spanning a factor of 20. The law is a statement about the forcing: a second theorem extends it to any affine live subsystem with m units, with constants uniform to $O(\sqrt{m})$. A pre-registered ladder follows, ending in two confirmatories. A 400-run MLP reproduces resurrection, the area in fact turning positive in 13 of 16 returns, hard ReLU structurally zero. A transformer confirmatory of 800 pre-registered runs transfers the class to the feed-forward block through the unit's own channel ($\hat{p} = 0.440$) and reproduces a second, upstream channel in all four arms: hard-ReLU units return 20 of 100, against 0 with attention frozen.

Keywords: entry–exit function; geometric singular perturbation; dead neurons; smoothed ReLU; gradient flow; loss of plasticity.

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1 Introduction

Write $\Phi : \mathcal{W} \to \mathcal{F}$ for the parametrization of a network and $J = D\Phi$ for its Jacobian. A neuron whose pre-activation margin is negative on every datum contributes an identically zero column to J: it sits on a stratum of the singular locus where J drops rank. This is exact for ReLU and exponentially thin for softplus-β; it does not hold for GELU at all, whose output is nonzero for every finite negative margin (Section 6.6). Under a hard ReLU the gradient of the loss in the directions of that unit is exactly zero, so the stratum is absorbing.

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A dead unit stays dead. This is the mechanism behind the empirically documented dormant-neuron and loss-of-plasticity phenomena [1, 2, 3]. The question of this paper is what replaces the absorbing stratum when the activation is smoothed, as it is in every softplus or GELU [4] network actually trained.

Our answer is an exact gating law and its numerical arc.

Theorem (informal)**.** In the sharp-activation limit of the smoothed flow, a dead unit does not sit on an absorbing stratum: it freezes and is released exactly when the signed time integral of the residual that excludes it returns to zero. The stratum is a memory gate rather than a trap. The convergence to that hybrid limit is uniform away from the two switching instants, at the rate $O(\log\beta/\beta)$ in the activation stiffness β, and the release time carries the first-order coefficient $-1/(cR(t_{\text{exit}}))$, in which the arrival speed has cancelled: the residual at death drops out, while the coefficient still depends on the datum constant c and on the residual at the exit. The precise statement is Theorem 2.

Scope of the proved object. Theorem 2 covers the minimal case: one dead unit, one live unit, one active datum at a time, a *fixed* readout and a one-dimensional input. Theorem 6 removes the second of those restrictions — the live side may be any affine subsystem of any order m — and in doing so it changes the status of the transfer from the proved object to the measured one, which in the previous version of this paper was simply empirical. The honest statement is now a gradation in three levels, and we keep the levels separate throughout.

(P1) **Proved, affine class.** For a network with fixed readout whose live units keep a constant activation pattern on the window, the entry–exit law holds for every m, with constants growing at most like $\sqrt{m}$ and threshold $\beta_0 = O(m)$ (Theorem 6 and Proposition 8). Nothing here is conjectural and nothing is fitted.

(P2) **Corollary under hypotheses verifiable a posteriori.** For an MLP of the affine class — readout held fixed — the law at a given resurrection episode is a *corollary* of Theorem 6, not an analogy, once (H1′)–(H2′) are checked on the hybrid trajectory and the window of (H4′) is identified. Both checks are one hybrid integration (Section 3.6); neither is guaranteed a priori by any initialization criterion we know: in our sweeps the acceptance rate is about 3% of random draws in the NTK-readout family and about 20% (179/900) in the unit-readout family.

(P3) **Empirical.** A trainable readout v_k, which is what the experiments of Section 6 use and what takes the live field outside the gradient-of-a-convex-function class; several data active at once; and the transformer of Section 6.5. For these the transfer is measured, not deduced, and we say so wherever a number from them is reported.

The contributions are:

- **A theorem (Section 3 and Theorem 2).** In the sharp limit $\beta \to \infty$ of a shallow softplus network under gradient flow, the death and resurrection of a neuron is the

entry–exit problem of a slow–fast system [5]: the dead unit freezes and reactivates when $\int R\,dt = 0$, where R is the residual excluding that unit. The uniform rate is $O(\log\beta/\beta)$ and the release time carries the first-order coefficient $-1/(cR(t_{\text{exit}}))$, from which the arrival speed has cancelled. The proof is organized as six lemmas (Section A), each with an independent executable symbolic or numerical verification. A consequence of the same construction is that the two limit orders, smooth-then-sharp against the pointwise generalized-gradient flow, do not commute (Section 4): the field-first order is the frozen branch of Eberle, Jentzen, Riekert and Weiss [6], our flow-first limit resurrects, and the autodiff convention that fixes the branch is discussed in Section 5. For the erf smoothing we verify numerically that the release criterion is unchanged; the proof is written for softplus, and GELU is outside the class because its derivative is non-monotone.

- **A second theorem: the law belongs to the forcing (Section 3.1 and Theorem 6).** The residual R is treated as a *prescribed forcing* and the live side is allowed to be any affine subsystem of order m — any number of live units with a constant activation pattern and a fixed readout — instead of the single live neuron of Theorem 2. The same law holds, with the same first-order coefficient $-1/(cR^{\text{hyb}}(t_{\text{exit}}))$, which is structurally free of m. The estimate that carries the live subsystem is replaced by an energy estimate using only $G_B \succeq 0$, so the constant $M_2 = C_f C_4/c$ carries no factor $e^{L_B H}$ and there is **no restriction of the form $m \cdot H$ small**: every explicit constant is uniform to $O(\sqrt{m})$ and the threshold is $\beta_0 = O(m)$ (Proposition 8), both at fixed forcing. The proof is Section B; the verification, on $66 + 23 + 67$ instances and a joint regression over 133, is Section 3.6.

- **A sweep from one instance to a law (Section 3.5).** Every constant is verified on a reference two-neuron instance, and a sweep of 600 random instances then tests the release law against the dynamics on the 57 that satisfy the hypotheses: it holds in all 57, with a median relative error of 2.68% over predicted coefficients ranging from -91.8 to -4.5.

- **A pre-registered experimental ladder (Section 6).** Organized by prediction, with the null controls and the negatives shown: a reduced deterministic-schedule model falsified under SGLD (Section 6.1); two instrument designs recorded as negatives because a calibrated null fired (Section 6.2); the mechanism first observed in an MLP pressure pilot (Section 6.3); and a confirmatory MLP run of 400 runs in which the area law reproduces with confidence intervals, hard ReLU acts as the zero-gradient control, and the return probability is monotone in β (Section 6.4 and Figure 5). The scoping of the transformer (Section 6.6) exposed two new mechanisms, a GELU sign-flip barrier and an upstream channel, and its own pre-registered confirmatory of 800 runs was executed: the theorem class transfers to the feed-forward block through the own channel, the upstream channel reproduces in all four arms including hard ReLU, and the barrier remains confounded with injection depth.

- **A confirmatory in a transformer block (Sections 6.5 and 6.6 and Section D).** The 800 pre-registered runs close their five estimands: the theorem class transfers to the feed-forward block through the unit's own channel (22/50 sustained returns, the area opening before 21/22 of them); the upstream channel reproduces in all four arms, hard ReLU included (20/100 against 0/100 with attention frozen); and the GELU barrier is left confounded with injection depth under the pre-registered three-way label. Table 5 carries the five rows.

The theorem is proved for the minimal configuration above and its constants are verified on one reference instance and on a sweep of 600 drawn instances; the two confirmatories, the MLP (4a.3) and the transformer (`etapa-5`[1]), cover two small architectures at zero noise; the stochastic regime is open and, where we probed it, it breaks the deterministic schedule (Section 6.1).

The paper is organized as follows. Section 2 fixes the model and the four hypotheses. Section 3 states the entry–exit theorem, its forcing form (Section 3.1), the six lemmas that compose it and the numerical verification of its constants; the complete proofs of the lemmas are deferred to Section A and the proof of the forcing theorem to Section B. Section 4 makes the non-commutation of the two limit orders precise and connects it to the plasticity literature, which Section 5 surveys. Section 6 is the experimental ladder, from the falsified reduced model to the two executed confirmatories, the MLP and the transformer; Sections 7 to 9 give the limitations, the conclusion, and the reproducibility record.

2 Setup and notation

Data $(x_i, y_i)_{i=1}^n \in \mathbb{R} \times \mathbb{R}$ in general position; shallow network with unit output weights

$$f_\theta(x) = \sum_k g_\beta(w_k x + b_k), \qquad g_\beta(z) = \beta^{-1} \log(1 + e^{\beta z}), \tag{1}$$

quadratic loss $L = \frac{1}{2n} \sum_i (f_\theta(x_i) - y_i)^2$ and gradient flow $\dot\theta = -\nabla L$ on the training horizon $[0, H]$. General monotone sigmoids are treated in Section 3. The parameter β is the *activation stiffness*: as $\beta \to \infty$, $g_\beta \to \mathrm{ReLU}$ and $g'_\beta \to \mathbf{1}_{\{z>0\}}$. We write H for the training horizon throughout, a time in the flow and a number of optimizer steps in the experiments, and $\mathcal{N}$ for the standard normal cumulative distribution function.

Fix one neuron A and one datum j. Its *margin* is $z = \theta_A \cdot (x_j, 1)$; write $c = \|(x_j, 1)\|^2/n$ and let

$$R(t) = y_j - f_{-A}\big(x_j; \theta_B(t)\big) \tag{2}$$

be the *residual excluding* A, where θ_B collects the other parameters. The *hybrid reference* θ^{hyb} runs the full ReLU flow until the first time t_{hit} with $z = 0$, then freezes θ_A and runs θ_B

[1] Artifact, commit and script names are in Portuguese, the project's lab language; they are kept verbatim because the commit hashes and manifests reference them.

alone; set R^{hyb} accordingly and

$$\mathcal{A}^{\mathrm{hyb}}(t) = \int_{t_{\mathrm{hit}}}^{t} R^{\mathrm{hyb}}(s)\, ds \tag{3}$$

for its signed area. The death of A is exactly the rank drop of J on the stratum $\{z \le 0\}$; the question is the fate of the trajectory on that stratum.

Hypotheses. On the episode we assume

$R^{\mathrm{hyb}}(t_{\mathrm{hit}}) < 0$ (transversal death); (H1)

$\mathcal{A}^{\mathrm{hyb}}$ has a first zero at $t_{\mathrm{exit}} < H$ with $R^{\mathrm{hyb}}(t_{\mathrm{exit}}) > 0$ and $\mathcal{A}^{\mathrm{hyb}} < 0$ on $(t_{\mathrm{hit}}, t_{\mathrm{exit}})$; (H2)

the other margins of A stay $\le -\delta_1 < 0$; (H3)

no other switching event occurs in $[0, H]$. (H4)

3 The entry–exit theorem

Definition 1 (hybrid limit)**.** θ^{∞} equals: the ReLU flow on $[0, t_{\mathrm{hit}}]$; on $[t_{\mathrm{hit}}, t_{\mathrm{exit}}]$, $\theta_A \equiv \theta_A(t_{\mathrm{hit}})$ (frozen) with θ_B the B-only flow; on $[t_{\mathrm{exit}}, H]$, the ReLU flow restarted at $(\theta_A(t_{\mathrm{hit}}), \theta_B^{\mathrm{hyb}}(t_{\mathrm{exit}}))$.

Theorem 2 (entry–exit law for the smoothed dead unit)**.** *Assume* (H1)–(H4)*. Then for every* $\delta > 0$

$$\sup_{t \in [0,H] \setminus (B_\delta(t_{\mathrm{hit}}) \cup B_\delta(t_{\mathrm{exit}}))} \left\| \theta^{\beta}(t) - \theta^{\infty}(t) \right\| = O_\delta\!\left(\frac{\log \beta}{\beta} \right), \tag{4}$$

and the first return $t^{\beta}_{\mathrm{release}}$ *of the margin* z *to* 0 *after death satisfies*

$$t^{\beta}_{\mathrm{release}} = t_{\mathrm{exit}} - \frac{1}{c\, R^{\mathrm{hyb}}(t_{\mathrm{exit}})} \cdot \frac{1}{\beta} + o(\beta^{-1}). \tag{5}$$

Remark 3 (what carries each constant). The $\log \beta$ in (4) is sharp and is carried solely by the normal coordinate of θ_A inside the episode, whose sticking depth is $z \sim -\beta^{-1} \log \beta$ (Lemma 12); θ_B obeys $O(\log \beta / \beta^2)$ (Lemma 10) and the tangential components of θ_A are exponentially frozen (Lemma 14). In the first-order coefficient of (5) the arrival speed cancels: in the product of the entry delay by the area it cuts, the residual at death drops out and only the reciprocal exit speed survives, so each corner contributes the reciprocal of its own transversal speed (Lemma 13). What does not cancel is the dependence on c and on $R^{\mathrm{hyb}}(t_{\mathrm{exit}})$, which are instance quantities. The remainder is $o(\beta^{-1})$; the numerics suggest $O(\log \beta / \beta^2)$, which would follow from quantifying the $o(1)$ of Lemma 11 with $\delta_z = \log \beta / \beta$, and we leave it as a stated refinement. Figure 1 draws the three stages this remark describes. The limit is that of the regularized flow; discrete ReLU gradient descent freezes the dead neuron forever (Section 4).

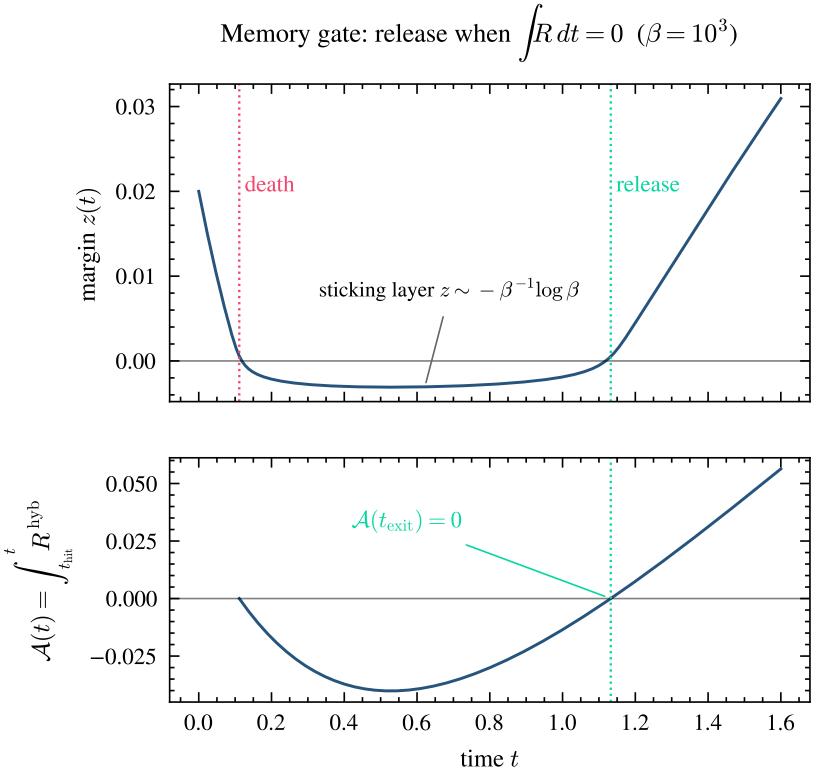


Figure 1: The memory gate at $\beta = 10^3$ (coupled system, reference instance). Top: the margin $z(t)$ dies at $t_{\rm hit}$, hugs the sticking layer of depth $\sim \beta^{-1}\log\beta$, and releases. Bottom: the hybrid area $\mathcal{A}(t) = \int_{t_{\rm hit}}^{t} R^{\rm hyb}$; release coincides with $\mathcal{A} = 0$ up to $O(\beta^{-1})$ (Theorem 2). Deterministic Euler integration ($dt = 10^{-5}$); the uncertainty is discretization only. Script `t2-prova/figuras_draft.py`.

Remark 4 (which hypotheses are essential). (H1)–(H2) are essential: they are the entry–exit geometry itself, and the law fails without the sign change of $R^{\rm hyb}$. (H3)–(H4) are there to simplify the proof: (H3) makes the other data contribute $O(e^{-\beta\delta_1})$ and (H4) isolates a single episode. We expect both to be removable at the cost of bookkeeping; we have not done it.

3.1 The forcing theorem: the law belongs to the forcing

Theorem 2 is stated for two neurons, and read literally it is a statement about a particular small network. It is not: nothing in the argument uses the live side except through the scalar R it generates. This subsection makes that reading a theorem. The residual becomes a *prescribed forcing*, the live side becomes an arbitrary affine subsystem of order m, and the law, the rate and the first-order coefficient are unchanged. The complete proof is Section B; the mathematical source is the standalone note `06-notas/t2-prova/teorema-forcing-draft.`

tex.

Network and live class. Keep the data and the loss of Section 2, but allow $m+1$ units with *fixed* readout weights, the readout of the unit that dies normalized to $+1$:

$$f_\theta(x) = g_\beta(w_0 x + b_0) + \sum_{k=1}^{m} a_k\, g_\beta(w_k x + b_k). \tag{6}$$

Unit $k=0$ is the unit A that dies. Only its readout enters c and the margin equation, so fixing it at $+1$ is a normalization and not a restriction; we reserve a_0 for the floor of the area, as in Section A. The remaining m units form the live subsystem B, with parameters $\theta_B \in \mathbb{R}^{2m}$, and R is the residual excluding A, exactly (2) with $f_{-A} = \sum_{k\geq 1} a_k g_\beta(w_k x + b_k)$. Write $\mathrm{res}_i = f_\theta(x_i) - y_i$ and $T_{\mathrm{ep}} = t_{\mathrm{exit}} - t_{\mathrm{hit}}$ for the length of the episode.

We call B *affine of order* m on a window if, on that window, the activation pattern of every live unit is constant and the readout is fixed. Then f_{-A} is linear in θ_B: with $\varphi_i \in \mathbb{R}^{2m}$ the feature vector of datum i, $(\varphi_i)_{2k-1} = a_k \mathbf{1}_{k,i} x_i$ and $(\varphi_i)_{2k} = a_k \mathbf{1}_{k,i}$, one has $f_{-A}(x_i;\theta_B) = \varphi_i \cdot \theta_B$ up to the softplus defect, and the ReLU B-field is $-G_B(\theta_B - \theta_B^*)$ with

$$G_B = \frac{1}{n}\sum_{i=1}^{n} \varphi_i \varphi_i^\top \in \mathbb{R}^{2m\times 2m}, \qquad G_B \succeq 0. \tag{7}$$

This is the only structural fact about B used below, and it holds for *every* m: a one-hidden-layer network with frozen activation pattern, fixed readout and only the first layer trained is linear in its trainable parameters, and G_B is then a Gram matrix, hence positive semidefinite. We write G for the full $(2m+2)\times(2m+2)$ Gram of the network with A alive and pattern frozen, and G_{BA} for its $B \times A$ block; G, and hence its principal submatrix G_B, is PSD.

Hypotheses. All of them are statements about the forcing R and about the hybrid reference trajectory, both computable by a single ReLU integration with A frozen. None constrains m, the dimension of θ_B, or the functional form of f_{-A} beyond the affine class.

(H0′) **Regularity of the forcing.** $R^{\mathrm{hyb}} \in C^1([t_{\mathrm{hit}}, H])$ with $\|\dot R^{\mathrm{hyb}}\|_\infty \leq C_R$.

(H1′) **Transversal entry.** $R^{\mathrm{hyb}}(t_{\mathrm{hit}}) < 0$; the arrival speed is $v = c\,|R^{\mathrm{hyb}}(t_{\mathrm{hit}})| > 0$.

(H2′) **Transversal exit at the zero of the area.** $\mathcal{A}^{\mathrm{hyb}}$ has a first zero at $t_{\mathrm{exit}} < H$, with $R^{\mathrm{hyb}}(t_{\mathrm{exit}}) > 0$, $\mathcal{A}^{\mathrm{hyb}} < 0$ on $(t_{\mathrm{hit}}, t_{\mathrm{exit}})$, and, for each $\delta > 0$, $|\mathcal{A}^{\mathrm{hyb}}| \geq a_0(\delta) > 0$ on $[t_{\mathrm{hit}} + \delta, t_{\mathrm{exit}} - \delta]$.

(H3′) **Lipschitz response.** There is Lip_R with $|R[\theta_B] - R[\theta_B']| \leq \mathrm{Lip}_R \|\theta_B - \theta_B'\|$, and the B-field is Lipschitz on a compact tube around the hybrid trajectory. In the affine class this is automatic, with $\mathrm{Lip}_R = \|\varphi_j\|$; we keep it as a hypothesis so that the statement survives outside the class.

(H4′) **Isolated episode, uniform live margins.** No unit other than A switches on $[0, H]$, and the live margins are *uniformly* bounded away from the kink,

$$\big|z_k(x_i; \theta_B(t))\big| \;\geq\; \delta_B > 0 \qquad \text{for all } k = 1, \dots, m,\ i = 1, \dots, n,\ t \in [0, H],$$

along both the hybrid and the β-trajectory. The other margins of A stay $\leq -\delta_1 < 0$. Finally, A itself switches *exactly twice* on $[0, H]$, at t_{hit} and at t_{release}; in particular $z \geq \delta_1' > 0$ on $[t_{\text{release}} + \eta, H]$ along both trajectories, for some $\eta, \delta_1' > 0$.

(H5′) **Bounded residuals.** $|\text{res}_i(t)| \leq C_{\text{res}}$ for all i and all $t \in [0, H]$, along both trajectories in the tube.

Remark 5 ((H4′) is the definition of the window, and why it had to be reinforced). Two things are going on in (H4′), and they should not be confused. The *uniform* margin floor δ_B is a genuine strengthening of (H4) of Section 2: in the two-neuron case the single live unit contributes one softplus-versus-ReLU discrepancy, absorbed silently into the $O(e^{-\beta\delta_1})$ bookkeeping; with m live units there are mn such discrepancies, and a per-instance margin is not enough — the bound must be uniform over units, data and time, or the sum of mn terms is not controlled by a single exponential. The constancy of the pattern, on the other hand, is not a debt but a *definition of scope*: Theorem 6 holds on any maximal window on which the live activation pattern is constant and which contains t_{hit} and t_{exit}, and such a window is identified a posteriori by the same hybrid integration that checks (H1′)–(H2′). Relaxing it is not free, and the price is exhibited in Section B.10: the two flows then switch at different instants, the difference of drifts acquires a term $-(G_B - G_B')\theta_B$ that is not of the form $-(\text{PSD})\Delta$, and one is back to a Grönwall factor $\exp(\sum_k \|\Delta G_k\| \|\theta_B\| / v_k)$ — exponential in m in the worst case, which is exactly what the estimate of Lemma 20 was introduced to remove.

Theorem 6 (entry–exit law, forcing form)**.** *Let A be a softplus-β unit with fixed readout $+1$ and margin z on the face j, coupled to a live subsystem B affine of order m, and let R be the residual (2) it generates. Assume (H0′)–(H5′). Then there is $\beta_0 < \infty$ such that for $\beta \geq \beta_0$:*

(i) (sticking) *on compact subintervals of $(t_{\text{hit}}, t_{\text{exit}})$ the margin satisfies $u = c\beta|\mathcal{A}^{\text{hyb}}(t)|(1 + o(1))$, that is $z_{\text{park}}(t) = -\beta^{-1} \log \beta\,(1 + o(1))$;*

(ii) (release) *the first return of z to 0 after death obeys*

$$t^{\beta}_{\text{release}} = t_{\text{exit}} - \frac{1}{c\, R^{\text{hyb}}(t_{\text{exit}})} \cdot \frac{1}{\beta} + o(\beta^{-1}), \tag{8}$$

where t_{exit} is the first zero of the hybrid area $\mathcal{A}^{\text{hyb}}$;

(iii) (uniform approximation) *for every $\delta > 0$,*

$$\sup_{t \in [0,H] \setminus (B_\delta(t_{\text{hit}}) \cup B_\delta(t_{\text{exit}}))} \big\|\theta^{\beta}(t) - \theta^{\infty}(t)\big\| = O_\delta\!\left(\frac{\log \beta}{\beta}\right), \tag{9}$$

the $\log \beta$ being carried solely by the normal coordinate of θ_A;

(ivA) (live subsystem, up to release)

$$\sup_{[0,t_{\text{release}}]} \left\|\theta_B^{\beta} - \theta_B^{\text{hyb}}\right\| \;\le\; 2M_2 \frac{\log\beta}{\beta^2}; \tag{10}$$

(ivB) (live subsystem, after release) *against the hybrid limit* θ^{∞} *of Definition 1, in which* A *is alive again,*

$$\sup_{[t_{\text{release}},H]} \left\|\theta_B^{\beta} - \theta_B^{\infty}\right\| \;\le\; \frac{M_3}{\beta}, \qquad M_3 = H\,\|G_{BA}\|\,C_A + 2M_2\frac{\log\beta}{\beta} + \frac{C_f H}{v'}, \tag{11}$$

so that in particular $\sup_{[0,H]} \|\theta_B^{\beta} - \theta_B^{\infty}\| = O(\beta^{-1})$*: the live subsystem never carries the* $\log\beta$*.*

Here $v' = cR^{\text{hyb}}(t_{\text{exit}}) > 0$ *is the exit speed,* C_A *is the constant of* (30)*, and*

$$M_2 = \frac{C_f C_4}{c}, \qquad C_4 = \frac{2}{c\min\big(|R^{\text{hyb}}(t_{\text{hit}})|, R^{\text{hyb}}(t_{\text{exit}})\big)} + \frac{T_{\text{ep}}}{a_0}, \qquad C_f = \frac{\|\varphi_j\|}{n}. \tag{12}$$

The word "in particular" in (ivB) is legitimate because the two references agree to the required order where the two clauses are glued: $\theta^{\infty} \equiv \theta^{\text{hyb}}$ on $[0, t_{\text{exit}}]$ *by definition* (Definition 1), and on the short interval $[t_{\text{exit}}, t_{\text{release}}]$, of length $O(\beta^{-1})$ by (8), they differ only by the drift that A's revival induces in θ^{β}, itself $O(\beta^{-1})$ because $z = O(\beta^{-1})$ there; the product is $O(\beta^{-2})$. Hence (10) may be read against either reference on $[0, t_{\text{release}}]$, which is also what licenses $\|\Delta(t_{\text{release}})\| \le 2M_2 \log\beta/\beta^2$ as the initial datum of step 5.

Remark 7 (why item (iv) changes reference at t_{release}). The two clauses cannot be merged into a single $\sup_{[0,H]} \|\theta_B^{\beta} - \theta_B^{\text{hyb}}\|$. The hybrid θ^{hyb} keeps θ_A frozen at $z = 0$ *forever*, so after t_{exit} its residuals never see A; the β-flow, and θ^{∞}, do. On $[t_{\text{release}}, H]$ the B-gap against θ^{hyb} is therefore $\Theta(1)$, not $O(\log\beta/\beta^2)$, and a statement of item (iv) with θ^{hyb} over all of $[0, H]$ would be false. The correct post-release reference is θ^{∞}, and the rate there is $O(\beta^{-1})$ — one power worse than during the episode, because the $O(\beta^{-1})$ mismatch of the release *instant* feeds into B through the cross block G_{BA} over a horizon of length H, but still one $\log\beta$ better than item (iii). The constant in (10) is $2M_2$, not M_2: M_2 is what the improvement step of Lemma 24(4.1) returns ($\frac{3}{2}M_2$ in fact), while $2M_2$ is what the closed induction propagates.

What is new, and what it means. Three things in Theorem 6 are new relative to Theorem 2. First, B is any affine subsystem of any order m, not one live neuron. Second, M_2 carries *no* factor $e^{L_B H}$: the Grönwall estimate of Lemma 10 is replaced by an energy estimate that uses only $G_B \succeq 0$, which is what removes the only exponential in m from the constants and, with it, any restriction of the form $m \cdot H$ small. Third, $C_f \le \|(x_j, 1)\|(\sum_k a_k^2)^{1/2}/n$ grows at most like $\sqrt{m}$, so every constant in (12) is polynomial in m (Section B.8). The reading is that *the law is a statement about the forcing, not about whatever generates it*: the

first-order coefficient $-1/(cR^{\mathrm{hyb}}(t_{\mathrm{exit}}))$ is built from c and from the hybrid forcing at one instant, and the cancellation that produces it is arithmetic in R (Remark 25), so structurally m cannot appear in it. Any m-dependence must live in the remainder, that is in β_0 — and the next proposition says how much.

Proposition 8 (polynomial threshold)**.** *Under (H0′)–(H5′), the threshold of Theorem 6 may be taken to be $\beta_0 = \max(\beta_1, \dots, \beta_5) \lesssim O(m)$ at fixed forcing — that is, with a_0, $R^{\mathrm{hyb}}(t_{\mathrm{hit}})$, $R^{\mathrm{hyb}}(t_{\mathrm{exit}})$, T_{ep}, δ_B, δ_1, C_{res}, $X_{\max} = \max_i \|(x_i, 1)\|$ and H held fixed while m grows — with the implied constant a fixed rational expression in a_0^{-1}, $R^{\mathrm{hyb}}(t_{\mathrm{exit}})^{-1}$, $|R^{\mathrm{hyb}}(t_{\mathrm{hit}})|^{-1}$, T_{ep}, H, $X_{\max}$, C_{res}, δ_B^{-1} and c. The five conditions and the identification of the binding one are Table 6.*

In particular β_0 is polynomial, indeed linear, in the size of the live subsystem: the theorem is not vacuous for large m. What is *not* controlled is the dependence of β_0 on the forcing itself, which enters through $1/a_0$ and $1/R^{\mathrm{hyb}}(t_{\mathrm{exit}})$, and which is exactly what the numerics of Section 3.6 see.

3.2 The six lemmas

The proof of Theorem 2 reduces the episode to a scalar entry–exit problem through the six lemmas below (their forcing-form counterparts are in Section B); they are proved in Section A. Each also carries a committed executable symbolic or numerical verification (`passo0--6`, `prova_L1--L6`), whose outcome is reported at the end of each proof.

Lemma 9 (normal form and additive quadrature)**.** *Let $w = e^{\beta z}$ and $u = 1/w$, and set $\Psi(u) = \log u + u$. In fast time $\tau = \beta t$ the margin obeys*

$$\frac{dw}{d\tau} = c\,R(t)\,\frac{w^2}{1+w}\,(1+E_\beta), \qquad |E_\beta| \le \frac{\log(1+w)}{\beta|R|}, \tag{13}$$

the quadratic normal form of De Maesschalck and Schecter [5, equations (1.9)–(1.10)] with $\varepsilon = \beta^{-1}$, and one has the exact additive identity

$$\frac{d\Psi(u)}{dt} = -c\,\beta\,R(t) + c\log(1+w) + \mathcal{E}_{(\mathrm{H3})}(t), \qquad \left|\mathcal{E}_{(\mathrm{H3})}\right| \le C_3\,\beta\,e^{-\beta\delta_1}. \tag{14}$$

Under (H1)–(H2) the additive correction satisfies $\int_{t_{\mathrm{hit}}}^{t_{\mathrm{release}}} c\log(1+w)\,ds = O(\log\beta/\beta)$.

The bound $|E_\beta| \le \log(1+w)/(\beta|R|)$ in (13) degenerates at the interior zero of R that (H1)–(H2) force, since R changes sign inside the episode; that is precisely why the proof switches to the quadrature Ψ of (14), which never divides by R.

Lemma 10 (B-subsystem and feedback)**.** *Under (H4) the B-field is affine with Lipschitz constant $L_B = \lambda_{\max}(G_B)$, where G_B is the Gram matrix of the active data, and R^{hyb} is*

affine in θ_B. Integral Grönwall consuming the bound of Lemma 9 gives, conditionally on the layer being maintained (supplied by Lemma 13),

$$\sup\|\theta_B^\beta - \theta_B^{\mathrm{hyb}}\| = O\Big(\frac{\log\beta}{\beta^2}\Big), \tag{15}$$

one full order better than the naive $O(\beta^{-1})$: in the bulk $u = O(\beta|\mathcal{A}|)$, so the coupling is $O(\beta^{-2})$ there and $O(\beta^{-1})$ only in the $O(\beta^{-1})$-wide corner windows.

Lemma 11 (entry corner)**.** *Let $v = c\,|R^{\mathrm{hyb}}(t_{\mathrm{hit}})|$ be the arrival speed, transversal by (H1). The exact identity $(1-\sigma(s))/\sigma(s) = e^{-s}$ collapses the crossing-time difference to an elementary integral and gives*

$$t_{\mathrm{cross}}^\beta = t_{\mathrm{hit}} + \frac{1}{\beta v}\big(1 + O(\delta_z) + o(1)\big), \qquad \big\|\theta^\beta(t_{\mathrm{cross}}) - \theta^\infty(t_{\mathrm{hit}})\big\| = O(\beta^{-1}), \tag{16}$$

where t_{cross}^β is the first time with $w = w_0$. The exponentially weighted approach integral is $\frac{3/2}{\beta v}(1+o(1))$, the factor $3/2$ coming from $\int_0^\infty e^{-s}(1+e^{-s})\,ds$.

Lemma 12 (sticking and release for prescribed R)**.** *Let $R \in C^1$ be prescribed and satisfy (H1)–(H2), and write $\mathcal{A}(t) = \int_{t_{\mathrm{hit}}}^t R$. Then*

(i) $\Psi(u(t)) = \Psi(u_0) - c\beta\mathcal{A}(t) + \mathrm{corr}(t)$ with $0 \le \mathrm{corr}(t) = \int c\log(1+w)\,ds = O(\log\beta/\beta)$; on compact subintervals of $(t_{\mathrm{hit}}, t_{\mathrm{exit}})$, $u = c\beta|\mathcal{A}(t)|\big(1 + O(\log\beta/(\beta a_0))\big)$, so the sticking depth is $z \sim -\beta^{-1}\log\beta$;

(ii) the release obeys the closed formula

$$t_{\mathrm{release}} - t_{\mathrm{exit}} = \frac{\mathrm{corr}(\beta)}{c\,\beta\,R(t_{\mathrm{exit}})}\,(1+o(1)) = O\Big(\frac{\log\beta}{\beta^2}\Big), \tag{17}$$

so u returns to $O(1)$ at $t_{\mathrm{exit}} + O(\beta^{-1})$, with no $\log\beta$ factor;

(iii) a perturbation $\tilde R = R + \delta$ shifts t_{exit} by at most $\delta\,(t_{\mathrm{exit}} - t_{\mathrm{hit}})/R(t_{\mathrm{exit}}) + O(\delta^2)$.

No division by R occurs anywhere: the layer is held by the area, not by the sign of R, and the interior zero of R forced by (H1)–(H2) is harmless.

Lemma 13 (coupled bootstrap)**.** *Under (H1)–(H4) the coupled system satisfies the two-term law*

$$t_{\mathrm{release}}^\beta = t_{\mathrm{exit}} - \frac{1}{c\,R^{\mathrm{hyb}}(t_{\mathrm{exit}})}\cdot\frac{1}{\beta} + \frac{\mathrm{corr}(\beta)}{c\,\beta\,R^{\mathrm{hyb}}(t_{\mathrm{exit}})} + o(\beta^{-1}), \qquad \mathrm{corr}(\beta) = O\Big(\frac{\log\beta}{\beta}\Big), \tag{18}$$

obtained by continuation with strict margins, with the constant $M_2 = e^{L_B H} C_f C_4/c$ independent of the continuation hypothesis for $\beta \ge \beta_0$. In the first-order coefficient the arrival speed cancels: in the product $K\,|R^{\mathrm{hyb}}(t_{\mathrm{hit}})|/R^{\mathrm{hyb}}(t_{\mathrm{exit}})$ of the entry delay by the area it cuts, the factor $|R^{\mathrm{hyb}}(t_{\mathrm{hit}})|$ cancels against $K = 1/(c|R^{\mathrm{hyb}}(t_{\mathrm{hit}})|)$ of Lemma 11, leaving a coefficient that still depends on c and on $R^{\mathrm{hyb}}(t_{\mathrm{exit}})$.

Lemma 14 (exit corner, branch selection and scalar reduction)**.** *Under* (H1)–(H4)*, after release: (i) the β-trajectory crosses the exit corner in time $\frac{1}{\beta v'}(1+o(1))$ with $v' = c\,R^{\mathrm{hyb}}(t_{\mathrm{exit}}) > 0$, the mirror of Lemma 11; (ii) the ReLU flow restarted at $(\theta_A(t_{\mathrm{hit}}), \theta_B^{\mathrm{hyb}}(t_{\mathrm{exit}}))$ has forward uniqueness on the activating branch, and that branch is selected by the limit itself, the frozen Filippov branch [7] being incompatible with release at $t_{\mathrm{exit}} + O(\beta^{-1})$; (iii) classical Grönwall glues on $[t_{\mathrm{release}}, H]$ at rate $O(\log\beta/\beta)$. Under* (H3) *the other data contribute $O(e^{-\beta\delta_1})$, the face-j field is rank one along $(x_j, 1)$, and the episode reduces to the scalar problem.*

3.3 Proof of the theorem

Proof of Theorem 2. We compose the six lemmas along the four stages of the episode.

By Lemma 14 the field on θ_A splits under (H3) into a rank-one part along $(x_j, 1)$ and terms of size $O(e^{-\beta\delta_1})$, so the tangential components of θ_A are frozen at an exponential rate and the normal component is exactly the scalar problem of Lemmas 9 to 13.

On $[0, t_{\mathrm{hit}} - \delta]$ both flows are the ReLU flow up to the softplus defect, which is bounded by $Ce^{-\beta z^0(t)}$; step 1 of the proof of Lemma 11 integrates that defect against the trajectory and gives a gap $O(\beta^{-1})$. Lemma 11 then delivers the entry corner: the crossing time is $t_{\mathrm{hit}} + 1/(\beta v)(1 + o(1))$ and the state at crossing is $O(\beta^{-1})$ from $\theta^\infty(t_{\mathrm{hit}})$.

On $[t_{\mathrm{hit}} + \delta, t_{\mathrm{exit}} - \delta]$, Lemma 13 closes the feedback loop: its continuation argument keeps the margin strictly inside the layer and θ_B within $O(\log\beta/\beta^2)$ of the hybrid, by consuming Lemma 10 (which needs the layer) and Lemma 12(iii) (which tolerates the resulting perturbation of the forcing). Inside the layer Lemma 12(i) gives the depth $z \sim -\beta^{-1}\log\beta$, and this is the largest of the three contributions to (4); θ_B contributes $O(\log\beta/\beta^2)$ and the tangential part of θ_A contributes $O(e^{-\beta\delta_1})$. Hence the supremum over this stage is $O_\delta(\log\beta/\beta)$, carried by the normal coordinate alone, which proves the sharpness asserted in Remark 3.

The terminal window and the release law are Lemma 13 step 5: the correction delay of Lemma 12(ii) is $O(\log\beta/\beta^2)$, the perturbation of the forcing shifts t_{exit} by $O(\log\beta/\beta^2)$ by Lemma 12(iii), and the area cut by the entry corner advances the exit by $-1/(cR^{\mathrm{hyb}}(t_{\mathrm{exit}})\beta)$. Summing the three gives (18), hence (5).

Finally Lemma 14 crosses the exit corner in time $O(\beta^{-1})$, selects the activating branch, and glues the two flows on $[t_{\mathrm{release}}, H]$ at rate $O(\log\beta/\beta)$. Taking the maximum of the four stage bounds gives (4). □

3.4 Numerical verification on one instance

The reference instance is $X = (0.5, 1.5)$, $Y = (0.5, 1.0)$, $\theta_A(0) = (-0.5, 0.27)$, $\theta_B(0) = (1.144, 0.261)$, giving $t_{\mathrm{hit}} = 0.1114$, $t_{\mathrm{exit}} = 1.1325$ and $\delta_1 = 0.508$. Every empirically measured constant is predicted by the proofs: the entry coefficient is $K_{\mathrm{teo}} = 7.04$ against 7.13 fitted (1.2%); the coupled two-term law $(-1 + \mathrm{corr})/(cR_{\mathrm{exit}})$ reproduces the fine-grid coupled sweep $-12.74/-13.93/-14.38/-14.68$ to 1.9%/0.7%/0.2%/0.1% for $\beta = 3\cdot 10^2 \ldots 10^4$, with the coefficient $1/(cR_{\mathrm{exit}}) = 14.815$ that the cancellation leaves (against -14.68 the leading

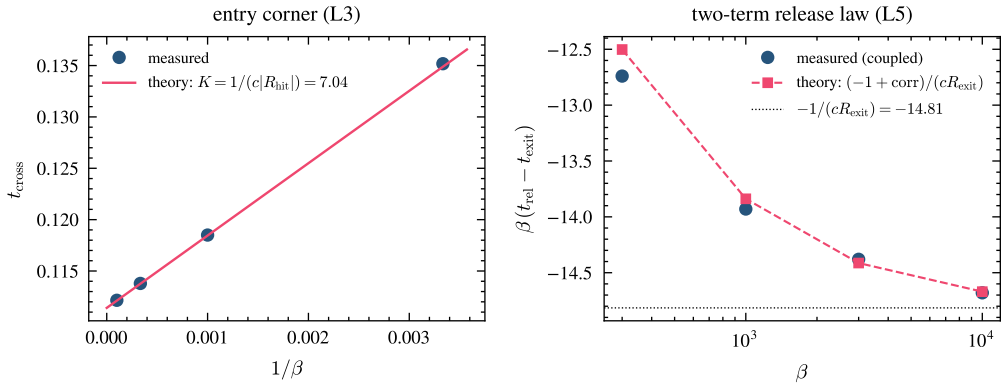


Figure 2: Predicted constants against measurements (deterministic runs; the uncertainty is discretization). Left: entry-corner crossing times against $1/\beta$; the line is the proved slope $K = 1/(c|R^{\rm hyb}(t_{\rm hit})|) = 7.04$ (Lemma 11), not a fit. Right: the two-term release law (Lemma 13, squares) against the coupled measurements (circles) over $\beta \in [3 \cdot 10^2, 10^4]$; the dotted line is the speed-independent first-order coefficient $-1/(cR^{\rm hyb}(t_{\rm exit})) = -14.81$. Every coupled run uses the explicit Euler step $dt = \min(10^{-5}, 0.02/\beta)$, so $dt = 10^{-5}$ at $\beta = 3 \cdot 10^2$ and 10^3, $6.7 \cdot 10^{-6}$ at $3 \cdot 10^3$ and $2 \cdot 10^{-6}$ at 10^4; Table 1 refines that step at $\beta = 10^4$. Script `t2-prova/figuras_draft.py`.

coefficient alone is at 0.9%); the tangential drift is at machine zero. This subsection is one instance; Section 3.5 tests the same law on a sweep of drawn instances.

The measured coefficient is a discretization, not a fit, so it must be shown to converge. Table 1 is that test: it refines the Euler step at the stiffest point of the sweep, $\beta = 10^4$, keeping everything else fixed, and asks whether the measured offset approaches the *proved* limit -14.815 of (18) rather than some nearby number.
It converges to the proved coefficient: 0.1% at the finest step. The driver reuses `roda_acoplado` verbatim from `t2-prova/prova_L5_verificacao.py`, overriding only the step.

3.5 From one instance to a law

A single instance cannot distinguish a law from a coincidence of constants, so we ran the same test over drawn instances. The script `sweep_instancias.py` samples 600 triples $(X, Y, \theta(0))$ from a prior fixed in advance and documented in the script, integrates each with the same dt, the same integrator and the same $\beta = 10^4$ as the canonical run — the coupled driver is copied verbatim from `passo4_verificacao.py` — and compares the measured release offset $\beta(t_{\rm release} - t_{\rm exit})$ against the prediction $-1/(c\,R^{\rm hyb}(t_{\rm exit}))$ formed from that instance's own constants. As a sanity check the reference instance of Section 3.4 is reproduced inside the sweep, at 2.9%.

Of the 600 draws, 57 satisfy (H1)–(H4). The exclusions are accounted for exactly: 340 have no death within the horizon, 162 fail (H2) because the area never returns to zero, and

Table 1: A convergence test for the two-term law, not a table of measurements: the proved coefficient $-1/(cR^{\mathrm{hyb}}(t_{\mathrm{exit}})) = -14.815$ is the limit, and the rows are successive refinements of the Euler step at $\beta = 10^4$ that must approach it. They do, monotonically, closing to 0.1% at the finest step; a law with the wrong constant would leave a floor here instead. The correction term of Lemma 13 is corr = 0.00977 at every step, so what is refining is the quadrature and not the law. At $dt = 10^{-5}$ the measurement is quantized in units of $\beta\, dt = 0.1$, which is the whole of that row's error. The sweep of Section 3.4 and Figure 2 uses $dt = \min(10^{-5}, 0.02/\beta)$, that is $dt = 2 \cdot 10^{-6}$ at $\beta = 10^4$.

dt	$\beta(t_{\mathrm{release}} - t_{\mathrm{exit}})$	deviation from -14.815
10^{-5}	-14.000	5.5%
$2 \cdot 10^{-6}$	-14.680	0.9%
10^{-6}	-14.760	0.4%
$5 \cdot 10^{-7}$	-14.805	0.1%

41 die outside the detector's window. On the 57 admissible instances the resurrection occurs in 57/57 — the release always precedes the hybrid t_{exit} — and the relative error of the release coefficient against the instance's own $-1/(c\, R^{\mathrm{hyb}}(t_{\mathrm{exit}}))$ has median 2.68%, ninetieth percentile 7.26% and maximum 12.01%. The predicted coefficients themselves range over $[-91.8, -4.5]$, a factor of 20, so the agreement is not the agreement of one number repeated. The worst quartile, those with |predicted| between 4.5 and 7.1, has median error 6.7%.

Two caveats belong with those numbers. First, the residual error is not centred: the median ratio of measured to predicted is 0.975 and 55 of the 57 instances have |measured| < |predicted|. That systematic shortfall is the $o(\beta^{-1})$ remainder of Theorem 2 at a finite β, not sampling noise, and it is reported as a bias rather than absorbed into a tolerance. Second, the fraction 57/600 is a property of the prior and measures nothing about how large the set (H1)–(H2) is. The geometric number is the conditional one: of the 219 deaths that fall inside the detector's window, 57, that is 26%, satisfy (H2). The memory gate is a real and restrictive condition, not a formality that a generic trajectory satisfies.

3.6 Verification of the forcing theorem

The sweep above tests Theorem 2 on two-neuron instances. Theorem 6 claims something the sweep cannot see, namely that the number of live units does nothing, so it needs its own campaigns. Five were run, in `05-experimentos/t2-n-neuronios/` (scripts `verifica_forcing.py`, `parecer14.py` and `beta.py`, the fifth being the pre-hit campaign whose own scripts are named with it below); none of them is part of the proof, and we say what each does and does not establish.

(1) The m-sweep with NTK readout. Sixty-six instances pass (H1′)–(H4′) — 23 at $m = 4$, 21 at $m = 8$, 22 at $m = 16$ — against a 57-instance two-neuron baseline, at $\beta = 10^4$, explicit Euler with $dt_0 = 10^{-5}$ on the hybrid and $dt = 2 \cdot 10^{-6}$ on the coupled flow, $H = 1.8$. No free parameter is fitted: the predicted coefficients span $[-109, -14]$, a factor of 7.6, the

measured ones $[-132, -14]$, with $R^2 = 0.950$ against the unfitted law, and an OLS slope of 1.154 ($R^2 = 0.968$). Median relative errors are 0.80%/6.85%/5.58% at $m = 4/8/16$, against 2.68% for the baseline. In an OLS of the error on $(\log_2 m,\ 1/R^{\mathrm{hyb}}(t_{\mathrm{exit}}),\ T_{\mathrm{ep}})$ the coefficient of $\log_2 m$ is not significant ($t = 0.71$) while $1/R^{\mathrm{hyb}}(t_{\mathrm{exit}})$ is ($t = 2.56$); in the joint regression over both readout families ($n = 133$, over which $\lambda_{\max}(G_B)$ ranges from 1.4 to 56.1, so that $e^{L_B H}$ would vary by 42 orders of magnitude) only $1/R^{\mathrm{hyb}}(t_{\mathrm{exit}})$ is significant ($t = 7.44$), with $\log_2 m$ at $t = -0.09$ and $\log \lambda_{\max}$ at $t = -0.47$. This is consistent with the remainder being governed by exit transversality rather than by m; it is not conclusive.

(2) Bit-identical cloning (23 instances). Each accepted $m = 4$ instance is replayed with every live unit split into p copies carrying readout $a_k/\sqrt{p}$ and parameters $(w_k, b_k)/\sqrt{p}$, so that f_B, the NTK and the whole flow — hence $R(t)$ — are unchanged while $m \to pm$. The measured errors are identical to machine precision at $m = 4, 8, 16$ (median 0.80%, maximum 16.17% in all three columns), and $\lambda_{\max}(G_B)$ is likewise invariant. This isolates the claim that at fixed forcing the number of live units does nothing, and it is the closest numerical analogue of Remark 25.

(3) The unit-readout family (67 instances: 22/24/21 at $m = 4/8/16$, median error 1.16%/4.52%/5.35%, maximum $\approx 28\%$ at $m = 8, 16$ and $\approx 20\%$ at $m = 4$). Here $\lambda_{\max}(G_B)$ grows with m and the published constant $e^{L_B H} C_f C_4/c$ would be exponential. The law survives, with the same qualitative degradation pattern; under the PSD constant (12) this is expected, under $e^{L_B H} C_f C_4/c$ it would not be. This is the empirical case for Lemma 20 being the right estimate.

(4) The β sweep on the worst instances. The six worst $m = 8$ instances of item (1), with errors 11.9–20.5% at $\beta = 10^4$, were rerun at $\beta = 3 \cdot 10^4$ with the same integrator and the same predicted coefficients, which are properties of the hybrid reference and therefore unchanged. The errors collapse to $20.45 \to 0.35$, $20.20 \to 4.93$, $17.30 \to 0.20$, $13.52 \to 4.23$, $12.62 \to 0.30$, $11.88 \to 0.50$ (per cent), on every instance. This is the sharpest available evidence that the $m = 8, 16$ degradation of item (1) is a β_0 effect — the asymptotic regime bites later — and not a failure of the law.

The two exponents of the pre-hit gap, measured. Step 4 of Lemma 20 predicts two *different* rates at the hit: $\rho_A(t_{\mathrm{hit}}) = O(\beta^{-1})$, because the A-block of the defect is a derivative defect and carries no β^{-1}, and $\rho_B(t_{\mathrm{hit}}) = O(\beta^{-2})$, because the B-block sees only A's *output* defect, which does carry a β^{-1}. The one-power separation is the whole content of the step and is falsifiable directly. Twelve instances (4 at each of $m = 4, 8, 16$, drawn from the accepted set of item (1)) were integrated from $t = 0$ to t_{hit} twice, once with the ReLU field and once with the softplus field, with an RK4 stepper at $dt = 2 \cdot 10^{-6}$ so that the integration error sits well below β^{-2} (scripts `parecer15_o5.py`, `parecer15_o5b.py`). Sweeping $\beta \in \{10^3, 2 \cdot 10^3, 4 \cdot 10^3, 8 \cdot 10^3\}$, both compensated quantities *saturate*, which is the signature of the exponent being exactly right: $\beta\rho_A$ rises to a plateau near 0.53 (for instance $0.404 \to 0.501 \to 0.526 \to 0.528$ on one $m = 4$ instance and $0.426 \to 0.510 \to 0.529 \to 0.530$ on one $m = 8$ instance), and $\beta^2 \rho_B$ likewise flattens. Instances with small $\beta t_{\mathrm{hit}} v$ saturate later, as they must: the estimate is asymptotic in the number of e-foldings available before

Table 2: The pre-hit B-gap at $\beta = 8 \cdot 10^3$, compensated by the exponent and the arrival speed that step 4 of Lemma 20 predicts. Over the twelve instances $\beta^2 v\, \rho_B = 0.198 \pm 0.046$, with no trend in m: the $m = 4$ mean is the largest and the $m = 8$ mean the smallest. What sets the constant is the transversality of the entry, not the size of the live subsystem. Scripts `parecer15_o5.py`, `parecer15_o5b.py`.

	$\beta^2 v\, \rho_B(t_{\mathrm{hit}})$, per instance				mean
$m = 4$	0.202	0.253	0.249	0.274	0.244
$m = 8$	0.203	0.176	0.175	0.119	0.168
$m = 16$	0.135	0.206	0.212	0.169	0.180

the hit. The constant is predicted to be governed by the arrival speed $v = c|R^{\mathrm{hyb}}(t_{\mathrm{hit}})|$ and *not* by m; Table 2 is that prediction measured.

The gap between the numerics and the theorem. In every instance generated by these designs, R^{hyb} changes sign *once* inside the episode, because the acceptance filter selects a single entry–exit episode with $\mathcal{A}^{\mathrm{hyb}} < 0$ throughout. Lemma 23 imposes no such restriction: (H2′) constrains the zeros of the *area*, not of R, so the theorem covers forcings that oscillate arbitrarily often inside the episode and the numerics do not test that regime — the numerical filter is strictly stronger than the hypothesis. Conversely, the acceptance rate of (H1′)–(H2′) is about 3% of random initializations in the NTK-readout family and about 20% (179/900) in the unit-readout family, so the hypotheses are verifiable a posteriori by one hybrid integration but are not guaranteed a priori by any initialization criterion we know. Finally, the filter of `verifica_forcing.py` rejects any instance in which A dies more than once, so the last clause of (H4′) is not merely verifiable in principle: it is the condition the experiments were already screening on.

3.7 Independence of the smoothing

The release criterion depends only on $\int R\, dt = 0$ and not on the sigmoid. For the erf class, $g'_\beta = \mathcal{N}(\beta z)$, the coupled system releases at the same t_{exit} with an $O(\beta^{-1})$ gap; this is verified numerically and not proved. GELU itself, $z\,\mathcal{N}(z)$, has a non-monotone derivative and is not a member of the family for any β; Section 6.6 shows that this is a distinct mechanism rather than a technicality.

4 Non-commutation and the plasticity bridge

Theorem 2 is the flow-first limit: one takes $\beta \to \infty$ of trajectories. The field-first order instead flows the pointwise-limit generalized gradient, and it freezes degenerate neurons forever. That branch is the rigorous one of Eberle, Jentzen, Riekert and Weiss [6, Theorem 3.3], whose selected solution is unique and has a degenerate set that is non-decreasing in

time; their degenerate stratum is the fully-zero one rather than our functionally dead neuron, but the convention $\mathrm{ReLU}'(0) = 0$ and the no-revival mechanism are the same.

Proposition 15 (the two limit orders do not commute)**.** *Let L_∞ be the loss of the hard-ReLU network and let $\nabla^0 L_\infty$ be its pointwise-limit gradient, with the convention $\mathrm{ReLU}'(0) = 0$; let θ^{FF} solve $\dot\theta = -\nabla^0 L_\infty(\theta)$ from $\theta^{\mathrm{hyb}}(t_{\mathrm{hit}})$, which exists and is unique by [6, Theorem 3.3]. Assume (H1)–(H4). Then $\theta_A^{\mathrm{FF}} \equiv \theta_A(t_{\mathrm{hit}})$ on $[t_{\mathrm{hit}}, H]$, whereas θ^∞ of Definition 1 releases A at $t_{\mathrm{exit}} < H$. The two limits therefore disagree on $(t_{\mathrm{exit}}, H]$.*

Proof. The A-block of $\nabla^0 L_\infty$ at θ is $-\frac{1}{n}\sum_i \mathrm{res}_i\, \mathbf{1}_{\{z_{A,i}>0\}}(x_i, 1)$, which vanishes identically whenever every margin of A satisfies $z_{A,i} \le 0$. At t_{hit} the margin on datum j is zero and, by (H3), the remaining margins are $\le -\delta_1 < 0$, so the A-block vanishes and $\dot\theta_A^{\mathrm{FF}} = 0$. The margins of A are functions of θ_A alone, so the condition $z_{A,i} \le 0$ for all i is preserved as long as θ_A is constant, and the set of such θ is therefore forward invariant: θ_A^{FF} stays at $\theta_A(t_{\mathrm{hit}})$ on all of $[t_{\mathrm{hit}}, H]$. By Definition 1 and (H2), θ^∞ restarts the active flow at $t_{\mathrm{exit}} < H$ with $\dot z(t_{\mathrm{exit}}^+) = cR^{\mathrm{hyb}}(t_{\mathrm{exit}}) > 0$, so $z > 0$ immediately after t_{exit}. □

Proposition 16 (the bridge to an MLP of the affine class, as a corollary)**.** *Let a one-hidden-layer network with $m+1$ units,* fixed *readout and a one-dimensional input be trained by gradient flow on $[0, H]$, and let A be one of its units. Suppose that on a window $W \subseteq [0, H]$ containing the death and the return of A the live activation pattern is constant with margin floor δ_B, that the other margins of A stay $\le -\delta_1 < 0$, that A switches exactly twice on $[0, H]$, and that the hybrid reference computed with θ_A frozen satisfies (H1′)–(H2′) and (H0′), (H5′). Then Theorem 6 applies verbatim: for $\beta \ge \beta_0 = O(m)$ the unit is released at $t_{\mathrm{exit}} - 1/(c\beta R^{\mathrm{hyb}}(t_{\mathrm{exit}})) + o(\beta^{-1})$, where t_{exit} is the first zero of the area of the residual excluding A.*

Proof. The hypotheses of the proposition are (H0′)–(H5′) with (H4′) read as the definition of the window (Remark 5), and a one-hidden-layer network with fixed readout, constant live pattern and only the first layer trained is affine of order m in the sense of Section 3.1, with G_B the Gram (7). Apply Theorem 6. □

The content of Proposition 16 is that it is not an analogy. In the previous version of this paper the passage from the theorem to the MLP measurements of Section 6 was declared empirical without qualification; with Theorem 6 it splits into the three levels (P1)–(P3) of Section 1. For an MLP of the affine class — readout held fixed — the law is deduced, conditionally on hypotheses that are checked a posteriori by one hybrid integration and not assumed: (H1′)–(H2′) on the hybrid trajectory and the window of (H4′). What remains empirical is precise, and it is the part that carries the measurements actually reported here: the readout v_k is trainable in every experiment of Section 6, which takes the live field outside the class in which $-G_B\Delta$ is an identity — this is the open case of the trainable readout, (O6) of Section B.10, where an $e^{L_B T_{\mathrm{ep}}}$ factor, with the episode length rather than the horizon, is expected to return; several data are active at once; and the transformer of Section 6.5 is a different architecture altogether. We do not claim the corollary covers any number reported in Section 6. It covers the object those numbers are evidence about.

Returning to the non-commutation itself: in machine-learning terms it is the statement behind an empirical trend. Discrete ReLU units that die stay dead — an exact statement about a unit's *own* channel under a fixed input, which is the setting of Proposition 15 — and this is the setting in which the dormant-neuron and loss-of-plasticity phenomena are usually read [1, 2, 3], mitigated in practice by recycling dead units or by migrating to smooth activations; the smoothed flow resurrects them by the area law. The channel that reopens a smoothed unit is $e^{\beta z}$, which closes as $\beta \to \infty$. Section 6.4 measures the resulting staircase directly. The qualification matters: once the input of the layer is itself moving, terminality is no longer implied, and in our transformer data 20% of injected hard-ReLU units return (20/100 against 0/100 with the upstream block frozen); Section 6.6 isolates that second channel.

5 Related work

Entry–exit and geometric singular perturbation. The release criterion $\int R\,dt = 0$ is the classical entry–exit function of slow–fast systems [5, Theorem 1.1], applied for instance to a neurotransmitter release model in [8], and Lemma 9 places our episode in the quadratic normal form of [5, equations (1.9)–(1.10)] with $\varepsilon = \beta^{-1}$. Our switching face is degenerate in a way that literature does not cover: it carries an identically zero field on the dead side. Generic piecewise-smooth exit theory [9, 10, 11] assumes a nonzero one-sided field, and the canonical Sotomayor–Teixeira regularization [12, 13] requires both one-sided normal components to be nonzero. The degeneracy is of the exponentially flat type treated by Kristiansen [14] with the blow-up method of Krupa and Szmolyan [15], but there the forcing is autonomous, whereas here the residual is generated by the very trajectory it gates. We are not aware of an analogue of that coupling in the entry–exit literature.

Gradient flows of ReLU networks. Existence, uniqueness and convergence rates for gradient flows of ReLU networks are established in [6], whose selected solution is the field-first branch of Proposition 15: once a neuron is degenerate it stays degenerate. Our result is the complementary, flow-first, selection, and the two disagree. The branch is fixed in practice by a convention of the differentiator, and that convention has a name: in the framework of Bolte and Pauwels [16] the derivative returned by automatic differentiation is a definable selection of a conservative field for the loss, and $\mathrm{ReLU}'(0) = 0$ is one such selection. Their Remark 13 also shows that the nondifferentiability locus $z = 0$ is reached with nonzero probability, so which branch the implementation picks is not a detail of measure zero: it is the choice that Proposition 15 shows the two limits disagree about.

Dormant neurons and loss of plasticity. Dead or dormant units are a measured pathology of trained networks: the dormant neuron phenomenon in deep reinforcement learning [1], the loss of plasticity under distribution shift [2], and its persistence in deep continual learning [3]. Death at initialization is a separate and better understood matter: Lu, Shin, Su and Karniadakis [17] prove born-dead results for ReLU networks at initialization

and study the initializations that avoid them. We invoke them only for that regime; every confirmatory count in this paper conditions on *dynamical* deaths, units that die during training, and born-dead units are excluded by the instrument. The practical mitigations are recycling dead units and smoothing the activation. Theorem 2 and Proposition 15 say what the second mitigation buys: a channel of width $e^{\beta z}$ and a return time set by an area.

Dormancy is not death. The two objects should not be conflated. Dormancy in [1] is a normalized activation score falling below a threshold: a unit can be dormant and still have a nonzero column in J. Our object is the exact stratum on which the unit's column of J is identically zero on every datum. The sets are different, and the apparent tension with deep reinforcement learning — whose input is non-stationary by construction, and where dormancy nonetheless persists — dissolves once they are kept apart: a low-score unit need not be on the stratum, so nothing in this paper predicts that it must return. Our results separate two independent channels. The *own* channel requires smoothness of the activation, and its width is the factor $e^{\beta z}$, which the staircase of Figure 5 measures arm by arm; for hard ReLU that width is exactly zero. The *upstream* channel requires only that the layer's input move, and it operates for every activation we tested, hard ReLU included.

6 Experiments

The claims are tested by prediction, each with its null control, and the failed designs are reported alongside the successful ones. Every confirmatory analysis was frozen by a pre-registration commit before the run, and hypotheses that fell were archived rather than rewritten. The four instruments that carry these claims are stated as pseudocode in Section C. The commits are: pressure pilot `b48b527`; rate pilot `995a87c`; 4a.3 pre-registration `a8f8f8c`; 4a.3 confirmatory `5a961cb`; transformer pre-registration `1b45498`.

Statistics. Proportions are reported as a point estimate with a two-sided 95% Clopper–Pearson interval [18]; the variability captured is the Bernoulli count of returns among deaths within a single arm, with the denominator given in each cell. Event rates λ, dynamical deaths per run, are reported with a 95% percentile bootstrap interval over $B = 2 \cdot 10^4$ resamples clustered by seed, so that the variability captured is seed-to-seed and not episode-to-episode. Nothing is pooled across arms. We write T for the SGLD temperature, that is, the scale of the injected noise; $T = 0$ is deterministic gradient descent, and H remains the training horizon in optimizer steps.

6.1 The deterministic schedule is falsified under SGLD

Before any confirmatory claim we tested whether the deterministic entry–exit schedule survives noise. A reduced "race" model, pre-registered at commit `190d94a` after two adversarial-review rounds, predicts the capture (resurrection) probability of a dead unit from the mean residual schedule; the falsifying run of 2700 runs under SGLD [19] rejected it.

The predicted-zero window was violated decisively: 139 late captures occurred where the model gives exactly zero, and the bin $[0, 0.1)$ saw 150 captures against 1.4 predicted (Poisson $z = +126$ against the model's own expected count, under the model's own independence assumption). The model is well calibrated where it says capture is likely, in the $(0.5, 0.9)$ bin, with $z = -0.26$; the entire error is on the "impossible" side. The theorem-part of the model held: in all nine cells the observed censoring of 2–7% matched the censoring rate that the reduced model itself predicts, from a Brownian first-passage approximation to the margin inside the cone of the deterministic schedule. The failure is therefore of the hypothesis, a deterministic schedule, and not of the instrument.

A follow-up note records what happened next. The natural fix, a stochastic Ornstein–Uhlenbeck schedule with no free parameter, predicts p(capture) nearly identical to the deterministic one, with differences ≤ 0.013, and a late-capture rate of 0.1–1.5% against the $\sim 13.8\%$ observed, which is insufficient by an order of magnitude. That model was therefore refuted at the predictor stage and never run; the leading candidate became the back-reaction of the reactivating unit on the shared residual. The stochastic gate remains open and is out of scope for the theorem, whose regime is deterministic gradient flow.

6.2 Two instrument designs failed their own null control

This subsection is the most transferable part of the paper: any pipeline that counts rare unit-level events during training — dormancy dashboards included — can commit the two errors documented here, and the calibrated null that catches them is reusable as stated in Section C. The first attempt to see the law in a real MLP failed on the instrument, and the failure was caught by a pre-registered null (the first MLP stage, labelled `etapa-4a` in the repository; commit `6411ac8`, 600 runs, MLP with $m \in \{16, 64, 256\}$, SGLD with $\eta = 3 \cdot 10^{-4}$). The pure-diffusion arm *fired the calibrated null* ($m = 16$, $p = 0.002$) — the control that must stay silent did not; by the pre-registered rule the ruler is then biased and no scientific verdict on the substantive predictions may be claimed, so the run is invalid. The diagnosis is in Figure 3: the ruler was measuring boundary chatter rather than the physical death of the theorem. Of all episodes 59% lasted ≤ 5 steps, and hard ReLU "resurrected" at the same rate and with the same duration distribution as softplus at $T > 0$, which is impossible by gradient, since a dead ReLU unit has exactly zero gradient; these were noise threshold-crossings. The real physics lives at $T \to 0$: the only genuinely long episodes sit in the $T = 0$ arm, where just 2 of 461 episodes completed inside the 10^5-step horizon (672 and 1548 steps; 99.6% censored). The calibrated null (Algorithm 2, whose retention step is the clean-episode filter of that ruler) held its false-positive rate near α in i.i.d. worlds, but the real world has per-neuron heterogeneity.

A redesign, `etapa-4a.2`, was itself rejected at the design stage by the same process (commit `a413b19`): at $T = 10^{-5}$ the ReLU-to-softplus ratio stayed ≈ 0.95 after debounce, because gradient starvation at $\beta = 300$ masked the effect, and the pooled null broke α by 30–92$\times$ in heterogeneous and autocorrelated worlds. The inheritance was explicit: a hysteresis and dwell ruler, a per-neuron stratified null, and a move to the deterministic regime $T = 0$ with a much longer horizon. These negatives falsify neither the area law nor

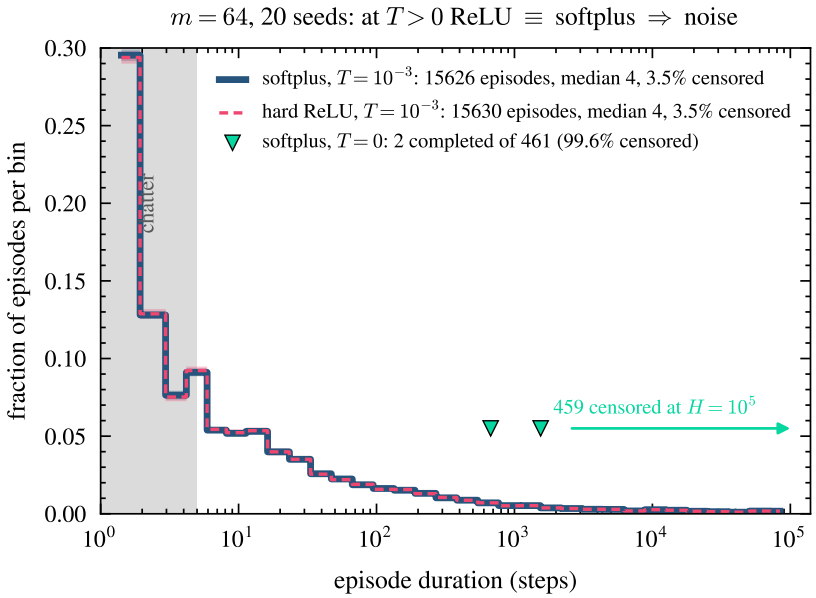


Figure 3: Why `etapa-4a` is an instrument negative (post-hoc diagnostic, $m = 64$, 20 seeds). Episode durations as the fraction of episodes per logarithmic bin; the bands are ± 1 standard error across the 20 seeds. At $T = 10^{-3}$ hard ReLU (dashed) lies on top of softplus (solid) — same median of 4 steps inside the shaded chatter band, same tail — which cannot be the zero-gradient theorem at work, since a dead ReLU unit has exactly zero gradient; these are noise threshold-crossings. The $T = 0$ arm completed only 2 of its 461 episodes within the same horizon, so those two are drawn as individual markers rather than as a density, and the remaining 99.6% are censored at $H = 10^5$; that is where the real physics sits and why the confirmatory moved to $T = 0$ with a longer horizon. Script `etapa4-mlp-populacao/figuras_diagnostico.py`.

the toy calibration. They falsify two rulers, and they are what forced the working design.

6.3 A softplus unit dies and returns in a real MLP, and the area opens first

Moving to $T = 0$ and to a sub-parametrized regime made the mechanism visible. The regime is the *pressure* pilot: a plateau at ~ 0.15 on the rich target $y = \sin(2\pi x) + 0.3\sin(5\pi x)$ with $n = 16$, $\eta = 10^{-3}$ and $H = 2 \cdot 10^5$; 64 runs, commit `b48b527`, exploratory. Hard ReLU gave a structural zero: 0 resurrections out of 25 deaths across 16 seeds, so the frozen branch of Proposition 15 is exact in a real network at $T = 0$. Softplus resurrected by mechanism: at $\beta = 10$ a unit died at step 781 and returned at 139 179; at $\beta = 300$ one died at 3315 and returned at 152 977, having parked shallow enough that the $e^{\beta z}$ channel did not vanish. Both dynamical resurrections showed the ordered triple $t_{\text{death}} < t_{\text{area}} < t_{\text{res}}$, so the area law

Table 3: Confirmatory 4a.3: 400 runs, 100 per arm, $T = 0$, $H = 5 \cdot 10^5$. λ is the number of dynamical deaths per run. $\hat{p}_{\text{death}}$ conditions on all dynamical deaths in the arm; $\hat{p}_{\text{elig}}$ conditions on the deaths that occur inside the eligibility window $[0, 307\,536]$, which is the only conditioning for which the return is observable within the horizon (for sp-β3 the eligible denominator is $N_{\text{elig}} = 30$ with 15 returns). Brackets are 95% Clopper–Pearson intervals, except the λ column, which carries a 95% bootstrap interval clustered by seed. ReLU is the validity control: any ground-truth return there would invalidate the run.

arm	dyn. deaths	λ/run [boot]	returns	$\hat{p}_{\text{death}}$	$\hat{p}_{\text{elig}}$
sp-β3	35	0.350 [0.220, 0.500]	16	0.457 [0.288, 0.634]	0.500 [0.313, 0.687]
sp-β10	27	0.270 [0.160, 0.390]	10	0.370 [0.194, 0.576]	0.385 [0.202, 0.594]
sp-β300	19	0.190 [0.100, 0.290]	4	0.211 [0.061, 0.456]	0.222 [0.064, 0.476]
ReLU	17	0.170 [0.090, 0.260]	0	0.000 [0.000, 0.195]	0.000 [0.000, 0.195]

held 2/2. The gap between gate and return is the finite-β rise time, $\sim |z_{\text{park}}|/\text{drift}$, rather than the asymptotic $O(1/\beta)$ correction. The bottleneck was quantified as an event rate of ~ 0.1 per run, and that number set the entire sizing of the confirmatory.

Figure 4 shows the $\beta = 10$ episode as it happened, in the three coordinates the theorem talks about: time, margin, accumulated area. The qualitative shape of Figure 1 — dive, sticking layer, debt, zero of the area, release — appears in a trained network with no scalar reduction and no smoothing of the data, only with the return lagging the gate by the finite-β rise time instead of by $O(1/\beta)$. The scope of this transfer is narrow: in this episode 68% of the lifetime elapses *after* the gate opens, so at $\beta = 10$ the asymptotic release formula is far from its regime. What transfers to the trained network is the *ordering* $t_{\text{death}} < t_{\text{area}} < t_{\text{res}}$ and the gate's necessity, not the $O(1/\beta)$ release-time expansion.

6.4 The area law reproduces under a pre-registered confirmatory

The confirmatory, `etapa-4a.3`, was pre-registered at commit `a8f8f8c` after two further adversarial rounds, with every design number derived programmatically from a 240-run rate pilot (commit `995a87c`); the run is commit `5a961cb`. Design grid: {sp-β3 (primary), sp-β10, sp-β300, ReLU} $\times\, m \in \{4, 6\} \times 50$ held-out seeds (1900–1949, not used in any pilot), for 400 runs at $T = 0$ and $H = 5 \cdot 10^5$. The eligibility window is $[0, 307\,536]$, that is $H - \tau$ with τ the $q_{0.9}$ return time of the pilot. The instrument is a single ground-truth clock with bit-exact parity to the pilot motor (Algorithm 1), read by the pre-registered decision procedure of Algorithms 4 and 5, and the area of the leader-datum residual is accumulated on that clock, so temporal ordering is structural and the only empirical failure mode of the area test is a return without area.

The verdicts (Tables 3 and 4 and Figure 5) are as follows.

- **C1, existence with rate, primary arm sp-β3: reproduces.** With $N_{\text{elig}} = 30$ against the floor $n^* = 10$, there are 15 eligible returns and $\hat{p}_{\text{elig}} = 0.500\,[0.313, 0.687]$, well above the kill threshold 0.076. Over all 35 dynamical deaths the same 16 returns

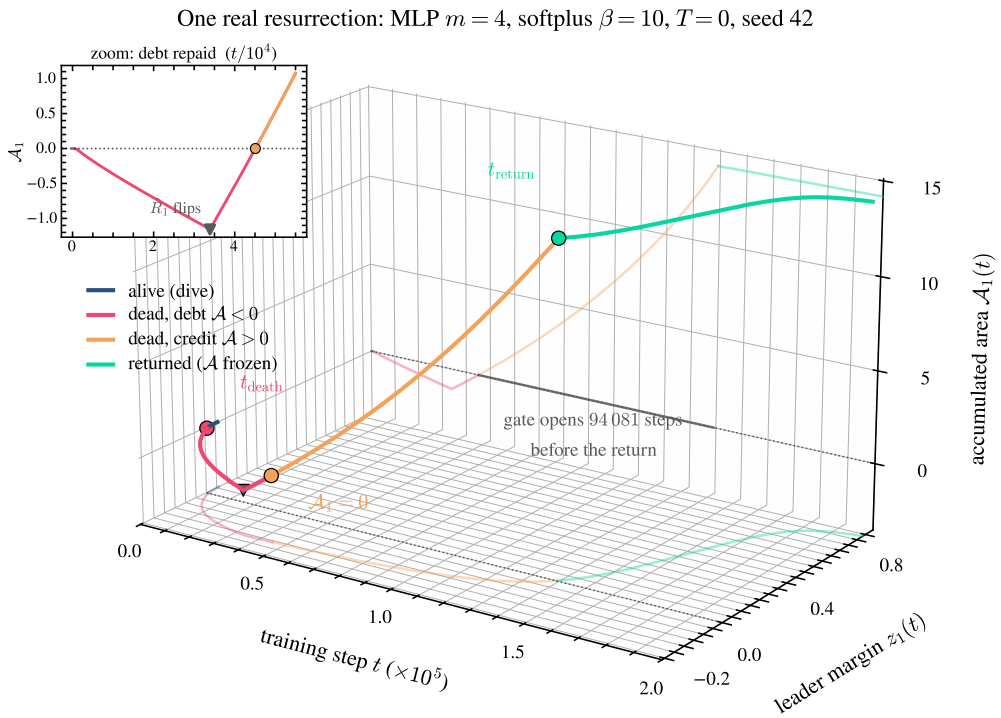


Figure 4: One real resurrection, drawn from the recorded trajectory of the run described above (pressure pilot, softplus $\beta = 10$, $m = 4$, $n = 16$, $\eta = 10^{-3}$, $T = 0$, $H = 2 \cdot 10^5$, seed 42, unit $k = 1$). The axes are the training step t, the leader margin $z_1(t)$ whose sign decides life or death, and the ruler's accumulated area $\mathcal{A}_1(t) = \eta \sum_{s=t_{\text{hit}}}^{t} \hat{R}_1(s)$; the pale curves on the two walls are the (t, z_1) and $(t, \mathcal{A}_1)$ projections. The unit dives and dies at $t = 781$ (t_{hit}). The area then runs into debt while the projected residual $\hat{R}_1$ is adverse, turns at $t = 33\,905$ where $\hat{R}_1$ changes sign (inset), and comes back to zero at $t = 45\,098$: the gate. The margin only then crawls up out of the sticking layer and crosses zero at $t = 139\,179$, so the gate opens 94 081 steps before the return — the ordering $t_{\text{hit}} < t_{\text{area}} < t_{\text{res}}$ of Theorem 2, with the lag being the finite-β rise time. The ruler stops accumulating at the return, which is why $\mathcal{A}_1$ is flat afterwards. Nothing is fitted or smoothed: the curve is the trajectory itself, sub-sampled every 40 steps, and the four marked steps are exact. Script `figura_trajetoria3d.py`, which re-runs the cell and refuses to plot unless the committed triple (781, 45 098, 139 179) is reproduced exactly.

Table 4: The pre-registered questions of 4a.3, with the condition that kills each one. All thresholds were derived programmatically from the rate pilot and frozen in `config_4a3.json` at commit `a8f8f8c`, before the run.

label	estimand	condition that kills	verdict
C1	$p(\text{ret} \mid \text{eligible})$, sp-$\beta 3$	CP upper < 0.076, $N_{\text{elig}} \geq 10$	reproduces
C2	returns with area first, sp-$\beta 3$	CP lower < 0.5, ≥ 6 returns	sustains
C3	rise gap against park depth	none (estimation)	not aggregated
C4	$p(\text{ret} \mid \text{death})$ against β	none (estimation)	monotone in β
V	returns in the ReLU arm	any single return	clean, 0/17

give $\hat{p}_{\text{death}} = 0.457\,[0.288, 0.634]$; the two numbers are two conditionings of one count, not two estimates of one quantity.

- **C2, the area law, sp-$\beta 3$: sustains.** The leader-drive area opens before the return in 13/16 dynamical resurrections, with lower Clopper–Pearson bound $0.544 > 0.5$ on ≥ 6 returns. The ordering itself is partly structural: the clock accumulates the area only on dead steps and closes at the return, so $t_{\text{area}} < t_{\text{res}}$ cannot fail whenever the area opens at all (Section C.1). The empirical content is that the area in fact turned positive in 13/16 cases; the 3 returns without area are real and are counted against the law: they are returns driven off the leader datum, which the single-datum hypothesis (H3) does not cover; a per-datum area $\mathcal{A}_i$ is the natural extension. What this instrument can say about the gate stops there, as the post-hoc aggregation of Section E makes explicit.

- **C3, the rise gap: estimation, not aggregated.** C3 was pre-registered as an estimation-only quantity with no kill condition, relating the gap $t_{\text{res}} - t_{\text{area}}$ to the park depth z_{park} and to the channel $\sigma(\beta z_{\text{park}})$. Those quantities are recorded per episode by the instrument, but the frozen analysis script contains no aggregate for C3. A post-hoc aggregation, produced after the confirmatory data were seen and labelled as such, is reported in Section E: it finds no evidence that the opening of the area predicts the return time beyond the onset of the episode. The episode's mean drive was never persisted by the instrument, which is why the question cannot be pushed further; persisting the drive series is inherited as a requirement for future experiments.

- **C4, channel staircase: estimation, monotone in β.** $\hat{p}_{\text{death}}$ falls $0.457 > 0.370 > 0.211 > 0.000$ from $\beta = 3$ to hard ReLU, with the intervals of Table 3. The pairwise intervals overlap; the pre-registered protocol declared C4 an estimation with no kill condition, so no trend test was frozen in advance. Conditioning on death when β also moves the death rate (35/27/19/17 dynamical deaths across the four arms) introduces a collider bias in $\hat{p}_{\text{death}}$; C4 was already estimation-only, and this note makes the caveat explicit. As *post-hoc supporting analysis* we report the Cochran–Armitage trend test across the four ordered arms: $z = -3.49$, $p = 4.9 \cdot 10^{-4}$ (pairwise Fisher, $\beta = 3$ vs. ReLU: $p = 8.1 \cdot 10^{-4}$; $\beta = 3$ vs. $\beta = 300$ alone: $p = 0.086$, not significant —

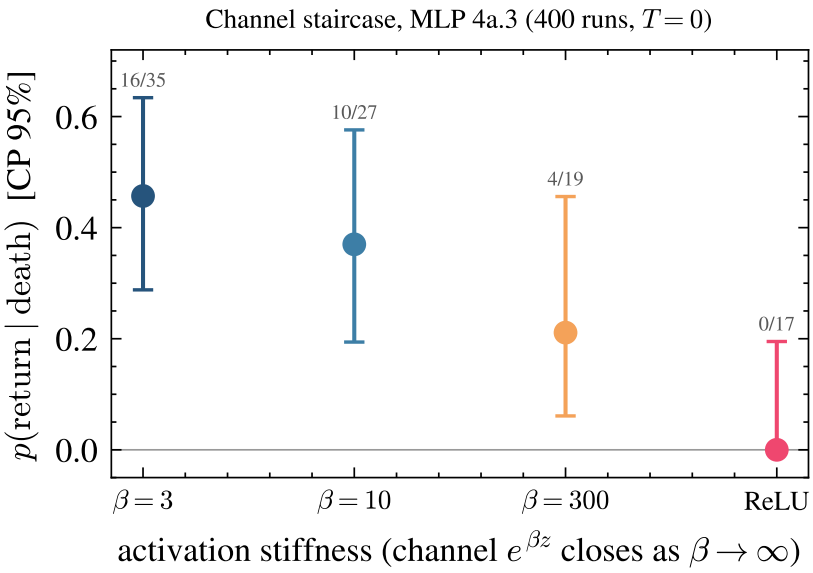


Figure 5: The channel staircase (C4, confirmatory 4a.3, 400 runs, $T = 0$). $\hat{p}_{\text{death}}$ is monotone in the activation stiffness and reaches a structural zero at hard ReLU, where the gradient vanishes identically; the smooth channel $e^{\beta z}$ closes as $\beta \to \infty$. The fractions above each point are returns over deaths and the error bars are 95% Clopper–Pearson intervals. Script `figuras_paper.py`.

the trend test, not pairwise comparison, is the appropriate instrument for an ordered dose–response). Likewise for C2, a post-hoc exact binomial test of 13/16 against 0.5 gives one-sided $p = 0.011$ (two-sided 0.021); the pre-registered criterion was the Clopper–Pearson bound, which these tests corroborate but did not replace. The activation smoothness controls the return channel, and $\beta = 3$, the stiffness scale of the GELU family, is the arm with the highest return probability. This is the direct bridge to the plasticity practice of Section 4.

- **Validity: clean.** ReLU gave 0/17 dynamical returns, which is Proposition 15 measured with $N = 100$ runs.

The limits of the confirmatory are $T = 0$, one small architecture, a one-dimensional rich target, and $m \in \{4, 6\}$. The stochastic regime is a paper of its own (Section 6.1). The kill condition of C1 is intentionally near-unreachable at this N, a consequence of the frozen rule that halves the pilot lower bound, so in practice C2 is the deciding test; this was signalled in the pre-registration rather than discovered afterwards.

6.5 The theorem class transfers to a transformer feed-forward block

The same question in a one-layer transformer, for the feed-forward block over an attended token, does not reduce to the MLP. The scoping run (exploratory, commit `d4e4c35`) exposed two mechanisms that reshaped the design, a GELU sign-flip barrier and an upstream resurrection channel; both are stated in Section 6.6. The confirmatory therefore tests the theorem class, softplus-β, on the block, and treats the barrier and the upstream channel as estimands of their own.

A rate pilot of 160 runs (commit `2dba27d`) then fixed the design. Natural FFN death is ≈ 0, since units with $d = 16$ input do not die on their own, so the confirmatory is necessarily injection-conditional; the non-stationary channel chatters, forcing the hysteresis instrument of Algorithm 3, with $\delta_h \approx 0.0476$ calibrated from data; and the return time is slow, $\tau \approx 60\,000$–$66\,000$ steps, forcing $H = 2 \cdot 10^5$. The upstream channel was already clean in the pilot: the difference between training and frozen attention was $+6/+6/+6/+8$ returns, and ReLU went from 0 to 8 with own-move 0.

The pre-registered protocol. A confirmatory run for the transformer was pre-registered at commit `1b45498`: 800 runs over {sp-β3, sp-β10, GELU, ReLU} $\times$ {attn-train, attn-frozen}$\times$ injection $\{-0.3, -2.0\} \times 50$ seeds 2000–2049, with $H = 2 \cdot 10^5$ and $T = 0$. The grid completed with 800/800 runs (manifest `manifest-e5.json`), with no change to the estimands. The decision procedure is Algorithms 4 and 5. The five pre-registered estimands E-A to E-E are stated verbatim in Section D, so that the protocol is on record ahead of the data.

The verdicts. Table 5 carries the frozen analysis, one row per estimand: the validity control is clean, the theorem class transfers through the unit's own channel, the upstream channel reproduces in all four arms, the β ordering of the MLP staircase repeats inside the block, and the GELU barrier is left consistent-with but confounded with depth. Section D keeps each estimand as it was frozen, with its result attached; Figure 6 shows the whole grid at a glance.

6.6 GELU sits outside the class and the moving input opens a second channel

The scoping run of the block (commit `d4e4c35`) produced two facts that are mechanisms in their own right, and that the confirmatory above carries as E-D and E-B:

- **S1, a GELU sign-flip barrier.** GELU$'(z)$ changes sign at $z^* = -0.7518$, verified analytically: for $z < z^*$ it is negative, so a deep-dead unit is pushed away from the return by its own gradient. Self-resurrection is possible only in the shallow band $(-0.752, 0)$; Figure 7 draws the barrier and both injection depths. The MLP's blind area port, which omits g', gets the sign wrong on 12/12 GELU episodes; the correct instrument uses the slope-weighted area $\mathcal{A}_{\text{drive}}$. GELU is therefore genuinely outside

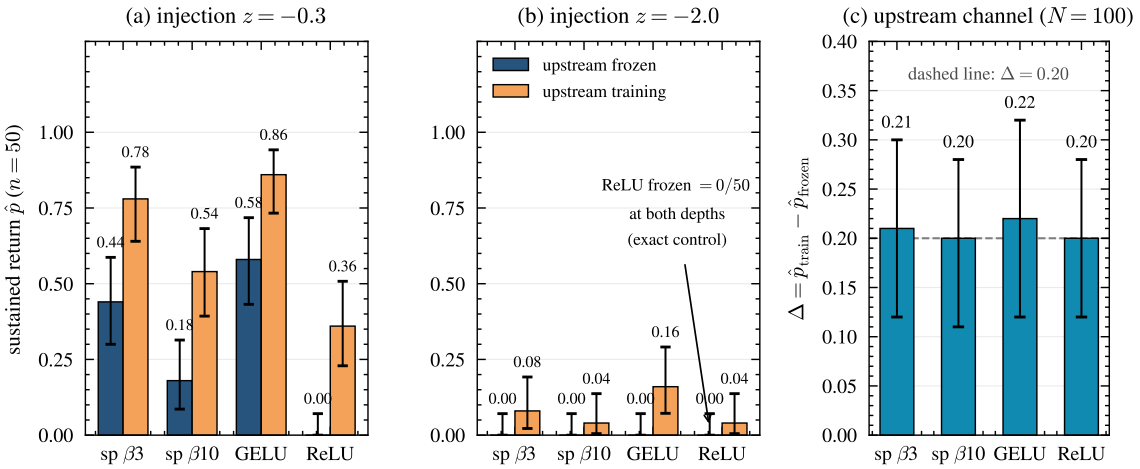


Figure 6: The confirmatory grid on the transformer feed-forward block (800 runs; 50 per activation-by-depth-by-regime cell; hysteresis $\delta_h = 0.0476$, horizon $H = 2 \cdot 10^5$). (a), (b) Sustained-return rate $\hat{p}$ with 95% Clopper–Pearson intervals at shallow ($z = -0.3$) and deep ($z = -2.0$) injection. (c) The upstream channel: $\Delta = \hat{p}_{\text{train}} - \hat{p}_{\text{frozen}}$ pooled over both depths (100 runs per arm, 95% bootstrap-by-seed intervals). Three readings: Δ is uniform at ≈ 0.20 across all four activations, so the upstream channel is not an artefact of any one nonlinearity; relu-frozen is exactly 0/50 at both depths, the designed negative control; and shallow GELU is the highest cell (0.580 frozen, 0.860 training), consistent with GELU having no dead stratum (Figure 7). Script `figura_grade_e5.py`.

the monotone-sigmoid class of Theorem 2. There is a prior point: GELU has no dead stratum to begin with. A unit at margin -0.3 emits $\text{GELU}(-0.3) = -0.115 \neq 0$ and keeps contributing to the residual, so $\{z \leq 0$ on every datum$\}$ is not a singular stratum for GELU but a region of weak, sign-inverted coupling. The sign-flip barrier is the symptom that the definition of death does not apply, and it is consistent with GELU being the most revivable arm we measured (shallow frozen 0.580); we note that against sp-β3-frozen (29/50 against 22/50) the difference is not statistically separable, two-sided Fisher exact test, $p = 0.23$.

- **S2, an upstream channel.** The input of the block, $h = e + \text{attn}(e)$, is non-stationary, so a hard-ReLU unit whose own gradient is identically zero can still revive because h moves. With attention frozen the control is exact: the margin change is 0.000 and the own-move is 0. With attention training the own-move is still 0, yet the margin climbs by $+0.69$ on average and by up to $+1.32$. The MLP statement "dead ReLU is terminal" holds only when the input is fixed; the transformer adds a channel the MLP does not have.

Neither mechanism is a technicality of the instrument. The barrier says that the MLP mechanism does not port unchanged to a GELU block, since the sign-flip obstructs self-

Table 5: Transformer confirmatory (`etapa-5`): 800 runs, $T = 0$, $H = 2 \cdot 10^5$, pre-registered at commit `1b45498`. One row per frozen estimand; the wording of each target is the pre-registration's, reproduced verbatim in Section D. Brackets are 95% Clopper–Pearson intervals for proportions and 95% bootstrap intervals for the differences Δ. Denominators are 50 runs per arm-by-depth cell and 100 per arm when the two depths are pooled.

estimand	pre-registered target	result	interval or test
E-A, validity	no sustained return in the relu-frozen arm	valid: 0/100 sustained returns, both depths	— (a single return would have invalidated the run)
E-B, upstream channel	$\Delta = p_{\text{sust}}(\text{train}) - p_{\text{sust}}(\text{frozen})$ with positive bootstrap lower bound, per arm	reproduced in all four arms: 0.210 (sp-β3), 0.200 (sp-β10), 0.220 (GELU), 0.200 (ReLU)	$[0.120, 0.300]$; $[0.110, 0.280]$; $[0.120, 0.320]$; $[0.120, 0.280]$. ReLU contrast 20/100 against 0/100, two-sided Fisher exact test, $p = 6.6 \cdot 10^{-7}$
E-C, transfer of the theorem class	at least $n^* = 6$ sustained returns, sp-β3 frozen, shallow injection	transferred: 22/50 sustained returns, 3.7× the gate; the leader-drive area opened before the return in 21/22	$\hat{p} = 0.440\,[0.300, 0.587]$; gate $n^* = 6$
E-D, GELU barrier	shallow injection far above deep in the own channel, against a depth-matched control	consistent with the barrier, confounded with depth (the frozen three-way label): GELU 29/50 shallow against 0/50 deep, but sp-β3 deep also gives 0/50	S1 signature, $\text{sign}(\mathcal{A}_{\text{drive}}) \neq \text{sign}(\mathcal{A}_{\text{mlp}})$, in 92% of deep against 4% of shallow episodes
E-E, β staircase	estimation across $\beta \in \{3, 10\}$, frozen and training, no kill condition	$\beta = 3$ above $\beta = 10$ in both factors: frozen 0.220 against 0.090, training 0.430 against 0.290; GELU is the most revivable arm	GELU shallow frozen $0.580\,[0.432, 0.718]$; with the upstream channel $0.860\,[0.733, 0.942]$

resurrection below z^*; the upstream channel says that terminality of a discretely dead unit is a statement about that unit's own channel under a stationary input, and nothing more. In the confirmatory the barrier's terminality could not be separated from injection depth, and it is reported that way (E-D of Table 5); the upstream channel is established in all four arms (E-B).

The deep zero is what the channel law predicts. The own channel of sp-β3 is $\sigma(3z)$, which is 0.289 at the shallow injection $z = -0.3$ and 0.00247 at the deep one $z = -2.0$: a ratio of about 117. The return time measured in the shallow cells is $\tau \approx 60\,000$–$66\,000$ steps, so scaling it by that ratio projects a deep return at roughly $7 \cdot 10^6$ steps, about 35 times the horizon $H = 2 \cdot 10^5$. The 0/50 of sp-β3 frozen at $z = -2.0$ is therefore predicted by the channel law itself, at this horizon, and is not a failure of the transfer. It also disarms part of the E-D confound: the deep suppression of softplus is starvation of a quantitatively predicted size, so the depth-matched control gives 0/50 for a reason that has nothing to do

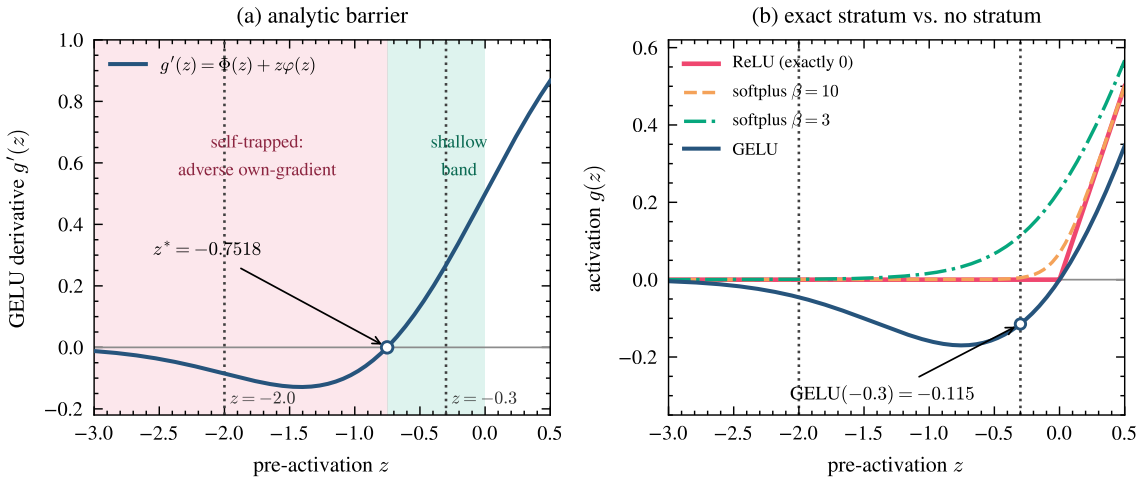


Figure 7: The analytic barrier and the non-stratum, both exact. Left: the GELU derivative $g'(z) = \Phi(z) + z\varphi(z)$ changes sign at $z^* = -0.7518$; below z^* (shaded) the unit's own gradient is adverse and the region is self-trapping, while on the shallow band $(z^*, 0)$ the own gradient still points back toward activity. The two injection depths of the confirmatory, $z = -0.3$ and $z = -2.0$, sit on opposite sides of the barrier. Right: the activations themselves. ReLU vanishes identically for $z < 0$, an exact dead stratum; softplus-β is exponentially small but strictly positive; GELU is strictly negative, with GELU$(-0.3) = -0.115$, so for GELU there is no dead stratum at all and the deep-injection suppression of Figure 6 must come from the sign of g', not from a loss of rank. Script `figura_barreira_gelu.py`.

with the GELU barrier, and the two mechanisms remain separable only by the depth sweep around $z^* = -0.752$ that Section D specifies.

6.7 Deviations from the pre-registered protocol

(i) None for 4a.3: the analysis frozen at `a8f8f8c` was run unchanged at `5a961cb`, and the validity control passed. (ii) The eligibility window $[0, 307\,536]$, the kill threshold 0.076, the floor $n^* = 10$ for C1 and the floor of ≥ 6 returns for C2 were all derived programmatically from the rate pilot (`995a87c`) and frozen in `config_4a3.json` before the run; none was chosen after seeing the confirmatory data. (iii) C3 was pre-registered as estimation-only; the frozen script reports no aggregate for it, and the post-hoc aggregation of Section E is labelled as such (Section 6.4). (iv) For the transformer, the pre-registered package was revised once before freezing: an earlier candidate was rejected in adversarial review for a confound between injection depth and the GELU barrier, and was replaced by the package of commit `1b45498`, which adds a depth-matched control. No run was executed under the rejected package.

7 Limitations

(i) The theorem is proved for the minimal configuration: one unit, one active datum at a time, a fixed readout and a one-dimensional input. Every constant is verified on a reference instance and the release law holds on 57/57 of the admissible instances of a 600-draw sweep (Section 3.5), with a residual bias, 55/57 measured offsets smaller in modulus than predicted, that we read as the $o(\beta^{-1})$ remainder at finite β. Theorem 6 removes the restriction to one live unit — the live side may be any affine subsystem of order m — but not the others: the readout is still fixed, and the experiments train v_k. The resulting gradation is (P1)–(P3) of Section 1, and Proposition 16 is the level at which the transfer is deduced rather than measured; the transfer to the experiments as run remains empirical, and we are explicit about this rather than treating the two as the same object. **(i′)** One step of the proof is open, and it is inherited rather than created by the forcing form: the $o(1)$ in the entry-corner estimate of Lemma 11 is not quantified, so the remainder of (5) is $o(\beta^{-1})$ and not the $O(\log\beta/\beta^2)$ the numerics suggest. It has a concrete empirical signature in the measured regression slope, and the competing explanation was excluded numerically; the argument and the numbers are (O3) of Section B.10. **(i″)** Two further constraints of Theorem 6 are scope and not debt, and are declared as such: the constancy of the live activation pattern (Remark 5), whose relaxation costs an exponential in m ((O2) of Section B.10), and the threshold β_0, exhibited as a chain of five explicit conditions (Table 6) with the binding ones linear in m and deliberately not composed into a single number. What is genuinely uncontrolled there is not the m-dependence but the *forcing* dependence, as (O1) of Section B.10 states. **(i‴)** The trainable readout is the one restriction that separates the proved object from the experiments as run; it is (O6) of Section B.10. **(ii)** The regime is $T = 0$ deterministic gradient flow throughout the theorem and the confirmatory; the stochastic gate is open and, where probed, falsifies the deterministic schedule (Section 6.1), since late captures recur past the deterministic window at a rate the reduced stochastic model under-predicts, leaving the reactivating unit's back-reaction as the leading candidate driver. **(iii)** Scale: the confirmatories are small networks ($m \le 6$ MLP units, $m_{\mathrm{ffn}} = 4$ block units) with rich one-dimensional targets, chosen so that a single unit can be tracked with a ground-truth clock. **(iv)** GELU is outside the monotone class; its own-channel revival is blocked below $z^* = -0.752$, and in the confirmatory the barrier could not be separated from depth (the pre-registered three-way label), because at the deep injection $z = -2.0$ the softplus own channel is starved to 0.9% of its shallow value — a factor of about 117, which projects the deep return past 35 horizons (Section 6.6) — while GELU′ is adverse, so both mechanisms predict the same zero at this horizon; the separating design, a sweep of injection depths around $z^* = -0.752$, is stated in Section D and left as future work. The upstream channel is established; the barrier's terminality remains open. **(v)** The cone apex, of corank ≥ 2, where weight decay produces genuine sliding toward the most singular stratum, is untouched. **(vi)** The independence of the law from the smoothing is proved for softplus and verified numerically for erf; it is not proved for the whole monotone-sigmoid class.

8 Conclusion

We derived an entry–exit law for the death and resurrection of a smoothed unit, verified each of its constants on a two-neuron instance, and tested the release law over a sweep of 600 drawn instances, where it holds on all 57 that satisfy the hypotheses, with a median relative error of 2.7%. We measured the law in a small MLP under a pre-registered protocol: resurrection occurs, the residual area opens before the return in 13 of 16 cases, hard ReLU gives a structural zero, and the return probability is monotone in the activation stiffness. In a pre-registered transformer confirmatory the theorem class transferred to the feed-forward block through the own channel (22/50, area before the return in 21/22), a second upstream channel reproduced in all four arms, and GELU's sign-changing derivative left its signature in 92% of deep episodes while its terminality remained confounded with depth — the pre-registered three-way label, reported as frozen.

What this is, and is not. The value of the law is understanding: it replaces "dead units are terminal" with a mechanism — a gate with memory — in the regime where the mechanism is provable. It is not a training tool. The asymptotics live on the channel $e^{\beta z}$ being exponentially small, and one would expect an adaptive optimizer that divides by the running RMS of the gradient to rescale precisely that smallness. An exploratory pilot on the toy instances (13 instances, `e7-adam-portao` in the repository) measured this expectation instead of assuming it, and it fails in an instructive way: under Adam the gate survives, parked deeper by the optimizer's floor — the release moves to where the channel crosses ϵ, empirically $\beta z_{\text{park}} \approx -\ln(1/\epsilon) + \text{const}$ — and the area law itself reappears as ϵ grows, with relative error 0.24% at $\epsilon = 1$ against 0.017% for plain gradient descent. This is a pilot, not a confirmatory, and momentum-rich production regimes remain unmeasured. What does transfer is smaller and sharper: the sign structure of the activation derivative, which is an exact fact about GELU (Section 6.6) and, with the zero at $z^* = -1.2785$ instead of -0.7518, about SiLU; the calibrated-null instrument of Section 6.2, which any measurement of rare events in training can reuse; and the folklore correction that the terminality of discrete death is a statement about stationary input. The cheapest consequential experiment this paper points at is not a scaling of the theorem but a histogram: the mass of feed-forward pre-activations below z^* in production-scale models, and whether that mass is sticky across training.

9 Reproducibility

All code, configs, per-run manifests and aggregate JSON needed to reproduce every number and figure in this note are in a single Git repository. One policy governs the raw data, in both confirmatories: the per-episode trace files (`results-4a3/*.jsonl.gz`, `results-e5/*.jsonl.gz`) are kept out of version control, while the manifests that checksum them (`results-4a3/manifest-4a3.json`, `results-e5/manifest-e5.json`) are versioned. The traces are byte-reproducible from the pinned configs (deterministic single-threaded CPU runs), so the versioned SHA256 list is what makes a regenerated file verifiable. That

repository is private at the time of writing, so the commit hashes below, while they resolve in it and fix the state of every experiment, are not independently checkable today; public release under the MIT licence is a precondition for submission. All runs are deterministic on CPU with per-run manifests (SHA256, git hash, config), and pre-registration commits precede every confirmatory run in the repository history. No external data were used; every dataset in this note is generated by the scripts listed below.

- **Theorem constants and figures.** `06-notas/t2-prova/`: lemma verifications `passo0--6.py` and `prova_L1--L6.py`; figures via `figuras_draft.py`. The forcing theorem is drafted standalone in `06-notas/t2-prova/teorema-forcing-draft.tex`, which is the mathematical source of Sections B and 3.1; its two adversarial referee reports are `05-experimentos/t2-n-neuronios/parecer-14.md` and `05-experimentos/t2-n-neuronios/parecer-15.md`.

- **Forcing theorem, numerics (Section 3.6).** `05-experimentos/t2-n-neuronios/`: the m-sweep, the cloning replica and the unit-readout family are `verifica_forcing.py` (outputs `forcing_m.jsonl`, `controle_dt.jsonl`, log `run_forcing.log`); the regressions and the unit-readout family are `parecer14.py`, the β sweep is `beta.py`, and the two pre-hit exponents of Table 2 are `parecer15_o5.py` and `parecer15_o5b.py`. Those last three campaigns originally reported to standard output only; their outputs are serialized by the wrapper `gera_artefatos.py`, which invokes them unchanged and writes `unit_readout_m.jsonl` (unit-readout family, one record per evaluated instance, with the screening counts), `beta_sweep.jsonl` (the six worst $m = 8$ instances at $\beta = 10^4$ and $3 \cdot 10^4$) and `o5_prehit.jsonl` (the twelve instances of Table 2, all four β each).

- **MLP mechanism (pilot).** `etapa4-mlp-populacao/pilotos-4a2/pressao/` (the *pressure* regime: sub-parametrized, plateau ~ 0.15); commit `b48b527`. The trajectory of Figure 4 is re-run and plotted by `06-notas/paper/figura_trajetoria3d.py`, which asserts the committed triple before drawing anything; the diagnostic of Figure 3 is `etapa4-mlp-populacao/figuras_diagnostico.py`.

- **MLP confirmatory (4a.3).** `.../pressao/taxa/` (the event-rate pilot and the confirmatory built on it): pre-registration `a8f8f8c`, run `5a961cb`. `python run_4a3.py --workers 4` then `python analise_4a3.py`, which reads `results-4a3/` and `config_4a3.json`. Rate pilot `995a87c`. Staircase figure: `06-notas/paper/figuras_paper.py`.

- **Transformer (etapa-5).** Directory `05-experimentos/etapa5-transformer/`; scoping `d4e4c35`, rate pilot `2dba27d`, confirmatory pre-registration `1b45498`. `python run_e5.py --workers 4` then `python analise_e5.py`.

- **Negatives.** `05-experimentos/NEGATIVOS.md` (the register of negative results) and the `etapa-4a` note record the invalidated instrument (commit `6411ac8`) and the

rejected redesign (`a413b19`); the falsified schedule is `p5-relu-morte-v2/` (commit `190d94a`).

Compute. Every run is single-threaded CPU on an Intel Core i9-9900K, with 4 concurrent workers through Python `multiprocessing` at `nice 19` and no GPU; the only dependency is `numpy` (Python 3.14, numpy 2.4). The 4a.3 confirmatory, 400 runs at $H = 5 \cdot 10^5$, took 4.9 h wall-clock, that is 19.5 CPU-hours at 175 s per run. The transformer confirmatory, 800 runs at $H = 2 \cdot 10^5$, took 13.9 h wall-clock, which includes one stall of the runner and its resumption. The verification of the forcing theorem (Section 3.6) is cheap by comparison: the m-sweep, the cloning replica, the unit-readout family, the β sweep on the six worst instances and the twelve RK4 pre-hit integrations add about 1 h wall-clock in total on the same machine, single-threaded. The preliminary and discarded runs, namely `etapa-4a` (600 runs), the `4a.2` redesign, the pressure pilot (64 runs) and the two rate pilots (240 and 160 runs), are of the same order again and are not otherwise reported.

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Data availability

Everything needed to reproduce every number and figure — code, configs, per-run manifests and aggregate JSON — is held at the commits cited in Section 9, under the trace-file policy stated there. The repository is not public at the time of writing; no external data were used.

Declaration of LLM usage

Large language models were used throughout the work: to draft and copy-edit the text, to run the adversarial reviews, to write the experiment and verification code, and to work out the derivations of Section A. The hypotheses, the pre-registered kill conditions, the decision to declare a design or a hypothesis dead, and the reading of the cited sources are the author's.

A review that only re-read the derivation would be circular, so the checks that carry weight here are the ones that do not read it. Each lemma is re-derived symbolically from scratch by a committed script (SymPy), which recomputes the identity rather than confirming the prose; the convergence table Table 1 refines the integrator until the measured coefficient meets the proved one; and the sweep of Section 3.5 confronts the release law with 600 independently drawn trajectories. These routes test the law against the dynamics, not against the text that describes it, and any of them can fail while the prose stands. The

central identity, Lemma 9, is in addition a computation of about ten lines and is presented in that form so that a reader can redo it by hand.

References

[1] G. Sokar, R. Agarwal, P. S. Castro, U. Evci, *The dormant neuron phenomenon in deep reinforcement learning*, in: Proceedings of the 40th International Conference on Machine Learning, PMLR 202 (2023) 32145–32168. arXiv:2302.12902.

[2] C. Lyle, Z. Zheng, E. Nikishin, B. Avila Pires, R. Pascanu, W. Dabney, *Understanding plasticity in neural networks*, in: Proceedings of the 40th International Conference on Machine Learning, PMLR 202 (2023) 23190–23211. arXiv:2303.01486.

[3] S. Dohare, J. F. Hernandez-Garcia, Q. Lan, P. Rahman, A. R. Mahmood, R. S. Sutton, *Loss of plasticity in deep continual learning*, Nature 632 (2024) 768–774. doi:10.1038/s41586-024-07711-7.

[4] D. Hendrycks, K. Gimpel, *Gaussian error linear units (GELUs)*, preprint (2016), arXiv:1606.08415.

[5] P. De Maesschalck, S. Schecter, *The entry–exit function and geometric singular perturbation theory*, J. Differential Equations 260 (2016) no. 8, 6697–6715. doi:10.1016/j.jde.2016.01.008.

[6] S. Eberle, A. Jentzen, A. Riekert, G. S. Weiss, *Existence, uniqueness, and convergence rates for gradient flows in the training of artificial neural networks with ReLU activation*, Electron. Res. Arch. 31 (2023) no. 5, 2519–2554. doi:10.3934/era.2023128. arXiv:2108.08106.

[7] A. F. Filippov, *Differential Equations with Discontinuous Righthand Sides*, F. M. Arscott (ed.), Mathematics and Its Applications, Kluwer, Dordrecht, 1988. doi:10.1007/978-94-015-7793-9.

[8] M. Sensi, M. Desroches, S. Rodrigues, *Slow–fast dynamics in a neurotransmitter release model: delayed response to a time-dependent input signal*, Physica D 455 (2023) 133887. doi:10.1016/j.physd.2023.133887. arXiv:2302.12852.

[9] M. R. Jeffrey, *Exit from sliding in piecewise-smooth flows: deterministic vs. determinacy-breaking*, Chaos 26 (2016) no. 3, 033108. doi:10.1063/1.4943386.

[10] N. Guglielmi, E. Hairer, *Classification of hidden dynamics in discontinuous dynamical systems*, SIAM J. Appl. Dyn. Syst. 14 (2015) no. 3, 1454–1477. doi:10.1137/15100326X.

[11] N. Guglielmi, E. Hairer, *An efficient algorithm for solving piecewise-smooth dynamical systems*, Numer. Algorithms 89 (2022) no. 3, 1311–1334. doi:10.1007/s11075-021-01154-1.

[12] J. Sotomayor, M. A. Teixeira, *Regularization of discontinuous vector fields*, in: L. Magalhães, C. Rocha, L. Sanchez (eds.), Equadiff 95: Proceedings of the International Conference on Differential Equations (Lisboa 1995), World Scientific, Singapore, 1998, pp. 207–223.

[13] J. Llibre, P. R. da Silva, M. A. Teixeira, *Regularization of discontinuous vector fields on* $\mathbb{R}^3$ *via singular perturbation*, J. Dynam. Differential Equations 19 (2007) no. 2, 309–331. doi:10.1007/s10884-006-9057-7.

[14] K. U. Kristiansen, *Blowup for flat slow manifolds*, Nonlinearity 30 (2017) no. 5, 2138–2184. doi:10.1088/1361-6544/aa6449. arXiv:1603.01821.

[15] M. Krupa, P. Szmolyan, *Extending geometric singular perturbation theory to nonhyperbolic points—fold and canard points in two dimensions*, SIAM J. Math. Anal. 33 (2001) no. 2, 286–314. doi:10.1137/S0036141099360919.

[16] J. Bolte, E. Pauwels, *Conservative set valued fields, automatic differentiation, stochastic gradient method and deep learning*, Math. Program. 188 (2021) no. 1, 19–51. doi:10.1007/s10107-020-01501-5. arXiv:1909.10300.

[17] L. Lu, Y. Shin, Y. Su, G. E. Karniadakis, *Dying ReLU and initialization: theory and numerical examples*, Commun. Comput. Phys. 28 (2020) no. 5, 1671–1706. doi:10.4208/cicp.OA-2020-0165. arXiv:1903.06733.

[18] C. J. Clopper, E. S. Pearson, *The use of confidence or fiducial limits illustrated in the case of the binomial*, Biometrika 26 (1934) no. 4, 404–413. doi:10.1093/biomet/26.4.404.

[19] M. Welling, Y. W. Teh, *Bayesian learning via stochastic gradient Langevin dynamics*, in: Proceedings of the 28th International Conference on Machine Learning (ICML 2011), 2011, pp. 681–688.

A Proofs of the six lemmas

Each proof closes with the outcome of its committed executable verification. The scripts are symbolic where an identity is claimed and numerical where a rate or a constant is claimed; they are listed in Section 9. Throughout, $\sigma(s) = 1/(1+e^{-s})$, $w = e^{\beta z}$, $u = 1/w$ and $\Psi(u) = \log u + u$, and we use freely the elementary identities $\sigma(\beta z) = 1/(1+u)$, $\beta\, g_\beta(z) = \log(1+w)$ and $(1-\sigma(s))/\sigma(s) = e^{-s}$, all of which are verified symbolically in `passo0_verificacao.py`.

A.1 Normal form and additive quadrature

Lemma 9 (normal form and additive quadrature). *Let $w = e^{\beta z}$ and $u = 1/w$, and set $\Psi(u) = \log u + u$. In fast time $\tau = \beta t$ the margin obeys*

$$\frac{dw}{d\tau} = c\,R(t)\,\frac{w^2}{1+w}\,(1+E_\beta), \qquad |E_\beta| \le \frac{\log(1+w)}{\beta|R|}, \tag{13}$$

the quadratic normal form of De Maesschalck and Schecter [5, equations (1.9)–(1.10)] with $\varepsilon = \beta^{-1}$, and one has the exact additive identity

$$\frac{d\Psi(u)}{dt} = -c\,\beta\,R(t) + c\log(1+w) + \mathcal{E}_{(\mathrm{H3})}(t), \qquad \big|\mathcal{E}_{(\mathrm{H3})}\big| \le C_3\,\beta\,e^{-\beta\delta_1}. \tag{14}$$

Under (H1)–(H2) the additive correction satisfies $\int_{t_{\mathrm{hit}}}^{t_{\mathrm{release}}} c\log(1+w)\,ds = O(\log\beta/\beta)$.

Proof of Lemma 9. Step 1 (margin equation, exact under (H3)). $\dot z = (x_j, 1)\cdot\dot\theta_A$, and the field on θ_A is $-\frac{1}{n}\sum_i \mathrm{res}_i\,\sigma(\beta z_i)(x_i,1)$. The term $i=j$ contributes $-c\,(g_\beta(z)-R)\,\sigma(\beta z)$, because $\mathrm{res}_j = g_\beta(z) - R$ by the definition (2) of R; the terms $i \ne j$ are bounded by $C_3' e^{-\beta\delta_1}$ by the scalar reduction of Lemma 14. Hence $\dot z = -c(g_\beta(z)-R)\sigma(\beta z) + O(e^{-\beta\delta_1})$, which in the gauge w and fast time $\tau = \beta t$ is (13).

Step 2 (exact chain rule). $\dot u = -\beta u\,\dot z$ and $\Psi'(u) = (1+u)/u$; together with $\sigma(\beta z) = 1/(1+u)$,

$$\frac{d\Psi}{dt} = \frac{1+u}{u}\cdot(-\beta u)\cdot\Big[-c\big(g_\beta(z)-R\big)\tfrac{1}{1+u}\Big] = c\,\beta\,\big(g_\beta(z)-R\big).$$

Step 3 (additive identity). Since $\beta\,g_\beta(z) = \log(1+e^{\beta z}) = \log(1+w)$, substitution gives $d\Psi/dt = -c\beta R + c\log(1+w)$, plus the (H3) error of Step 1 carried through the same factors, which is bounded by $C_3\beta e^{-\beta\delta_1}$. This is (14). No division by R occurs, so the interior zero of R forced by (H1)–(H2) through the intermediate value theorem is harmless.

Step 4 (bound on the correction, in three windows). On the episode $w = 1/u$ with $u = c\beta|\mathcal{A}(t)|(1+o(1))$ away from neighbourhoods of the endpoints (Lemma 12(i)). In the bulk, where $|\mathcal{A}| \ge a_0(\delta)$, one has $\log(1+w) \le w \le 1/(c\beta a_0)$, so that stretch contributes at most $T_{\mathrm{ep}}/(\beta a_0) = O(\beta^{-1})$, where $T_{\mathrm{ep}} = t_{\mathrm{exit}} - t_{\mathrm{hit}}$ is the duration of the episode, not the training horizon H: the bulk stretch lies inside the episode by construction. Near t_{hit} and t_{exit} the function $\mathcal{A}$ has transversal zeros, with derivatives $cR(t_{\mathrm{hit}}) \ne 0$ and $cR(t_{\mathrm{exit}}) \ne 0$, so $|\mathcal{A}(t)| \ge \kappa|t - t_*|$ locally and

$$\int \log(1+w) \;\lesssim\; \int_{1/\beta}^{\epsilon} \frac{ds}{c\beta\kappa s} = \frac{\log(\beta\epsilon)}{c\beta\kappa};$$

on the $O(\beta^{-1})$-wide stretches glued to the endpoints one has $w = O(1)$ and $\log(1+w) \le \log 2 + O(1)$, contributing $O(\beta^{-1})$. Summing the three windows gives $O(\log\beta/\beta)$ with constant $C_4 = 2/\big(c\min(|R(t_{\mathrm{hit}})|, R(t_{\mathrm{exit}}))\big) + T_{\mathrm{ep}}/a_0$. □

Remark 17 (consequence used downstream). Integrating (14) gives $\Psi(u(t)) = \Psi(u_0) - c\beta\mathcal{A}(t) + O(\log\beta/\beta)$ uniformly on the episode. This is the form that sustains the depth $u = c\beta|\mathcal{A}|(1+o(1))$ on compacts and the release at $t_{\text{exit}} + O(\beta^{-1})$, and it is what Lemmas 12 and 13 consume.

Verification. `prova_L1_verificacao.py` confirms the additive identity symbolically, and the integral correction with the normalized metric is stable in $[8.2, 10.6]$, below the leading theoretical term $2/\big(cR^{\text{hyb}}(t_{\text{exit}})\big) = 2/(0.625 \times 0.108) = 29.6$, for $\beta \in [3 \cdot 10^2, 10^4]$. The full theoretical constant adds the bulk term: with $T_{\text{ep}} = t_{\text{exit}} - t_{\text{hit}} = 1.1325 - 0.1114 = 1.021$ and $a_0 \approx 0.02$ in the instance, $C_4 = 29.6 + 1.021/0.02 = 80.7$. Using the training horizon $H = 1.7$ in place of T_{ep} would give 114.6 instead; the bound of Lemma 9 is over the episode, so T_{ep} is the correct length.

A.2 The B-subsystem and the feedback

Lemma 10 (B-subsystem and feedback). *Under (H4) the B-field is affine with Lipschitz constant $L_B = \lambda_{\max}(G_B)$, where G_B is the Gram matrix of the active data, and R^{hyb} is affine in θ_B. Integral Grönwall consuming the bound of Lemma 9 gives, conditionally on the layer being maintained (supplied by Lemma 13),*

$$\sup\big\|\theta_B^{\beta} - \theta_B^{\text{hyb}}\big\| = O\left(\frac{\log\beta}{\beta^2}\right), \tag{15}$$

one full order better than the naive $O(\beta^{-1})$: in the bulk $u = O(\beta|\mathcal{A}|)$, so the coupling is $O(\beta^{-2})$ there and $O(\beta^{-1})$ only in the $O(\beta^{-1})$-wide corner windows.

Proof of Lemma 10. Step 1 ((H4) makes the field linear). With the activation pattern constant, the B-field is $-G_B(\theta_B - \theta_B^*)$ with G_B the Gram matrix of the active data, hence C^∞ with Lipschitz constant $L_B = \lambda_{\max}(G_B)$, equal to 2.13 in the reference instance. Moreover $R^{\text{hyb}} = y_j - (x_j, 1)\cdot\theta_B$ is affine in θ_B and C^1 in t, with $\|\dot{R}^{\text{hyb}}\|_\infty \le \|(x_j,1)\|\,\|G_B\|\,\text{diam} =: C_R$ explicit.

Step 2 (coupling bound). The perturbation caused by A in res_i is $g_\beta(z_{A,i})$. Under (H3) the terms $i \neq j$ contribute $C_3 e^{-\beta\delta_1}$ and the term $i = j$ contributes $g_\beta(z)$; the Gram factors give a constant C_f, so $\|G_A(t)\| \le C_f\, g_\beta(z(t)) + C_3 e^{-\beta\delta_1}$.

Step 3 (integral Grönwall). Writing $\Delta = \theta_B^{\beta} - \theta_B^{\text{hyb}}$ we have $\dot{\Delta} \le L_B\Delta + \|G_A\|$, hence $\Delta(t) \le \int e^{L_B(t-s)}\|G_A(s)\|\,ds$, and therefore

$$\sup_{[t_{\text{cross}}, t_{\text{release}}]} \|\Delta\| \;\le\; e^{L_B H}\left[\tfrac{C_f}{c\beta}\int c\log(1+w)\,dt + H\,C_3 e^{-\beta\delta_1}\right].$$

Since $g_\beta = \beta^{-1}\log(1+w)$ and Lemma 9 gives $\int c\log(1+w)\,dt = O(\log\beta/\beta)$, the total is $O(\log\beta/\beta^2)$, which is (15). Consequently $R^{\beta} = R^{\text{hyb}} + O(\log\beta/\beta^2)$, which is the perturbation size consumed by Lemma 13. □

Verification. `prova_L2_verificacao.py` confirms that the field and R are affine with $L_B = 2.133$, that the margins of (H4) hold with $\min z_B = 0.39$, and that the refined rate holds: the metric $\sup \cdot \beta^2/\log\beta$ is stable in $[4.2, 5.7]$ over $\beta \in [3 \cdot 10^2, 3 \cdot 10^3]$. The exponent is what the metric discriminates: under a pure $O(\beta^{-1})$ the same metric would grow like $\beta/\log\beta$, by a factor of about 6.5 per decade, and it grew by a factor of 1.24.

A.3 The entry corner

Lemma 11 (entry corner). *Let $v = c\,|R^{\mathrm{hyb}}(t_{\mathrm{hit}})|$ be the arrival speed, transversal by (H1). The exact identity $(1-\sigma(s))/\sigma(s) = e^{-s}$ collapses the crossing-time difference to an elementary integral and gives*

$$t^{\beta}_{\mathrm{cross}} = t_{\mathrm{hit}} + \frac{1}{\beta v}\big(1 + O(\delta_z) + o(1)\big), \qquad \big\|\theta^{\beta}(t_{\mathrm{cross}}) - \theta^{\infty}(t_{\mathrm{hit}})\big\| = O(\beta^{-1}), \tag{16}$$

where $t^{\beta}_{\mathrm{cross}}$ is the first time with $w = w_0$. The exponentially weighted approach integral is $\frac{3/2}{\beta v}(1+o(1))$, the factor $3/2$ coming from $\int_0^{\infty} e^{-s}(1+e^{-s})\,ds$.

Proof of Lemma 11. Step 1 (pre-corner with exponential weight). For $z \geq \delta_z$ one has $|g_\beta(z) - z| \leq \beta^{-1}e^{-\beta z}$ and $1 - \sigma(\beta z) \leq e^{-\beta z}$, so the inhomogeneity of the Grönwall comparison against the ReLU flow is $h(t) \leq Ce^{-\beta z^0(t)}$. Near the hit, transversality gives $z^0(t) = v\,(t_{\mathrm{hit}} - t)(1 + O(\delta_z))$, hence

$$\int h\,dt \;\leq\; C\int e^{-\beta v(t_{\mathrm{hit}}-t)}\,dt \;\leq\; \frac{C}{\beta v}.$$

Along the real trajectory the weight carries in addition the factor σ from the layer, and the exact value is $\int e^{-\beta z}dt = \frac{1}{\beta v}\int_0^{\infty} e^{-s}(1+e^{-s})\,ds\,(1+o(1)) = \frac{3/2}{\beta v}(1+o(1))$.

The weight is self-consistent by a monotone route, with no exponential fixed point. Since $e^{-\beta z}$ is decreasing in z, the flow whose inhomogeneity is evaluated at any sub-solution $z_- \leq z^{\beta}$ majorizes the true inhomogeneity. Iterate: $z^{(0)} = z^0$, and let $z^{(k+1)}$ solve the problem with weight $e^{-\beta z^{(k)}}$. The iteration is monotone decreasing and bounded below by the ReLU sub-solution with slack $2C/(\beta v)$: the increment of each step is at most $C/(\beta v)\cdot \sup e^{\beta(z^{(k)}-z^{(k+1)})}$, and while $|z^{(k)} - z^{(k+1)}| \leq 1/\beta$ that factor is at most e, a bound maintained by the size of the increment itself, by induction on k. The iteration therefore converges to z^{β} with $\Delta\theta(t_\delta) = O(1/(\beta v))$, without requiring a fixed-point inequality of the form $C_2 e^{c'K} \leq K$.

Step 2 (window-crossing identity). On $[0, \delta_z]$ the crossing speeds are $v_0(z) = v(1 + O(\delta_z))$ for ReLU and $v_\beta(z) = v_0(z)\sigma(\beta z) + O(\beta^{-1})$ for softplus, so the difference of crossing times is

$$\begin{aligned}\int_0^{\delta_z}\Big(\frac{1}{v_\beta} - \frac{1}{v_0}\Big)dz &= \frac{1}{v}\int_0^{\delta_z}\frac{1-\sigma(\beta z)}{\sigma(\beta z)}\,dz\,(1+O(\delta_z)) \\ &= \frac{1}{v}\int_0^{\delta_z} e^{-\beta z}\,dz\,(1+O(\delta_z)) = \frac{1-e^{-\beta\delta_z}}{\beta v}(1+O(\delta_z)),\end{aligned}$$

where the middle equality is the exact identity $(1-\sigma(s))/\sigma(s) = e^{-s}$. With $w_0 = 1$ there is no further shift, since $\log w_0 = 0$. Adding Steps 1 and 2 gives $t_{\text{cross}} - t_{\text{hit}} = \frac{1}{\beta v}(1 + O(\delta_z) + o(1))$, and the speeds are bounded on the window, so the state at crossing is within $O(\beta^{-1})$ of $\theta^{\infty}(t_{\text{hit}})$. □

Verification. `prova_L3_verificacao.py` confirms the two identities symbolically and both constants numerically: $K_{\text{teo}} = 1/v = 7.042$ against $K_{\text{emp}} = 7.127$ refitted from the measured crossing times, a deviation of 1.2%; and the exponentially weighted approach integral with the exact factor 3/2 gives 10.563 in theory against 10.52 measured, below 1%.

A.4 Sticking and release for prescribed forcing

Lemma 12 (sticking and release for prescribed R). *Let $R \in C^1$ be prescribed and satisfy (H1)–(H2), and write $\mathcal{A}(t) = \int_{t_{\text{hit}}}^{t} R$. Then*

(i) $\Psi(u(t)) = \Psi(u_0) - c\beta\mathcal{A}(t) + \text{corr}(t)$ with $0 \le \text{corr}(t) = \int c\log(1+w)\,ds = O(\log\beta/\beta)$; on compact subintervals of $(t_{\text{hit}}, t_{\text{exit}})$, $u = c\beta|\mathcal{A}(t)|\big(1 + O(\log\beta/(\beta a_0))\big)$, so the sticking depth is $z \sim -\beta^{-1}\log\beta$;

(ii) the release obeys the closed formula

$$t_{\text{release}} - t_{\text{exit}} = \frac{\text{corr}(\beta)}{c\,\beta\,R(t_{\text{exit}})}\,(1 + o(1)) = O\!\left(\frac{\log\beta}{\beta^2}\right), \tag{17}$$

so u returns to $O(1)$ at $t_{\text{exit}} + O(\beta^{-1})$, with no $\log\beta$ factor;

(iii) a perturbation $\tilde{R} = R + \delta$ shifts t_{exit} by at most $\delta\,(t_{\text{exit}} - t_{\text{hit}})/R(t_{\text{exit}}) + O(\delta^2)$.

No division by R occurs anywhere: the layer is held by the area, not by the sign of R, and the interior zero of R forced by (H1)–(H2) is harmless.

Proof of Lemma 12. Step 1 (monotone comparison). The right-hand side of $\dot{\Psi} = -c\beta R + c\log(1+w(\Psi))$ is decreasing in Ψ, because $w = 1/u$ and u is increasing in Ψ. The Chaplygin comparison principle therefore applies: $R_1 \le R_2$ implies $\Psi_1 \ge \Psi_2$, and piecewise-constant envelopes of R give sub- and super-solutions that are exactly integrable through the same primitive Ψ, hence with explicit constants.

Step 2 (depth, item (i)). Integrate the exact identity (14); the correction is bounded by Lemma 9, in the same three windows. On compact subintervals of $(t_{\text{hit}}, t_{\text{exit}})$ one has u large, so $\Psi(u) = u(1 + o(1))$ and the depth law $u = c\beta|\mathcal{A}(t)|(1 + o(1))$ follows, that is $z \sim -\beta^{-1}\log\beta$.

Step 3 (release, item (ii)). At release Ψ returns to its entry value $\Psi(u_0)$, so the constant $\Psi(u_0)$ cancels and no spurious term survives; item (i) then gives $c\beta\mathcal{A}(t_{\text{release}}) = \text{corr}(t_{\text{release}})$. By (H2) the zero of $\mathcal{A}$ at t_{exit} is transversal, with $\mathcal{A}(t) = R(t_{\text{exit}})(t - t_{\text{exit}})(1 + o(1))$, and solving for t_{release} gives (17). The delay of the release is exactly the correction integral of Lemma 9, reparametrized by the exit speed.

Step 4 (perturbation, item (iii)). Apply the implicit function theorem to $\mathcal{A}_{\tilde{R}}(t) = 0$ with denominator $R(t_{\text{exit}}) \neq 0$, splitting the episode into the same three windows: near the hit

$|\mathcal{A}| \geq |R(t_{\rm hit})|s/2$ for $s = t - t_{\rm hit}$ small, in the bulk $|\mathcal{A}| \geq a_0(\delta)$, and near the exit the zero is transversal. This gives the shift $\leq \delta\,(t_{\rm exit} - t_{\rm hit})/R(t_{\rm exit}) + O(\delta^2)$. □

Remark 18 (the corollary that explains the coupled constant). In the coupled system the initial datum comes from the entry corner, so by Lemma 11 the episode starts with a delay K/β that cuts an initial negative area $\approx R(t_{\rm hit})K/\beta$. Item (ii) then gives $\beta\,(t_{\rm release} - t^{\rm hyb}_{\rm exit}) \to -K\,|R(t_{\rm hit})|/R(t_{\rm exit})$. With the theoretical $K = 1/(c|R(t_{\rm hit})|)$ this is exactly $-1/(cR(t_{\rm exit})) = -14.81$, the coefficient of (5) left by the cancellation; with the empirical $K = 7.13$ it is $-(7.13 \times 0.2272)/0.1080 = -15.0$, against -14.38 measured, a deviation of 4% whose residue is the positive correction of item (ii).

Verification. `prova_L4_verificacao.py` measures $t_{\rm release}$ and corr on the same run and tests $\Delta t = {\rm corr}/(c\beta R_{\rm exit})$: the formula closes to 0.2%/1.0%/9.5% at $\beta = 3\cdot 10^2/10^3/3\cdot 10^3$ on a grid of $2\cdot 10^{-6}$, the last figure having collapsed from 35% to 9.5% under a fivefold grid refinement, which identifies the residue as discretization. The corollary of Remark 18 closes at -14.99 against -14.38 measured (4.2%).

A.5 The coupled bootstrap

Lemma 13 (coupled bootstrap). *Under* (H1)–(H4) *the coupled system satisfies the two-term law*

$$t^{\beta}_{\rm release} = t_{\rm exit} - \frac{1}{c\,R^{\rm hyb}(t_{\rm exit})}\cdot\frac{1}{\beta} + \frac{{\rm corr}(\beta)}{c\,\beta\,R^{\rm hyb}(t_{\rm exit})} + o(\beta^{-1}), \qquad {\rm corr}(\beta) = O\!\left(\frac{\log\beta}{\beta}\right), \tag{18}$$

obtained by continuation with strict margins, with the constant $M_2 = e^{L_B H}C_f C_4/c$ independent of the continuation hypothesis for $\beta \geq \beta_0$. In the first-order coefficient the arrival speed cancels: in the product $K\,|R^{\rm hyb}(t_{\rm hit})|/R^{\rm hyb}(t_{\rm exit})$ of the entry delay by the area it cuts, the factor $|R^{\rm hyb}(t_{\rm hit})|$ cancels against $K = 1/(c|R^{\rm hyb}(t_{\rm hit})|)$ of Lemma 11, leaving a coefficient that still depends on c and on $R^{\rm hyb}(t_{\rm exit})$.

Proof of Lemma 13. The argument is a continuation with strict margins.

Step 1 (the bootstrap property). For s in $[t_{\rm cross}, t_{\rm exit} - \varepsilon_\beta]$ let $\mathcal{P}_s$ be the property that, on $[t_{\rm cross}, s]$, (a) $\sup\|\theta^\beta_B - \theta^{\rm hyb}_B\| < 2M_2\log\beta/\beta^2$ and (b) $z(t) < \beta^{-1}\log w_0 - \beta^{-1}$ for $t \geq t_{\rm cross} + C_{\rm collar}/\beta$. The collar is necessary: at $t_{\rm cross}$ the margin crosses $\beta^{-1}\log w_0$ continuously, so the layer can only be strict after an $O(\beta^{-1})$ entry window. The constant is $M_2 = e^{L_B H}C_f C_4/c$, explicit through the chain Lemma 10 composed with Lemma 9.

Step 2 (nonempty and open). Lemma 11 supplies the initial datum with slack: the state is within $O(\beta^{-1})$ in θ and the layer is reached transversally, so $\mathcal{P}_s$ holds for s just past the collar. All quantities in $\mathcal{P}_s$ are continuous in s, since the trajectories are C^1, so the strict inequalities propagate to a neighbourhood.

Step 3 (improvement, hence closed). Assume $\mathcal{P}_s$ with the constant $2M_2$. Since R is affine in θ_B (Lemma 10(a)), the forcing satisfies $R^\beta = R^{\rm hyb} + \delta$ with $\|\delta\| \leq {\rm Lip}_R\cdot 2M_2\log\beta/\beta^2$. The sandwich of Lemma 12 with perturbed forcing then keeps $u(t) \geq c\beta(|\mathcal{A}(t)| - \|\delta\|H)/2 > u_0$

for $t \le t_{\rm exit} - \varepsilon_\beta$ with $\varepsilon_\beta = C_\varepsilon \log\beta/\beta$ and $C_\varepsilon = 4C_4/(cR_{\rm exit})$, so the layer is maintained with a margin. With the layer in hand, the Grönwall of Lemma 10 is unconditional and returns M_2, half of what was assumed, for an explicit $\beta \ge \beta_0$. The threshold β_0 is finite and traceable through the chain of constants, though we do not exhibit it. Crucially M_2 does not depend on the hypothesis $2M_2$, because the chain of constants is unconditional; this is what closes the bootstrap.

Step 4 (continuation). The set of s for which $\mathcal{P}_s$ holds is nonempty (Step 2), open (strict margins and continuity) and closed (the bounds pass to limits) in $[t_{\rm cross}, t_{\rm exit} - \varepsilon_\beta]$, hence it is the whole interval.

Step 5 (terminal window). On $[t_{\rm exit} - \varepsilon_\beta, t_{\rm release}]$ apply Lemma 12(ii)–(iii) with the forcing frozen at $R^{\rm hyb} + O(\log\beta/\beta^2)$. Three contributions add: the displacement caused by δ is $O(\log\beta/\beta^2)$ by item (iii); the correction delay is $+{\rm corr}/(c\beta R_{\rm exit})$ by item (ii); and the area cut at entry, $\int_{t_{\rm hit}}^{t_{\rm hit}+K/\beta} R \approx R(t_{\rm hit})K/\beta$, advances the exit by $-K|R(t_{\rm hit})|/(R(t_{\rm exit})\beta) = -1/(cR(t_{\rm exit})\beta)$, the arrival speed cancelling against $K = 1/(c|R(t_{\rm hit})|)$. Summing gives (18). □

Verification. `prova_L5_verificacao.py` checks that the two-term law reproduces the whole coupled sweep, with deviations 1.9%/0.7%/0.2%/0.1% for $\beta = 3\cdot 10^2 \dots 10^4$, and that the surviving coefficient is $1/(cR_{\rm exit}) = 14.815$; the chain of constants is exhibited with $M_2 = 440.8$ from the empirical $C_4 = 10.6$, against $M_2 \le 1233$ from the leading theoretical term $2/(cR^{\rm hyb}(t_{\rm exit})) = 29.6$ alone and $M_2 \lesssim 3300$ from the full theoretical $C_4 = 80.7$, which adds the bulk term $T_{\rm ep}/a_0$ with $T_{\rm ep} = t_{\rm exit} - t_{\rm hit} = 1.021$ and $a_0 \approx 0.02$ in the instance. The strict layer is confirmed at $\beta = 10^4$. One honest caveat: with the conservative C_ε the inner window of the layer check is empty for $\beta \le 3\cdot 10^3$, so the check is vacuous there, and at those β the maintenance of the layer is implied only by the detection of the release itself.

A.6 The exit corner, branch selection and the scalar reduction

Lemma 14 (exit corner, branch selection and scalar reduction). *Under (H1)–(H4), after release: (i) the β-trajectory crosses the exit corner in time $\frac{1}{\beta v'}(1+o(1))$ with $v' = c\,R^{\rm hyb}(t_{\rm exit}) > 0$, the mirror of Lemma 11; (ii) the ReLU flow restarted at $(\theta_A(t_{\rm hit}), \theta_B^{\rm hyb}(t_{\rm exit}))$ has forward uniqueness on the activating branch, and that branch is selected by the limit itself, the frozen Filippov branch [7] being incompatible with release at $t_{\rm exit} + O(\beta^{-1})$; (iii) classical Grönwall glues on $[t_{\rm release}, H]$ at rate $O(\log\beta/\beta)$. Under (H3) the other data contribute $O(e^{-\beta\delta_1})$, the face-j field is rank one along $(x_j, 1)$, and the episode reduces to the scalar problem.*

Proof of Lemma 14. Step 1 (item (i)). Mirror Lemma 11, replacing v by $v' = cR^{\rm hyb}(t_{\rm exit}) > 0$, which is transversal by (H2). The crossing identity is the same, and the weight is $\frac{3/2}{\beta v'}$.

Step 2 (item (ii), forward uniqueness and branch selection). At the restart point the active field is locally Lipschitz on the side $z \ge 0$ and $\dot z(t_{\rm exit}^+) = v' > 0$, so the solution enters $\{z > \epsilon t\}$ and Picard uniqueness holds forward from any $t > t_{\rm exit}$. The only Filippov

alternative [7], namely remaining in $z \le 0$ where the field is identically zero, is incompatible with the limit of the θ^β, since Lemma 13 places the release at $t_{\text{exit}} + O(\beta^{-1})$. The selection of the activating branch is therefore proved by Lemma 13 and not postulated in Definition 1.

Step 3 (item (iii), gluing). The initial datum of the gluing satisfies $\|\theta^\beta(t_{\text{release}}) - \theta^\infty(t_{\text{exit}})\| = O(\beta^{-1}\log\beta)$: the position of θ_A in the layer contributes $O(\log\beta/\beta)$, θ_B contributes the rate of Lemma 10, and the time displacement $O(\beta^{-1})$ contributes through bounded speeds. In the active interior the two fields are exponentially close, so classical Grönwall on $[t_{\text{release}}, H]$ keeps the gap at $O(\log\beta/\beta)$.

Step 4 (scalar reduction). The field on θ_A is $-\frac{1}{n}\sum_i \operatorname{res}_i \sigma(\beta z_i)(x_i, 1)$. Under (H3), that is $z_{A,i} \le -\delta_i < 0$ for $i \ne j$ on the limit episode, the terms $i \ne j$ are bounded by $Ce^{-\beta\delta_i}$ and are absorbed by the Grönwall estimates without changing any $O(\beta^{-1})$ rate. The term $i = j$ is rank one along $(x_j, 1)$, so the component of θ_A tangential to the face is frozen at the exponential rate, while the normal component is exactly the scalar problem of Lemmas 9 to 13. Hypothesis (H3) is an open condition on the data and the initialization, and it is verified numerically in the instance with an explicit margin δ_1. □

Verification. `prova_L6_verificacao.py` measures the gluing directly: for $\beta \in \{10^3, 3\cdot 10^3\}$ the supremum of $\|\theta^\beta - \theta^{\text{relu}}\|$ over $[t_{\text{release}} + 0.05, H]$ is $1.6\cdot 10^{-3}$ and $5.5\cdot 10^{-4}$, so the metric $\sup\cdot\beta/\log\beta$ is stable at 0.21–0.23, confirming the rate $O(\log\beta/\beta)$. `passo6_verificacao.py` confirms the scalar reduction: (H3) holds with $\delta_1 = 0.508$ and the tangential drift is below 10^{-10}, machine zero, at both values of β, consistent with $e^{-\beta\delta_1} \sim 10^{-65}$.

The debugging of this script is worth recording, because it exhibits the alternative branch. The first version of the reference restarted the ReLU flow at $z_A = -1.1\cdot 10^{-6}$, an overshoot of one Euler step of the pre-hit loop, and fell into the frozen Filippov branch, with a constant gap of ≈ 0.039. The correct reference requires exact projection onto the face together with an activating nudge larger than the overshoot. Branch selection is a real condition of the statement.

B Proof of the forcing theorem

This appendix proves Theorem 6. It is a rewrite of Lemmas 9 to 14 with the residual R treated as a prescribed forcing generated by an arbitrary live subsystem of the affine class of Section 3.1, instead of by a single live neuron. Each lemma below carries a *provenance* line saying what is inherited from Section A and what is new. Where the provenance says *identical with R generic*, we do not reproduce the proof: the audited claim is that the proof of Section A never uses the structure of B, only (H0′)–(H2′), and reproducing it verbatim would add nothing. Where the provenance says *new*, the proof is written out in full. Only one lemma, Lemma 20, changes substantively, and it changes in our favour.

Throughout, $X_{\max} = \max_i \|(x_i, 1)\|$, $\|a\|_2 = (\sum_{k\ge 1} a_k^2)^{1/2}$, $T_{\text{ep}} = t_{\text{exit}} - t_{\text{hit}}$, and t_{cross} is the first time at which $w = w_0$. We use repeatedly the elementary bounds, valid for all z,

$$0 \le g_\beta(z) - z_+ \le \beta^{-1}e^{-\beta|z|}, \qquad \left|\sigma(\beta z) - \mathbf{1}_{\{z>0\}}\right| \le e^{-\beta|z|}, \qquad \beta g_\beta(z) = \log(1+w), \tag{19}$$

with $w = e^{\beta z}$, $u = 1/w$, $\Psi(u) = \log u + u$, $\sigma(s) = 1/(1+e^{-s})$, $\sigma(\beta z) = 1/(1+u)$ and $(1-\sigma(s))/\sigma(s) = e^{-s}$.

B.1 L1′: normal form and additive quadrature

Lemma 19 (normal form and additive quadrature, forcing form)**.** *With $w = e^{\beta z}$, $u = 1/w$, $\Psi(u) = \log u + u$ and $\tau = \beta t$,*

$$\frac{dw}{d\tau} = c\,R(t)\,\frac{w^2}{1+w}\,(1+E_\beta), \qquad |E_\beta| \le \frac{\log(1+w)}{\beta|R|}, \tag{20}$$

and one has the exact additive identity

$$\frac{d\Psi(u)}{dt} = -c\beta R(t) + c\log(1+w) + \mathcal{E}(t), \qquad |\mathcal{E}(t)| \le C_3\beta e^{-\beta\delta_1}. \tag{21}$$

Under (H1′)–(H2′), $\int_{t_{\rm hit}}^{t_{\rm release}} c\log(1+w)\,ds \le C_4 \log\beta/\beta$ *with* C_4 *as in* (12).

Provenance: identical to Lemma 9 with R generic. Step 1 uses only $\mathrm{res}_j = g_\beta(z) - R$, which is the *definition* of R, and the scalar reduction (Lemma 26, step 4), which is a statement about the other margins of A; steps 2–3 are algebra in (Ψ, u, σ); step 4 bounds the correction in three windows using only $R(t_{\rm hit}) \neq 0$, $R(t_{\rm exit}) \neq 0$ and $|\mathcal{A}| \ge a_0$ in the bulk, and produces C_4 written purely in terms of R. No line mentions B. The proof is unchanged; we record only that $T_{\rm ep}$, not H, is the correct length in the bulk term, since the bound is taken over the episode.

B.2 L2′: the live subsystem, a PSD energy estimate

Lemma 20 (live subsystem and feedback, PSD form)**.** *Assume* (H4′), (H5′), *and, on* $[t_{\rm hit}, s]$, *the layer bound* $\int_{t_{\rm hit}}^{s} c\log(1+w)\,dt \le 2C_4\log\beta/\beta$ *supplied by Lemma 24. Put* $\Delta = \theta_B^\beta - \theta_B^{\rm hyb}$. *Then*

$$\sup_{[0,s]}\|\Delta\| \;\le\; \underbrace{\frac{C_f C_4}{c}\cdot\frac{2\log\beta}{\beta^2}}_{\text{episode}} + \underbrace{O(\beta^{-2})}_{\text{pre-hit}} + \underbrace{H\Big(C_3 e^{-\beta\delta_1} + C_5\sqrt{m}\,C_{\rm res}\,e^{-\beta\delta_B}\Big)}_{\text{beyond all orders}}, \tag{22}$$

so $\sup\|\Delta\| = O(\log\beta/\beta^2)$ *with constant* $M_2 = C_f C_4/c$, *and consequently*

$$R^\beta = R^{\rm hyb} + \delta, \qquad \|\delta\|_\infty \le \mathrm{Lip}_R \sup\|\Delta\| + \frac{m}{\beta}e^{-\beta\delta_B}. \tag{23}$$

No factor $e^{L_B H}$ occurs, and $\lambda_{\max}(G_B)$ does not appear. *Moreover the same PSD estimate applies after release: on* $[t_{\rm release}, H]$, *where both the β-flow and θ^∞ have A alive with the same frozen pattern, one has* (11), *again with no exponential factor (step 5).*

Provenance: new proof. Lemma 10 obtains the same rate through the integral Grönwall $\|\Delta(t)\| \le \int e^{L_B(t-s)}\|G_A(s)\|\,ds$, paying $e^{L_B H}$ with $L_B = \lambda_{\max}(G_B)$; with unit readouts $\lambda_{\max}(G_B)$ grows like m, so that route makes M_2 exponential in m and the theorem vacuous for large m. The observation below is that the sign of G_B is available and Grönwall throws it away. Step 2b, the live-unit discrepancies, has no counterpart in Lemma 10.

Proof. Step 1 (exact perturbed linear system on the episode). On $[t_{\mathrm{hit}}, H]$ the hybrid freezes θ_A at the value with $z = 0$, so A's contribution to the hybrid residuals is *exactly* zero: $\mathrm{ReLU}(z) = 0$ on the face j by construction, and $\mathrm{ReLU}(z_{A,i}) = 0$ on the other data because their margins are $\le -\delta_1 < 0$ by (H4′) — a ReLU is not approximately zero there, it is zero. The $O(e^{-\beta\delta_1})$ that appears below belongs to the β-flow, where the softplus of A leaks $\beta^{-1}e^{-\beta\delta_1}$ into each of those residuals; it is a term of G_A, not a defect of the hybrid. The hybrid B-flow is therefore exactly $\dot\theta_B^{\mathrm{hyb}} = -G_B\theta_B^{\mathrm{hyb}} + b$. In the β-flow the same computation applies with two extra sources: the output of A, which enters every residual, and the softplus defects of the live units themselves. Hence

$$\dot\Delta = -G_B\Delta + G_A(t) + G_S(t), \tag{24}$$

where, with φ_i the feature vectors of (7),

$$G_A(t) = -\frac{1}{n}\sum_{i=1}^{n} g_\beta\big(z_{A,i}(t)\big)\varphi_i, \qquad G_S(t) = \text{(live softplus defect, bounded in step 2b)}.$$

Equation (24) is an identity, not an estimate: the affine class makes the drift exactly linear with the *same* matrix G_B for both flows, which is why the difference of the drifts is $-G_B\Delta$ with no remainder.

Step 2a (bound on G_A). Write $C_f = \|\varphi_j\|/n$. By (H4′) the margins $z_{A,i}$, $i \ne j$, are $\le -\delta_1$, so those terms contribute at most $C_3 e^{-\beta\delta_1}$; the term $i = j$ is $C_f g_\beta(z)$. Hence

$$\|G_A(t)\| \le C_f\, g_\beta(z(t)) + C_3 e^{-\beta\delta_1} = \frac{C_f}{\beta}\log\big(1 + w(t)\big) + C_3 e^{-\beta\delta_1}. \tag{25}$$

Nothing in (25) depends on m except through C_f: the size of the perturbation that A injects is a property of A.

Step 2b (the live-unit discrepancies; new term). The β-flow's B-field is, per unit k,

$$-\frac{1}{n}\sum_i \mathrm{res}_i\, a_k\, \sigma(\beta z_{k,i})\,(x_i, 1),$$

whereas the hybrid uses $a_k \mathbf{1}_{\{z_{k,i}>0\}}(x_i, 1)$; and res_i itself contains $\sum_k a_k g_\beta(z_{k,i})$ where the hybrid has $\sum_k a_k (z_{k,i})_+$. By (H4′) every live margin satisfies $|z_{k,i}| \ge \delta_B$, so (19) gives, per unit and per datum, a derivative defect $\le e^{-\beta\delta_B}$ and an output defect $\le \beta^{-1}e^{-\beta\delta_B}$. Using (H5′) to bound $|\mathrm{res}_i| \le C_{\mathrm{res}}$ and $\|(x_i, 1)\| \le X_{\max}$, the $2m$-vector G_S obeys

$$\|G_S(t)\| \;\le\; \underbrace{\Big(\sum_{k=1}^{m} a_k^2\Big)^{1/2} X_{\max}}_{\le \|a\|_2 X_{\max}} \Big(C_{\mathrm{res}} + \tfrac{m}{\beta}\max_k |a_k| X_{\max}\Big) e^{-\beta\delta_B}, \tag{26}$$

the crude count being mn scalar discrepancies, each $O(e^{-\beta\delta_B})$, recombined into a Euclidean norm.

We do *not* absorb the second bracket into a $(1+o(1))$. That step would be hiding a condition on β inside asymptotic notation: the bracket collapses to $C_{\mathrm{res}}(1+o(1))$ only when

$$\beta \;\geq\; \beta_2 := \frac{m \max_k |a_k| X_{\max}}{C_{\mathrm{res}}}, \tag{27}$$

that is only for $\beta \gtrsim m$. This is a genuine condition of the theorem and it is recorded as row 5 of Table 6; it is, together with row 4, what makes β_0 linear in m rather than logarithmic. Writing $C_5 := 2\|a\|_2 X_{\max}/\sqrt{m}$, an $O(1)$ quantity for both readout scalings of Section B.8, (26) reads $\|G_S\| \leq C_5\sqrt{m}\, C_{\mathrm{res}} e^{-\beta\delta_B}$ for $\beta \geq \beta_2$, and unconditionally with C_{res} replaced by $C_{\mathrm{res}} + m\max_k |a_k| X_{\max}/\beta$. The dependence on m is at worst m — $\sqrt{m}$ from the readout norm times $\sqrt{m}$ from the second bracket — and multiplies a quantity beyond all orders in β; for $\beta \geq \beta_1 := \delta_B^{-1}\log(m^2 H\beta^2)$ the whole term is $o(\beta^{-2})$.

Step 3 (energy estimate; this is where PSD replaces Grönwall). Let $\rho(t) = \|\Delta(t)\|$. From (24),

$$\frac{d}{dt}\tfrac{1}{2}\rho^2 = -\Delta^\top G_B \Delta + \Delta^\top (G_A + G_S) \;\leq\; \rho\,\big(\|G_A\| + \|G_S\|\big),$$

because $G_B \succeq 0$ by (7). Wherever $\rho > 0$ this gives $\dot\rho \leq \|G_A\| + \|G_S\|$; since ρ is Lipschitz, the same inequality holds for its upper Dini derivative everywhere, and comparison with $y' = \|G_A\| + \|G_S\|$, $y(t_{\mathrm{hit}}) = \rho(t_{\mathrm{hit}})$, yields

$$\rho(t) \;\leq\; \rho(t_{\mathrm{hit}}) + \int_{t_{\mathrm{hit}}}^{t} \big(\|G_A\| + \|G_S\|\big)\, ds. \tag{28}$$

Substituting (25) and using the layer bound of the hypothesis,

$$\int_{t_{\mathrm{hit}}}^{s} \|G_A\|\, ds \leq \frac{C_f}{c\beta}\int_{t_{\mathrm{hit}}}^{s} c\log(1+w)\, ds + HC_3 e^{-\beta\delta_1} \leq \frac{C_f C_4}{c}\cdot\frac{2\log\beta}{\beta^2} + HC_3 e^{-\beta\delta_1},$$

which is the first and third groups of (22), with $M_2 = C_f C_4/c$.

Step 4 (the pre-hit initial gap $\rho(t_{\mathrm{hit}})$). On $[0, t_{\mathrm{hit}}]$ both flows have A alive with frozen pattern, so the *full* system is affine with the full Gram $G \succeq 0$ and $\delta\theta = \theta^\beta - \theta^{\mathrm{ReLU}}$ obeys $\dot{\delta\theta} = -G\,\delta\theta + E$, where E collects the softplus defects. Repeating step 3 on the full vector, $\|\delta\theta(t)\| \leq \int_0^t \|E\|$. The A-block of E is $O(C_{\mathrm{res}} e^{-\beta z^0(t)})$ — a derivative defect, with no β^{-1}: it is $|\sigma(\beta z) - \mathbf{1}_{\{z>0\}}| \leq e^{-\beta|z|}$ multiplying res_j — and near the hit transversality gives $z^0(t) = v(t_{\mathrm{hit}} - t)(1 + O(\delta_z))$, so

$$\|\delta\theta_A(t)\| \;\leq\; \frac{C}{\beta v}\, e^{-\beta z^0(t)}\big(1+o(1)\big) \;+\; H\, C_5\sqrt{m}\, C_{\mathrm{res}}\, e^{-\beta\delta_B}, \tag{29}$$

which is step 1 of Lemma 22 in pointwise form; it integrates to $O(\beta^{-1})$ at $t = t_{\mathrm{hit}}$. The second term of (29) is the live-unit defect of step 2b, which is present pre-hit as well — the A-block of E sees the live softplus defects through res_j — and which the bookkeeping of

Section A.2 omits; it is beyond all orders and does not change the rate, but it must be carried.

Two provisos on (29). First, the exponential weight is written against z^0, the *ReLU* margin, and the estimate as displayed presumes z^0 decreasing on the approach — guaranteed on a fixed neighbourhood $[t_{\rm hit}-\eta, t_{\rm hit}]$ of the hit by transversality (H1′), where $\dot z^0 = -v < 0$. Away from that neighbourhood z^0 need not be monotone, and the correct statement is the crude one, $\|\delta\theta_A(t)\| \le C\,t\,e^{-\beta\delta_z}$ with $\delta_z = \min_{[0,t_{\rm hit}-\eta]} z^0 > 0$; this is again beyond all orders and is dominated by the collar term. Splitting $[0, t_{\rm hit}]$ at $t_{\rm hit}-\eta$ and using the two forms on the two pieces is what makes the integral below legitimate.

The B-block of E contains *no* derivative defect of A — B's gradients involve only the live units' σ's — but only A's output defect $\le C_f\beta^{-1}e^{-\beta z^0}$ and G_S. Splitting the B-block of the drift as $-G_B\Delta - G_{BA}\delta\theta_A$ and repeating step 3 on Δ alone, again using $G_B \succeq 0$ and treating the cross term as a source,

$$\rho(t_{\rm hit}) \le \int_0^{t_{\rm hit}} \Big(\|G_{BA}\|\,\|\delta\theta_A\| + \frac{C_f}{\beta}e^{-\beta z^0} + \|G_S\| \Big) ds \le \frac{C\|G_{BA}\|}{(\beta v)^2} + \frac{C_f}{\beta^2 v} + HC_5\sqrt{m}C_{\rm res}e^{-\beta\delta_B},$$

using $\int_0^{t_{\rm hit}} e^{-\beta v(t_{\rm hit}-s)}ds = (\beta v)^{-1}(1+o(1))$ twice. Hence $\rho(t_{\rm hit}) = O(\beta^{-2})$, which is the second group of (22) and is dominated by the episode term. Estimate (23) follows from (H3′) plus the $m\beta^{-1}e^{-\beta\delta_B}$ discrepancy between R^β and $R^{\rm hyb}$ at equal θ_B.

Step 5 (after release: the same PSD estimate, against θ^∞). This step proves clause (ivB), (11), of Theorem 6, clause (ivA) being the closed induction of Lemma 24. It is the exact repetition of step 3 on the interval $[t_{\rm release}, H]$, with θ^∞ in place of $\theta^{\rm hyb}$; we write it out because the reference trajectory has to change there (Remark 7) and because the rate it returns, $O(\beta^{-1})$, is not the episode rate.

On $[t_{\rm release}, H]$ the β-flow has A *alive* on the face j ($z > 0$ after release, Lemma 26(i)) and θ^∞ likewise runs the ReLU flow on the activating branch (Lemma 26(ii)). By (H4′) no live unit switches and the other margins of A stay $\le -\delta_1$, so on this interval *both* flows are affine with the *same* full Gram $G \succeq 0$ of Section 3.1 — the pattern is the same for both, the readouts are the same, and only the softplus-versus-ReLU defects differ. Writing $\delta\theta = \theta^\beta - \theta^\infty$ we have again the identity $\dot{\delta\theta} = -G\,\delta\theta + E$, with no remainder, exactly as in step 1: affineness of both drifts with a common matrix is what removes the Lipschitz constant. Here

$$\|E\| \le \underbrace{C_f\big(\beta^{-1}+C_{\rm res}\big)e^{-\beta z(t)}}_{\substack{A \text{ on the face } j\text{: output and derivative defects}}} + \underbrace{C_3e^{-\beta\delta_1}}_{A,\ i\neq j} + \underbrace{C_5\sqrt{m}\,C_{\rm res}e^{-\beta\delta_B}}_{\text{live units}},$$

the middle term being the same one that step 4 lists and that step 2a bounds; we carry it here only for consistency of bookkeeping, since it is beyond all orders and is absorbed into β_1 exactly as before. On $[t_{\rm release}+\eta, H]$ the first term is itself beyond all orders, by the clause $z \ge \delta_1' > 0$ of (H4′); the $O(\beta^{-1})$ below is produced entirely by the collar $[t_{\rm release}, t_{\rm release}+\eta]$

and by the release-instant mismatch. The A-block gives, by the same Dini comparison as in step 3 and using $z(t) \geq v'(t - t_{\text{release}})(1 + o(1))$ from Lemma 26(i),

$$\sup_{[t_{\text{release}}, H]} \|\delta\theta_A\| \;\leq\; \frac{C_A}{\beta}, \qquad C_A := \frac{C_f(1 + C_{\text{res}})}{v'} + c\,\big|t_{\text{release}} - t_{\text{exit}}\big|\beta, \tag{30}$$

where an unnamed factor of order one has been absorbed in passing from the scalar margin gap Δz to the parameter gap $\Delta\theta_A$: the two differ by the normalization $1/\|(x_j, 1)\|$, since $z = \theta_A \cdot (x_j, 1)$ and the field on the face is rank one along $(x_j, 1)$ (Lemma 26, step 4), so $\|\Delta\theta_A\| = |\Delta z|/\|(x_j, 1)\|$ on the normal component. It is $O(1)$ in β and in m, and we do not track it. The second contribution to C_A is the $O(\beta^{-1})$ mismatch of the release *instant* itself, which by Lemma 24 is $t_{\text{release}} - t_{\text{exit}} = -1/(cR^{\text{hyb}}(t_{\text{exit}})\beta) + o(\beta^{-1})$ and hence contributes $C_A \ni 1/R^{\text{hyb}}(t_{\text{exit}})$; this is the term that prevents the post-release rate from being $O(\beta^{-2})$.

For the B-block we split the drift as $-G_B\Delta - G_{BA}\delta\theta_A$ and repeat step 3 on $\Delta = \theta_B^\beta - \theta_B^\infty$ alone, using $G_B \succeq 0$ and treating the cross term as a source, exactly as in step 4:

$$\begin{aligned}\|\Delta(t)\| \;&\leq\; \|\Delta(t_{\text{release}})\| + \int_{t_{\text{release}}}^{t} \Big(\|G_{BA}\|\,\|\delta\theta_A\| + \|E_B\|\Big)ds \\ &\leq\; 2M_2\frac{\log\beta}{\beta^2} + H\|G_{BA}\|\frac{C_A}{\beta} + \frac{C_f H}{v'\beta},\end{aligned}$$

where $\|\Delta(t_{\text{release}})\| \leq 2M_2 \log\beta/\beta^2$ is (10) at $t = t_{\text{release}}$ — at t_{release} the two references agree to that order, θ^∞ having just restarted from $\theta_B^{\text{hyb}}(t_{\text{exit}})$ and $|t_{\text{release}} - t_{\text{exit}}| = O(\beta^{-1})$ — and E_B carries only A's *output* defect $C_f\beta^{-1}e^{-\beta z}$ plus the live-unit term, for the same reason as in step 4: B's gradients never see A's σ. This is (11) with M_3 as displayed there. Note the rate is $O(\beta^{-1})$ and *not* $O(\log\beta/\beta^2)$: the $\log\beta$ disappears, but one power of β is lost to the release-instant mismatch propagated through G_{BA} over the horizon H. By (U-new) of Section B.8, $\|G_{BA}\| \lesssim \sqrt{m}$, so M_3 is again polynomial in m. No factor $e^{L_B H}$ appears here either. □

Remark 21 (what the PSD step buys). Lemma 10 pays $e^{L_B H}$ because Grönwall's inequality is stated for $\|\dot\Delta\| \leq L_B\|\Delta\| + \|G_A\|$ and is blind to the fact that the linear part is *dissipative*. Here $-G_B$ contracts, so the homogeneous part can only help, and (28) is an equality-free statement of that. The price is that the argument is confined to the affine class: outside it, $-G_B\Delta$ acquires a remainder and one is back to a Lipschitz constant. Within the class, which is exactly the class the theorem is stated for, the improvement is unconditional and removes the only exponential in m from the constants.

B.3 L3′: the entry corner

Lemma 22 (entry corner, forcing form)**.** *Let $v = c|R^{\text{hyb}}(t_{\text{hit}})| > 0$, transversal by (H1′). Then*

$$t_{\text{cross}}^\beta = t_{\text{hit}} + \frac{1}{\beta v}\big(1 + O(\delta_z) + o(1)\big), \qquad \|\theta^\beta(t_{\text{cross}}) - \theta^\infty(t_{\text{hit}})\| = O(\beta^{-1}), \tag{31}$$

$t^{\beta}_{\mathrm{cross}}$ *being the first time with* $w = w_0$*; the exponentially weighted approach integral equals* $\frac{3/2}{\beta v}(1 + o(1))$*, the* $3/2$ *coming from* $\int_0^\infty e^{-s}(1 + e^{-s})\, ds$*.*

Provenance: identical to Lemma 11 with R generic. Both steps are scalar in z: step 1 is the monotone sub-solution iteration against the weight $e^{-\beta z^0}$, step 2 is the window-crossing identity $(1 - \sigma)/\sigma = e^{-s}$. R enters only through v, that is only through (H1′); B appears zero times. We used the pointwise form of step 1 in (29), which is the same integral read without integrating to the endpoint.

B.4 L4′: sticking and release for prescribed forcing

Lemma 23 (sticking and release for prescribed R)**.** *Let* $R \in C^1$ *be prescribed and satisfy* (H0′)–(H2′)*, and* $\mathcal{A}(t) = \int_{t_{\mathrm{hit}}}^{t} R$*. Then*

(i) $\Psi(u(t)) = \Psi(u_0) - c\beta\mathcal{A}(t) + \mathrm{corr}(t)$ *with* $0 \le \mathrm{corr}(t) = \int c\log(1 + w)\, ds = O(\log\beta/\beta)$*; on compacts of* $(t_{\mathrm{hit}}, t_{\mathrm{exit}})$*,* $u = c\beta|\mathcal{A}(t)|(1 + O(\log\beta/(\beta a_0)))$*, whence* $z \sim -\beta^{-1}\log\beta$*;*

(ii) $t_{\mathrm{release}} - t_{\mathrm{exit}} = \dfrac{\mathrm{corr}(\beta)}{c\beta R(t_{\mathrm{exit}})}(1 + o(1)) = O(\log\beta/\beta^2)$*;*

(iii) a perturbation $\tilde{R} = R + \delta$ *shifts* t_{exit} *by at most* $\|\delta\|_\infty(t_{\mathrm{exit}} - t_{\mathrm{hit}})/R(t_{\mathrm{exit}}) + O(\|\delta\|_\infty^2)$*.*

No division by R occurs: the layer is held by the area, not by the sign of R.

Provenance: identical to Lemma 12, which is already stated for prescribed R and never mentions B. Step 1 is the Chaplygin comparison, valid because the right-hand side of $\dot{\Psi} = -c\beta R + c\log(1 + w(\Psi))$ is decreasing in Ψ; steps 2–4 use (H1′)–(H2′) and the implicit function theorem at the transversal zero $\mathcal{A}(t_{\mathrm{exit}}) = 0$. In particular the lemma imposes *no* sign structure on R beyond the two transversal zeros of $\mathcal{A}$: R may change sign any number of times inside the episode. This is the point taken up at the end of Section 3.6.

B.5 L5′: coupled bootstrap by continuous induction

Lemma 24 (coupled bootstrap, forcing form)**.** *Under* (H0′)–(H5′) *there is* $\beta_0 < \infty$ *such that for* $\beta \ge \beta_0$ *the coupled system satisfies the two-term law*

$$t^{\beta}_{\mathrm{release}} = t_{\mathrm{exit}} - \frac{1}{cR^{\mathrm{hyb}}(t_{\mathrm{exit}})} \cdot \frac{1}{\beta} + \frac{\mathrm{corr}(\beta)}{c\beta R^{\mathrm{hyb}}(t_{\mathrm{exit}})} + o(\beta^{-1}), \qquad \mathrm{corr}(\beta) = O(\log\beta/\beta), \tag{32}$$

with $M_2 = C_f C_4/c$ *independent of the continuation hypothesis.*

Provenance: structure of Lemma 13; the continuation is rewritten in full because it is the step where the lemmas are mutually dependent, and because step 3 must now consume (23) *through* (H3′) *and the new live-unit term instead of "R is affine in* θ_B*".*

Proof. Step 0 (the dependency cycle, and how it is broken). The three lemmas are mutually dependent: the bound of Lemma 19 on $\int c\log(1+w)$ presupposes the depth law of Lemma 23(i), which presupposes the forcing actually felt by A, which is R^β and is controlled by Lemma 20, which presupposes the layer bound of Lemma 19. The cycle is not vicious because every implication is *causal in time*: each estimate on $[t_{\mathrm{hit}}, s]$ uses only data on $[t_{\mathrm{hit}}, s]$. We therefore run a continuous induction in s.

Step 1 (the induction hypothesis). Fix $\varepsilon_\beta = C_\varepsilon \log\beta/\beta$ with $C_\varepsilon = 4C_4/(cR^{\mathrm{hyb}}(t_{\mathrm{exit}}))$ and a collar constant C_{collar}. For $s \in I := [t_{\mathrm{cross}} + C_{\mathrm{collar}}/\beta,\ t_{\mathrm{exit}} - \varepsilon_\beta]$ let $\mathcal{P}_s$ be the conjunction of

$$\text{(a)}\quad \sup_{[0,s]} \|\theta_B^\beta - \theta_B^{\mathrm{hyb}}\| \ \le\ 2M_2 \frac{\log\beta}{\beta^2}, \tag{a}$$

$$\text{(b)}\quad u(t) \ \ge\ u_0 \quad \text{for all } t \in [t_{\mathrm{cross}} + C_{\mathrm{collar}}/\beta,\ s], \tag{b}$$

$$\text{(c)}\quad \int_{t_{\mathrm{hit}}}^{s} c\log(1+w)\,dt \ \le\ 2C_4 \frac{\log\beta}{\beta}. \tag{c}$$

The collar is necessary: at t_{cross} the margin crosses $\beta^{-1}\log w_0$ continuously, so the layer can be strict only after an $O(\beta^{-1})$ window. Let $S = \{s \in I : \mathcal{P}_s \text{ holds}\}$.

Step 2 ($S \neq \emptyset$). Lemma 22 supplies the entry datum with slack: $\|\theta^\beta(t_{\mathrm{cross}}) - \theta^\infty(t_{\mathrm{hit}})\| = O(\beta^{-1})$, the layer is entered transversally with speed v, and by step 4 of Lemma 20 the B-gap at t_{hit} is $O(\beta^{-2})$, an order below (a). On a collar of width $C_{\mathrm{collar}}/\beta$ the integrand of (c) is $O(1)$, so (c) holds with room. Hence $\mathcal{P}_s$ holds for s just past the collar and S contains a right-neighbourhood of the left endpoint of I.

Step 3 (S closed in I). All three quantities in (a)–(c) are continuous, indeed monotone non-decreasing, in s: sup and $\int$ over $[0,s]$, and (b) is a closed condition on a closed interval. Non-strict inequalities pass to limits, so S is closed.

Step 4 (S open in I: the improvement). Let $s \in S$. We show the three bounds hold on $[0,s]$ with the constants *halved*; openness then follows from continuity. Assume $\mathcal{P}_s$.

(4.1) From (c) and Lemma 20, whose hypothesis is exactly (c), we get $\sup_{[0,s]}\|\Delta\| \le M_2\log\beta/\beta^2 + O(\beta^{-2}) + H(C_3 e^{-\beta\delta_1} + C_5\sqrt{m}C_{\mathrm{res}}e^{-\beta\delta_B})$. For $\beta \ge \beta_0$ with β_0 large enough that the last two groups are below $\frac{1}{2}M_2\log\beta/\beta^2$, this is $\le \frac{3}{2}M_2\log\beta/\beta^2 < 2M_2\log\beta/\beta^2$: strict improvement of (a). *The improvement is unconditional*: $M_2 = C_f C_4/c$ is built from C_f, a property of A and of the features, from C_4, a property of R^{hyb}, and from c; the factor 2 in the hypothesis never enters M_2. This is what makes the induction non-circular.

(4.2) By (23) the forcing felt by A on $[0,s]$ is $R^\beta = R^{\mathrm{hyb}} + \delta$ with $\|\delta\|_\infty \le 2\mathrm{Lip}_R M_2\log\beta/\beta^2 + m\beta^{-1}e^{-\beta\delta_B}$. Enlarging β_0 so that $\|\delta\|_\infty H \le a_0/2$, the perturbed area satisfies $|\mathcal{A}^\beta(t)| \ge \frac{1}{2}|\mathcal{A}^{\mathrm{hyb}}(t)|$ on $[t_{\mathrm{hit}} + \delta, t_{\mathrm{exit}} - \varepsilon_\beta]$ and the two transversal zeros move by $O(\log\beta/\beta^2)$ by Lemma 23(iii). Feeding this perturbed forcing into Lemma 23(i) gives, on $[t_{\mathrm{cross}} + C_{\mathrm{collar}}/\beta,\ s]$,

$$u(t) \ \ge\ \tfrac{1}{2}c\beta\big(|\mathcal{A}^{\mathrm{hyb}}(t)| - \|\delta\|_\infty H\big) \ >\ u_0,$$

the last inequality because $|\mathcal{A}^{\mathrm{hyb}}(t)| \geq \kappa\, C_{\mathrm{collar}}/\beta$ just past the collar, by transversality at t_{hit}, and $|\mathcal{A}^{\mathrm{hyb}}(t)| \geq \kappa\varepsilon_\beta = \Theta(\log\beta/\beta)$ just before $t_{\mathrm{exit}} - \varepsilon_\beta$, by transversality at t_{exit}, while $\|\delta\|_\infty H = O(\log\beta/\beta^2)$. Choosing C_{collar} and C_ε as above makes the inequality strict with a fixed margin: strict improvement of (b).

(4.3) With (b) strict on $[0, s]$, the three-window computation of step 4 of Lemma 19 applies with R^β in place of R; by (4.2) its two transversality constants and its bulk constant a_0 are within a factor 2 of the hybrid ones, so it returns $\int_{t_{\mathrm{hit}}}^{s} c\log(1+w) \leq \frac{3}{2}C_4 \log\beta/\beta$, the factor $\frac{3}{2}$ absorbing the factor-2 degradation of the constants against the slack we left: strict improvement of (c).

Since (a)–(c) hold on $[0, s]$ with strict inequality and all three functionals are continuous in s, they persist on $[0, s+\eta)$ for some $\eta > 0$; hence S is open in I.

Step 5 (conclusion of the induction). I is a non-degenerate interval, hence connected; $S \subseteq I$ is non-empty, open and closed; therefore $S = I$. Equivalently, $s^* = \sup S$ cannot be interior, since at an interior s^* continuity plus the strict improvement of step 4 would extend S past s^*. All three bounds therefore hold on the whole episode up to $t_{\mathrm{exit}} - \varepsilon_\beta$.

Step 6 (terminal window and the two-term law). On $[t_{\mathrm{exit}} - \varepsilon_\beta, t_{\mathrm{release}}]$ apply Lemma 23(ii)–(iii) with the forcing frozen at $R^{\mathrm{hyb}} + O(\log\beta/\beta^2)$. Four contributions add:

1. the displacement caused by δ, $O(\log\beta/\beta^2)$ by (iii);
2. the correction delay $+\mathrm{corr}(\beta)/(c\beta R^{\mathrm{hyb}}(t_{\mathrm{exit}}))$ by (ii);
3. the area cut at entry. By Lemma 22 the episode starts with a delay K/β, $K = 1/(c|R^{\mathrm{hyb}}(t_{\mathrm{hit}})|)$, during which the forcing is $R^{\mathrm{hyb}}(t_{\mathrm{hit}}) < 0$, so an area $\approx R^{\mathrm{hyb}}(t_{\mathrm{hit}})K/\beta$ is removed from $\mathcal{A}$; by transversality at t_{exit} this advances the exit by
$$-\frac{K\,|R^{\mathrm{hyb}}(t_{\mathrm{hit}})|}{R^{\mathrm{hyb}}(t_{\mathrm{exit}})\,\beta} = -\frac{1}{c\,R^{\mathrm{hyb}}(t_{\mathrm{exit}})\,\beta},$$
the arrival speed cancelling against K;
4. the error of the freezing itself. Replacing R by its value at $t_{\mathrm{exit}} - \varepsilon_\beta$ on the terminal window of width $\varepsilon_\beta = C_\varepsilon \log\beta/\beta$ misstates the area by at most $\frac{1}{2}C_R\varepsilon_\beta^2 = O(\log^2\beta/\beta^2)$ by (H0′), which by transversality at t_{exit} displaces the exit by $O(C_R\varepsilon_\beta^2/R^{\mathrm{hyb}}(t_{\mathrm{exit}})) = O(\log^2\beta/\beta^2)$. This is $o(\beta^{-1})$, hence harmless, but it is a fourth contribution and not zero; it is of the same order as contributions 1 and 2 up to the extra $\log\beta$, and it is the largest of the three subleading terms.

Summing gives (32): contributions 2 and 3 are the two displayed terms of the law, while the remainder $o(\beta^{-1})$ collects contributions 1 and 4, both $O(\log^2\beta/\beta^2)$ or better, together with the unquantified $o(1)$ of Lemma 22, which is (O3) and is the dominant one. □

Remark 25 (the cancellation is arithmetic in R). Item 3 of step 6 is the entire content of the claim that the first-order coefficient does not know about B. Both factors, $K =$

$1/(c|R^{\mathrm{hyb}}(t_{\mathrm{hit}})|)$ from Lemma 22 and $|R^{\mathrm{hyb}}(t_{\mathrm{hit}})|$ from the cut area, are written in R^{hyb} and c alone; their product is $1/c$, and the division by the exit speed $cR^{\mathrm{hyb}}(t_{\mathrm{exit}})$ leaves $-1/(cR^{\mathrm{hyb}}(t_{\mathrm{exit}}))$. Nothing about the live subsystem — not m, not G_B, not the readout scale — survives to that coefficient. Any m-dependence must therefore appear in the *remainder* $o(\beta^{-1})$, that is in β_0, and not in the law.

B.6 L6′: exit corner, branch selection, scalar reduction

Lemma 26 (exit corner, branch selection and scalar reduction, forcing form)**.** *Under* (H0′)–(H5′)*, after release: (i) the β-trajectory crosses the exit corner in time $\frac{1}{\beta v'}(1+o(1))$ with $v' = cR^{\mathrm{hyb}}(t_{\mathrm{exit}}) > 0$; (ii) the ReLU flow restarted at $(\theta_A(t_{\mathrm{hit}}), \theta_B^{\mathrm{hyb}}(t_{\mathrm{exit}}))$ has forward uniqueness on the activating branch, and that branch is selected by the limit itself, the frozen Filippov branch being incompatible with release at $t_{\mathrm{exit}} + O(\beta^{-1})$; (iii) classical Grönwall glues on $[t_{\mathrm{release}}, H]$ at rate $O(\log\beta/\beta)$. Under* (H4′) *the other data of A contribute $O(e^{-\beta\delta_1})$, the face-j field is rank one along $(x_j, 1)$, and the episode reduces to the scalar problem.*

Provenance: identical to Lemma 14 with R generic, under the reinforced (H4′). (i) mirrors Lemma 22 with v'; (ii) needs the active field to be locally Lipschitz on the side $z \geq 0$ at the restart point, which with m live units holds provided no live unit sits exactly on its kink at t_{exit} — this is precisely the uniform margin δ_B of (H4′), and it is the second place, after step 2b of Lemma 20, where the reinforcement is used; the Filippov alternative is excluded by Lemma 24, independently of m; (iii) is classical Grönwall on the final interval, where all margins are bounded away from zero — but for the B-block we do *not* use it: step 5 of Lemma 20 replaces it by the PSD estimate on $[t_{\mathrm{release}}, H]$, which is what yields the sharper (11). Grönwall is retained only for the A-block, where it is harmless because the interval is what it is and the constant multiplies an already-$O(\beta^{-1})$ datum. Step 4, the scalar reduction, uses *only* that the other margins *of* A are $\leq -\delta_1$ — a statement about A, not about B — and is therefore verbatim for every m.

B.7 Proof of Theorem 6

Proof of Theorem 6. By step 4 of Lemma 26 the field on θ_A splits, under (H4′), into a rank-one part along $(x_j, 1)$ and terms of size $O(e^{-\beta\delta_1})$: the tangential components of θ_A are frozen at an exponential rate and the normal component is exactly the scalar problem of Lemmas 19 to 24. This is what licenses treating the episode as one-dimensional in z for any m.

On $[0, t_{\mathrm{hit}} - \delta]$ both flows are affine with the same PSD Gram, and step 4 of the proof of Lemma 20 bounds their gap by $O(\beta^{-1})$ in the A-block and $O(\beta^{-2})$ in the B-block. Lemma 22 then delivers the entry corner: the crossing time is $t_{\mathrm{hit}} + 1/(\beta v)(1+o(1))$ and the state at crossing is $O(\beta^{-1})$ from $\theta^{\infty}(t_{\mathrm{hit}})$.

On $[t_{\mathrm{hit}} + \delta, t_{\mathrm{exit}} - \delta]$, Lemma 24 closes the loop: its continuous induction, steps 1–5, maintains simultaneously the layer, the bound $\|\theta_B^{\beta} - \theta_B^{\mathrm{hyb}}\| \leq 2M_2 \log\beta/\beta^2$ and the correction integral. Inside the layer Lemma 23(i) gives $u = c\beta|\mathcal{A}^{\mathrm{hyb}}|(1+o(1))$, that is the sticking

depth $z = -\beta^{-1}\log\beta(1+o(1))$: this is item (i) of the theorem, and it is the largest of the three contributions to (9), the other two being $O(\log\beta/\beta^2)$ for θ_B, items (ivA)–(ivB), and $O(e^{-\beta\delta_1})$ for the tangential part of θ_A. Hence the supremum over this stage is $O_\delta(\log\beta/\beta)$, carried by the normal coordinate alone.

The terminal window and the release law are step 6 of Lemma 24: the correction delay of Lemma 23(ii) is $O(\log\beta/\beta^2)$, the perturbation of the forcing shifts t_{exit} by $O(\log\beta/\beta^2)$ by Lemma 23(iii), and the area cut by the entry corner advances the exit by $-1/(cR^{\mathrm{hyb}}(t_{\mathrm{exit}})\beta)$. Summing gives (32), hence item (ii), (8).

Finally Lemma 26 crosses the exit corner in time $O(\beta^{-1})$, selects the activating branch, and glues the two flows on $[t_{\mathrm{release}}, H]$ at rate $O(\log\beta/\beta)$. Taking the maximum of the four stage bounds gives (9), item (iii). Item (ivA) is the closed induction hypothesis (a) of Lemma 24, evaluated up to t_{release}, and item (ivB) is step 5 of Lemma 20, which repeats the PSD estimate on $[t_{\mathrm{release}}, H]$ against θ^∞; the change of reference at t_{release} is forced and is discussed in Remark 7. The threshold β_0 is the largest of the five thresholds of Table 6, namely those required in steps (4.1)–(4.3) of Lemma 24 and β_1, β_2 of step 2b of Lemma 20; by Proposition 8 it is $O(m)$ at fixed forcing. □

B.8 Uniformity in the size of the live subsystem

Remark 27 (what depends on m, and how). We collect the m-dependence of every constant that enters Theorem 6.

(U1) **The first-order coefficient is exactly m-free.** By Remark 25, $-1/(cR^{\mathrm{hyb}}(t_{\mathrm{exit}}))$ is built from c, a property of the datum j and of n, and from the hybrid forcing at one instant. The cancellation that produces it is arithmetic in R. Structurally, m cannot appear.

(U2) $C_4 = 2/(c\min(|R^{\mathrm{hyb}}(t_{\mathrm{hit}})|, R^{\mathrm{hyb}}(t_{\mathrm{exit}}))) + T_{\mathrm{ep}}/a_0$ is written purely in R^{hyb} and c: m-free *given the forcing*. The forcing of course depends on m; the point is that C_4 is a functional of the forcing only, so all m-dependence is routed through the observable quantities $R^{\mathrm{hyb}}(t_{\mathrm{hit}})$, $R^{\mathrm{hyb}}(t_{\mathrm{exit}})$, a_0, T_{ep}.

(U3) $C_f = \|\varphi_j\|/n \le X_{\max}\|a\|_2/n$. With unit readouts $a_k \equiv 1$ this is $\le X_{\max}\sqrt{m}/n$; with NTK readouts $a_k = m^{-1/2}$ it is $O(1)$. *At worst* $C_f \lesssim \sqrt{m}$.

(U4) $M_2 = C_f C_4/c \lesssim \sqrt{m}$. This is the substantive gain of Lemma 20: the constant of Lemma 10 is $e^{L_B H} C_f C_4/c$ with $L_B = \lambda_{\max}(G_B)$, and $\lambda_{\max}(G_B) \sim m$ for unit readouts, so that constant grows like $\sqrt{m}\, e^{\Theta(mH)}$. **There is no $m \cdot H$ restriction** in the PSD version: the horizon enters only through T_{ep}/a_0 inside C_4 and through H multiplying quantities that are beyond all orders in β.

(U5) The live-unit term (26) carries $\sqrt{m}\, C_{\mathrm{res}}$ — crudely mn scalar discrepancies — times $e^{-\beta\delta_B}$. It affects $\beta_1 = \delta_B^{-1}\log(m^2 H\beta^2)$, a *logarithmic* channel, and also, through the second bracket of (26), the threshold $\beta_2 \propto m$ of (27), which is a *linear* channel. The

Table 6: The five conditions that define β_0, and their order in m at fixed forcing. Rows 4 and 5 are the binding ones, both linear in m; row 4 is set in bold because the reading below identifies it as the one that binds first. All five constants blow up as $a_0 \to 0$ or $R^{\mathrm{hyb}}(t_{\mathrm{exit}}) \to 0$ — the forcing degenerating, the episode losing transversality — which is the structure the numerics of Section 3.6 see.

#	condition	used at	order in m
1	$\beta \geq \beta_1 = \delta_B^{-1} \log(m^2 H \beta^2)$	Lemma 20, step 2b	$\log m$
2	$H\big(C_3 e^{-\beta\delta_1} + C_5\sqrt{m}\, C_{\mathrm{res}} e^{-\beta\delta_B}\big) \leq \frac{1}{2} M_2 \log\beta/\beta^2$	Lemma 24 (4.1)	$\log m$
3	$\|\delta\|_\infty H \leq a_0/2$, with $\mathrm{Lip}_R M_2 \lesssim m$	Lemma 24 (4.2)	$\sqrt{m}$
4	$\kappa\, C_\varepsilon \dfrac{\log\beta}{\beta} \;\geq\; 2\,\mathrm{Lip}_R M_2 H \dfrac{\log\beta}{\beta^2}$	Lemma 24 (4.2), collar	**linear in m**
5	$\beta \geq \beta_2 = m \max_k \lvert a_k\rvert X_{\max}/C_{\mathrm{res}}$	(27), in (26)	linear in m

latter is easy to overlook because it hides inside a $(1 + o(1))$; we do not hide it. See row 5 of Table 6.

(U-new) **The cross block.** $G_{BA} = \frac{1}{n}\sum_i \varphi_i \psi_i$ with $\psi_i = \mathbf{1}_{A,i}(x_i, 1)$, so

$$\|G_{BA}\| \;\leq\; \frac{X_{\max}^2\, \|a\|_2}{n} \;\lesssim\; \sqrt{m}$$

with unit readouts, and $O(1)$ with NTK readouts. This is the constant that enters $\rho(t_{\mathrm{hit}})$ in step 4 of Lemma 20 and M_3 in step 5. Its growth is the same $\sqrt{m}$ as C_f, so it does not change the conclusion.

(U6) β_0 itself is *exhibited*, not left open: Table 6 lists the five conditions that define it and shows that the binding one is linear in m, whence $\beta_0 \lesssim O(m)$ at fixed forcing, with a_0, $R^{\mathrm{hyb}}(t_{\mathrm{exit}})$ and T_{ep} held fixed. What is *not* controlled is the dependence of β_0 on the forcing itself, which enters through $1/a_0$ and $1/R^{\mathrm{hyb}}(t_{\mathrm{exit}})$ and which is exactly what the numerics of Section 3.6 see.

Conclusion: under (H0′)–(H5′) every explicit constant of Theorem 6 depends on m at most like $\sqrt{m}$, the threshold β_0 at most like m, and the law itself not at all.

B.9 The threshold β_0, exhibited as a chain

The proof invokes "β large enough" five times. Rather than compose them into a single unenlightening number, we display them as a chain and read off the binding one. Each row of Table 6 is the inequality as it is actually used, the place where it is used, and its order in m at fixed forcing — that is, with a_0, $R^{\mathrm{hyb}}(t_{\mathrm{hit}})$, $R^{\mathrm{hyb}}(t_{\mathrm{exit}})$, T_{ep}, δ_B, δ_1, C_{res}, $X_{\max}$ and H held fixed while m grows. Proposition 8 is then the statement that the maximum of the five is $O(m)$; it is a bookkeeping statement, not a new argument, and its content is entirely in which row binds.

Reading the rows. Row 1 is written implicitly — β occurs on both sides of $\beta \geq \delta_B^{-1}\log(m^2H\beta^2)$ — and is to be resolved before the maximum of Proposition 8 is taken. It always resolves, and in two iterations: starting from $\beta^{(0)} = \delta_B^{-1}\log(m^2H)$ and setting $\beta^{(k+1)} = \delta_B^{-1}\log(m^2H(\beta^{(k)})^2)$, the map is a contraction on $[\beta^{(0)}, \infty)$ because its derivative is $2/(\delta_B\beta) \ll 1$ there, and the fixed point satisfies $\beta_1 = \delta_B^{-1}\log(m^2H) + 2\delta_B^{-1}\log\log(m^2H) + O(1)$. The double-logarithmic correction is the only trace of the implicitness, which is why row 1 is $\log m$ and not worse.

Rows 1 and 2 are logarithmic because the m's sit inside an exponential that is beyond all orders: doubling m costs $\delta_B^{-1}\log 2$ in β. Row 3 is $\sqrt{m}$ because $\mathrm{Lip}_R M_2 = \|\varphi_j\| \cdot C_f C_4/c$ carries $\|\varphi_j\| \lesssim X_{\max}\|a\|_2$ once and C_f once, that is m in the product for unit readouts, but enters the condition only through β^{-2} against a constant, giving $\beta \gtrsim \sqrt{m}$. Row 4 is the binding one: it compares two quantities that both carry a $\log\beta$, so the log cancels and one is left with $\beta \geq 2\mathrm{Lip}_R M_2 H/(\kappa C_\varepsilon)$, linear in m for unit readouts. Row 5 is the condition that (26) would have hidden inside a $(1 + o(1))$, and it is linear in m for the same reason, so it binds alongside row 4; row 4 is set in bold because it is the one that binds first in every instance we computed. Taking the maximum of the five proves Proposition 8.

B.10 Steps we did not close

Stated plainly, and separated from what is proved above.

(O1) β_0**: closed as a chain, not as a number.** Table 6 lists the five inequalities the proof imposes and identifies the binding one, row 4 with row 5 alongside, giving $\beta_0 \lesssim O(m)$ at fixed forcing (Proposition 8). We deliberately do *not* multiply the five constants together into a single expression: the product would be a large rational function of nine quantities, would be dominated by whichever of them is worst in a given instance, and would suggest a sharpness the argument does not have. What remains genuinely uncontrolled is not the m-dependence but the *forcing* dependence: every row degenerates as $a_0 \to 0$ or $R^{\mathrm{hyb}}(t_{\mathrm{exit}}) \to 0$, and we have no lower bound on those quantities from any initialization criterion.

(O2) (H4′) **is the definition of the window, not a debt.** The statement of scope is Remark 5: Theorem 6 holds on any maximal window on which the live activation pattern is constant and which contains t_{hit} and t_{exit}; such a window is identified a posteriori by a single hybrid integration, together with the margin floor δ_B, and there is nothing to verify that is not already computed when checking (H1′)–(H2′).

Why the relaxation is not free. Suppose we allow live units to switch inside the window. The β-flow and the hybrid then switch at *different* instants — the β-flow's units cross their kinks at times displaced by $O(\|\Delta\|/v_k)$, where v_k is the speed at which margin k crosses zero. On the intervening slivers the two drifts are governed by *different* Gram matrices, G_B and G'_B, and the difference of drifts is $-G_B\Delta - (G_B - G'_B)\theta_B$, whose second term is *not* of the form $-(\mathrm{PSD})\Delta$: it is a source of size $\|\Delta G\|\|\theta_B\|$ acting for a time $O(\|\Delta\|/v_k)$, hence a term *linear in* $\|\Delta\|$ *with no sign.* Step 3 of Lemma 20

then no longer yields $\dot{\rho} \le \|G_A\| + \|G_S\|$, and one is back to Grönwall with exponent $\sum_{\text{switchings}} \|\Delta G\|\,\|\theta_B\|/v_k$, that is a factor $\exp\big(\sum_k \|\Delta G_k\|\|\theta_B\|/v_k\big)$ in M_2. With up to mn switchings in the window, this is exponential in m in the worst case — precisely the $e^{L_B H}$ that the PSD step was introduced to remove. The identity of step 1 is not a convenience of exposition; it is the load-bearing element, and constancy of the pattern is exactly what makes it an identity.

Where the hypothesis is consumed. (H4′) is used at five places: step 2b of Lemma 20 (uniform live margins, for the beyond-all-orders term), step 1 of the same lemma (constant pattern, for the identity), step 5 of the same lemma (constant pattern again, for the post-release estimate), Lemma 26(ii) (branch selection: no live unit on its kink at t_{exit}), and — this one concerning A rather than B — step 5 of Lemma 20 again, which needs A to be alive throughout $[t_{\text{release}}, H]$ *in both flows*, since otherwise the two trajectories do not share a common Gram after release and the identity $\dot{\delta\theta} = -G\delta\theta + E$ fails exactly as it does under live switching. This is the clause of (H4′) that fixes the number of switchings of A at two, with margin floor δ_1' on $[t_{\text{release}} + \eta, H]$. The numerical campaigns already impose it: the acceptance filter of `verifica_forcing.py` counts the deaths of A along the β-run and rejects any instance with more than one, so every instance reported in Section 3.6 has A crossing zero exactly at the entry and the exit of a single episode and staying alive thereafter; the point is made with the numbers in Section 3.6.

It is worth noting that item (ivB) does *not* relieve the requirement on $[t_{\text{release}}, H]$: the PSD repetition of step 5 replaces Grönwall there, but it still needs both flows to be affine with a common Gram, which is constancy of the pattern. So (H4′) is required on all of $[0, H]$, and now for two reasons, item (ivB) and Lemma 26(iii), rather than one. Asking for it only on $[0, t_{\text{release}}]$ would leave item (iv) with a Grönwall factor on the tail; we prefer the hypothesis to the factor, and say so.

(O3) **The $o(1)$ of Lemma 22 is not quantified — and it is the named suspect for the measured slope bias.** The entry-corner estimate carries $O(\delta_z) + o(1)$ and we do not sharpen it to $O(\log\beta/\beta)$; consequently the remainder of (8) is $o(\beta^{-1})$ and not the $O(\log\beta/\beta^2)$ that the numerics suggest. This is inherited from Section A — it is the same unquantified $o(1)$ already present in Lemma 11 and flagged in Remark 3 — and it is the *only* open item here. It has a concrete empirical signature: the regression of item (1) of Section 3.6 returns a slope of 1.154 rather than 1, a 15% systematic bias in the regression slope: the law *under*-predicts the measured coefficient in modulus, the measured span being $[-132, -14]$ against a predicted $[-109, -14]$. The other candidate for that bias, the second-order term of Lemma 23(ii), was excluded numerically: its sign is symmetric across the sample, 36 of 66 instances positive with median $+0.042$, whereas the observed bias is one-signed. The entry corner is one-signed by construction — the 3/2 of Lemma 22 and the $O(\delta_z)$ both bias the effective delay K/β in the same direction — and item 3 of step 6 of Lemma 24 converts that delay directly into the coefficient of β^{-1}. Quantifying the entry corner is therefore the single most valuable

remaining piece of work on this theorem, and it would be testable against the measured 1.154.

(O4) **The measured degradation at $m = 8, 16$ is not explained by the constants.** Remark 27 predicts $M_2 \lesssim \sqrt{m}$, a mild growth, whereas the median error grows by a factor 4–8 from $m = 4$ to $m = 8$ in the NTK family. Three pieces of evidence say the discrepancy lives in β_0 and not in the law: the cloning experiment shows that at fixed forcing m does nothing; the regression shows the error tracking $1/R^{\mathrm{hyb}}(t_{\mathrm{exit}})$; and the β sweep drives the six worst instances from 12–20% to 0.2–4.9% by tripling β. But a bound that *predicts* the observed degradation from exit transversality alone is not in this paper. What (O1) supplies is the structural half of the story: every row of Table 6 carries $1/a_0$ or $1/R^{\mathrm{hyb}}(t_{\mathrm{exit}})$, so β_0 is expected to blow up as the exit loses transversality — which is what the regression measures — while remaining linear in m (Proposition 8). The remaining status of this item is therefore *empirical*, not logical: we have a certified polynomial bound on $\beta_0(m)$ and an unfitted, instance-by-instance account of β_0's dependence on the forcing. What is missing is a sharp constant, not a certification.

(O5) **Closed.** *(Was: step 4 of Lemma 20 is new and unaudited.)* The pre-hit initial gap has since been recomputed independently — the pre-hit system is an affine identity with $G \succeq 0$, the A-block of the defect is $O(C_{\mathrm{res}} e^{-\beta z})$ with no β^{-1} because it is a derivative defect, the B-block carries only A's output defect $C_f \beta^{-1} e^{-\beta z^0}$ since B's gradients never involve A's σ, the pointwise form follows from $\int e^{-\beta v s}\, ds = (\beta v)^{-1}$, and the cross term $G_{BA}\delta\theta_A$ is a legitimate source for the repetition of step 3 — and verified numerically: Section 3.6 and Table 2 exhibit the one-power separation directly, with $\beta\rho_A$ and $\beta^2\rho_B$ both saturating and $\beta^2 v \rho_B = 0.198 \pm 0.046$ invariant in m over twelve instances. Three pieces of bookkeeping that the first draft omitted are in the text above: the additive live-unit term in (29), the monotonicity proviso on z^0 away from the hit, and the bound $\|G_{BA}\| \lesssim \sqrt{m}$ of (U-new).

(O6) **The trainable readout.** Every experiment reported in this paper trains the readout v_k, and that is exactly what leaves the affine class of Section 3.1: with v_k moving, f_{-A} is bilinear rather than linear in the trainable parameters, the B-field is no longer $-G_B(\theta_B - \theta_B^*)$ with a *fixed* PSD Gram, and the identity of step 1 of Lemma 20 — affineness of both drifts with a common matrix — fails. The energy estimate that removed the exponential therefore does not apply, and an $e^{L_B T_{\mathrm{ep}}}$ factor is expected back in M_2, with the episode length T_{ep} rather than the horizon H in the exponent, because outside the episode the two flows share their branch and the gap is reset by the release. We have not carried out that estimate, and we do not know whether the first-order coefficient $-1/(cR^{\mathrm{hyb}}(t_{\mathrm{exit}}))$ survives it unchanged; the numerics of Section 6 are consistent with its surviving, but they are measurements and not a proof. This is the single restriction of Theorem 6 that separates the proved object from the object the experiments actually run, and it is why level (P3) of Section 1 exists.

C Algorithms

Four routines carry the experimental claims of Section 6: the ground-truth area clock (Algorithm 1), the truncated calibrated null (Algorithm 2), the hysteresis counter (Algorithm 3) and the pre-registered decision procedure (Algorithms 4 and 5). They are given here in full, because each of them is an instrument built for this paper and each of them has already falsified one of its own predecessors. Smaller routines are folded into the algorithm that consumes them: the clean-episode filter inside Algorithm 2, the Clopper–Pearson bisection and the gate n^* inside Algorithms 4 and 5.

C.1 The ground-truth area clock

The ruler of Section 6.2 was replaced, after it failed its own null, by a clock with no ruler at all: death and return are read directly off the margin over all data, with no threshold, no debounce and no dwell. The single free choice left is the accumulator, and it runs on the same clock, so the temporal ordering $t_{\text{hit}} \le t_{\text{area}} < t_{\text{res}}$ is structural rather than measured; the only empirical failure mode left is a return with no area.

One normalization is deliberate and is recorded here because it looks like an inconsistency. The descent step is taken on $L = \frac{1}{2n}\sum_j \text{res}_j^2$, so the parameter move carries a factor $1/n$, whereas the accumulated drive $\hat{\mathcal{A}}_k$ is built from the leader-datum quantity $\hat{R}_k$, which does not. The two differ by the fixed positive constant $1/n$, and every claim read off the clock — the sign of $\hat{\mathcal{A}}_k$, the first instant at which it turns positive (t_{area}) and hence the ordering $t_{\text{hit}} \le t_{\text{area}} < t_{\text{res}}$ — is invariant under multiplication by a positive constant. What is *not* invariant is the magnitude of $\hat{\mathcal{A}}_k$, which is therefore never compared against the theoretical area of Theorem 2 and is used only through its sign and its first zero crossing.

Two guards are asserted before any run is used. Bit-exact parity: the loss trace must agree with the pilot engine to 10^{-12}, so the clock is known to be a pure observer. And the episode count must equal the number of born-dead units plus the number of dynamic deaths, so no episode is silently dropped.

Executable: `pilotos-4a2/pressao/taxa/motor_4a3.py`, *config SHA in* `results-4a3/manifest-4a3.json`.

C.2 The truncated calibrated null

This is the methodological repair of Section 6.2, and it is the one piece of the paper we would offer for reuse. The instrument stops measuring the area at the resurrection, so $t_{\text{area}} \le t_{\text{res}}$ holds mechanically and a naive permuted-delay placebo, which does not inherit that truncation, inflates the false-positive rate by up to $58\times$ the nominal α. The repair is to put the truncation *inside* the null: fit the delay distribution treating non-firing as right censoring, then reproduce the truncation in every bootstrap replicate, so the structural bias sits on both sides of the comparison and cancels.

Under H_0 the delay is independent of the duration, which is exactly what the resampling imposes: the durations are never resampled, only the delays are, and they are drawn from

Algorithm 1 Ground-truth area clock (deterministic MLP, $T = 0$).

Require: width m, activation g_β, rate η, horizon H, seed; data (x, y) with n points
Ensure: for every episode: $(k, \text{class}, t_{\text{death}}, t_{\text{area}}, t_{\text{res}}, \text{revived}, z_{\text{park}})$

initialize (w, b, v) from the seed; $\text{open}_k \leftarrow$ FALSE for all k
for $t = 0$ **to** $H - 1$ **do**
  $z \leftarrow xw^\top + b$; $a \leftarrow g_\beta(z)$; $f \leftarrow av$; res $\leftarrow f - y$
  $z_k^{\max} \leftarrow \max_{j \le n} z_{jk}$; $j_k^* \leftarrow \arg\max_{j \le n} z_{jk}$
  $\text{dead}_k \leftarrow [\, z_k^{\max} < 0 \,]$ {ground truth: the unit is off on *every* datum}
  $\hat{R}_k \leftarrow v_k \big[(y - f)_{j_k^*} + v_k \, a_{j_k^* k} \big]$ {leader-datum residual drive; the $1/n$ of L is dropped (see below)}
  for all k with $\text{dead}_k \wedge \neg\text{open}_k$ **do**
    open an episode: $\text{open}_k \leftarrow$ TRUE, $\hat{\mathcal{A}}_k \leftarrow 0$, $t^k_{\text{area}} \leftarrow -1$, $t^k_{\text{death}} \leftarrow t$, $z^k_{\text{park}} \leftarrow z_k^{\max}$
    class $\leftarrow$ BORN if $t = 0$, else DYN
  end for
  for all k with $\text{open}_k \wedge \text{dead}_k$ **do**
    $\hat{\mathcal{A}}_k \leftarrow \hat{\mathcal{A}}_k + \hat{R}_k \, \eta$ {the area accrues only on dead steps}
    if $t^k_{\text{area}} < 0$ **and** $\hat{\mathcal{A}}_k > 0$ **then**
      $t^k_{\text{area}} \leftarrow t$ {the gate opens}
    end if
    $z^k_{\text{park}} \leftarrow \min(z^k_{\text{park}}, z_k^{\max})$
  end for
  for all k with $\text{open}_k \wedge (z_k^{\max} > 0)$ **do**
    close the episode with $t_{\text{res}} \leftarrow t$, revived $\leftarrow$ TRUE; $\text{open}_k \leftarrow$ FALSE
  end for
  $\theta \leftarrow \theta - \eta \nabla_\theta \frac{1}{2n} \sum_j \text{res}_j^2$ {plain gradient descent on the loss L of Section 2; the instrument never touches the dynamics}
end for
close every still-open episode as **censored**: $t_{\text{res}} \leftarrow$ NONE, revived $\leftarrow$ FALSE {never recorded as "permanent"}

the pooled $\hat{F}$. The level and the power are demonstrated on synthetic truncated worlds before the procedure is applied to data.
Executable: `etapa4-mlp-populacao/analise.py (p_calibrado, _km, _amostra_km)`, *config SHA in* `etapa4-mlp-populacao/results/manifest.json`.

C.3 The hysteresis counter

In the transformer the input of the block is non-stationary, so the margin chatters across zero and a bare threshold counts noise as resurrection. The counter therefore commits a transition only outside a band $\pm\delta_h$ calibrated from data, and a return counts only if the unit is still alive at the horizon.
Inside the band $[-\delta_h, +\delta_h]$ the state is held, which is what kills the chatter; the test that

Algorithm 2 Truncated calibrated null for the area-delay statistic.

Require: episodes grouped by seed; B replicates; cell key (i_b, i_m)

Ensure: p, T_{obs}, effective number of seeds

Retention (clean-episode filter). Keep an episode iff it completed ($t_{\text{res}} \geq 0$) and: at most q switching events of *other* units in $(t_{\text{death}}, t_{\text{res}}]$; the readout v_k did not change sign; $\min_t |v_k| \geq \varphi_v$, which excludes death by a vanishing readout; and no change of the leader j^*. Drop seeds with fewer than three surviving episodes.

for each episode: duration $d_i \leftarrow t_{\text{res}} - t_{\text{death}}$; delay $a_i \leftarrow t_{\text{area}} - t_{\text{death}}$ if the area fired, else -1; $f_i \leftarrow [\, a_i \geq 0 \,]$

Survival encoding. $\text{obs}_i \leftarrow a_i$ if f_i else d_i; $\text{ev}_i \leftarrow f_i$ {non-firing = right censoring at the episode's own duration}

$T_{\text{obs}} \leftarrow \textsc{Stat}(\{(d, a, f)\})$ with one independent stream per seed

$\hat{F} \leftarrow \textsc{KaplanMeier}(\text{obs}, \text{ev})$, pooled over the seeds of the cell

exceed $\leftarrow 0$

for $b = 1$ **to** B **do**

  for all seeds, holding the durations d *fixed* **do**

    draw $A_i \leftarrow \hat{F}^{-1}(u_i)$, $u_i \sim U[0,1)$, and set $A_i \leftarrow \infty$ when u_i exceeds $\hat{F}$ at the last observed event {the censored mass stays censored}

    $f_i^\dagger \leftarrow [\, A_i \leq d_i \,]$; $a_i^\dagger \leftarrow A_i$ if $f_i^\dagger$ else -1 {the truncation is reproduced, replicate by replicate}

  end for

  $T_b \leftarrow \textsc{Stat}(\{(d, a^\dagger, f^\dagger)\})$; exceed $+= [\, T_b \geq T_{\text{obs}} \,]$

end for

return $p = (1 + \text{exceed})/(B + 1)$

function STAT: for each seed, restrict to the fired episodes; if fewer than two, the seed contributes nothing. Otherwise draw for each i an index j_i uniform on the other fired episodes of that seed and put $W = \frac{1}{n} \sum_i \big[\, (d_i - a_i) < |d_i - a_{j_i}| \,\big]$. Return the mean of W over the surviving seeds.

the band is doing work is that the return count at $\delta_h = 0.05$ is strictly below the count at $\delta_h = 0$ on a seed known to chatter. One disclosure is on record: under TRAIN the same calibration gives $\delta_h \approx 0.07$–0.08, about $1.7\times$ larger, so every estimand that depends on δ_h being decidable is confined to the FROZEN cells.

Executable: `etapa5-transformer/motor_e5_conf.py`, *config SHA in* `results-e5/manifest-e5.json`.

C.4 The pre-registered decision procedure

The same procedure serves Section 6.4 and Section 6.5, with the estimand-specific thresholds read from the frozen config. Its one structural commitment is the last line: insufficient evidence returns INCONCLUSIVE and never a negative.

The thresholds are derived, not chosen: ϑ is half the pilot's Clopper–Pearson lower bound on the same proportion, and τ is the pilot's $q_{0.9}$ return time. Both are read from the config committed before the grid ran. One honest disclosure: at $N = 50$ the transformer's KILLS branch is nearly unreachable, since the Clopper–Pearson upper bound on 0/50 already exceeds ϑ; the operative decision there is TRANSFERS against INCONCLUSIVE, and it would take $N \gtrsim 800$ for the kill to become available.
Executables: `pilotos-4a2/pressao/taxa/analise_4a3.py` *and* `etapa5-transformer/analise_e5.py`, *config SHAs in* `results-4a3/manifest-4a3.json` *and* `results-e5/manifest-e5.json`.

D Pre-registered estimands of the transformer confirmatory

The following five estimands were frozen at commit `1b45498`, ahead of the data, and are reproduced here verbatim; Section 6.6 carries the two scoping findings S1 and S2 that motivated them.

- **E-A, validity.** Any sustained return in the relu-frozen arm will invalidate the run, since an own-gradient identically zero together with a fixed input makes death terminal. The pilot gave 0/40. *Result:* **VALID.** 0/100 sustained returns in relu-frozen across both injection depths.

- **E-B, upstream channel.** We will estimate $\Delta = p_{\text{sust}}(\text{train}) - p_{\text{sust}}(\text{frozen})$ per arm, and will call the effect reproduced if the bootstrap lower bound is positive. Under the hysteresis instrument the test is stricter than in the pilot. The hysteresis band was calibrated on frozen runs only; alive units in training runs fluctuate up to 1.7× wider, so in the training arms the band is narrow relative to the arm's own noise and crossings are easier — an anti-conservative direction for Δ. The sustained-return filter is the measured containment: five returns failed sustainment (hover), all five in training cells and none in frozen, and none entered the counts. The ReLU arm is, in addition, immune to that argument: its own-move is identically zero on both sides of the contrast, so if a band that is narrow relative to training noise were manufacturing spurious crossings, they would appear in relu-frozen as well — and relu-frozen gives 0/100. The 20/100 against 0/100 contrast (two-sided Fisher exact test, $p = 6.6 \cdot 10^{-7}$) is therefore the internal control that closes E-B without relying on the calibration of δ_h. *Result:* **REPRODUCED in all four arms.** $\Delta = 0.210\,[0.120, 0.300]$ (sp-β3), $0.200\,[0.110, 0.280]$ (sp-β10), $0.220\,[0.120, 0.320]$ (GELU), $0.200\,[0.120, 0.280]$ (ReLU, 20/100 against 0/100): the non-stationary FFN input is a real resurrection channel, of comparable size across activations.

- **E-C, transfer of the theorem class.** For sp-β3 frozen under shallow injection we will estimate $p(\text{sustained return} \mid \text{injection})$, and will call the class transferred if there are at least $n^* = 6$ sustained returns. This is the direct analogue of C2 in the MLP. *Result:* **TRANSFERRED.** 22/50 sustained returns, $\hat{p} = 0.440\,[0.300, 0.587]$, 3.7×

the gate $n^* = 6$; the area of the leader drive opened before the return in 21/22 of them. The memory gate of the theorem operates in a transformer FFN through the own channel.

- **E-D, GELU barrier.** For gelu-frozen we will compare deep injection (-2.0) against shallow (-0.3); the barrier predicts shallow far above deep in the own channel, with a depth-matched control against sp-β3. The pilot at -1.0 was a tie and did not decide. *Result:* **Consistent with the barrier, confounded with depth** (the three-way label frozen at pre-registration). Intra-GELU: shallow 29/50 vs. deep 0/50; but the depth-matched control sp-β3-frozen at -2.0 also gave 0/50, so depth alone accounts for the deep zero and the sustained-barrier claim is not licensed by this design. The S1 signature remains sharp: $\mathrm{sign}(\mathcal{A}_{\mathrm{drive}}) \neq \mathrm{sign}(\mathcal{A}_{\mathrm{mlp}})$ in 92% of deep vs. 4% of shallow episodes — the barrier is present in the drive field; its terminality is not separable from depth here. The floor is structural, not accidental: at $z = -2.0$ the own channel of sp-β3 is $\sigma(3z) = 0.0025$, that is 0.9% of its value at $z = -0.3$ (gradient starvation), while $\mathrm{GELU}'(-2.0) \approx -0.085$ is adverse — the depth -2.0 is deep for *both* mechanisms at once, so 0/50 against 0/50 (two-sided Fisher exact test, $p = 1.0$) carries no information by construction. The design that would separate them injects at $z \approx -0.9$, where $\mathrm{GELU}' \approx -0.055$ is already adverse but $\sigma(3z) \approx 0.063$ still leaves $\approx 22\%$ of the shallow softplus channel; a sweep over $\{-0.5, -0.75, -0.9, -1.2\}$ tests the differential prediction — an abrupt drop for GELU at $z^* = -0.7518$ against a smooth decay for softplus — a signature no depth confound reproduces. This is inherited as future work, not claimed here.

- **E-E, β staircase.** We will estimate p(sustained return) across $\beta \in \{3, 10\}$, frozen and training, as an estimation with no kill condition. *Result:* $\beta = 3$ above $\beta = 10$ in both factors (frozen $0.220\,[0.143, 0.314]$ against $0.090\,[0.042, 0.164]$; training $0.430\,[0.331, 0.533]$ against $0.290\,[0.204, 0.389]$). GELU itself is the most revivable arm (shallow frozen $0.580\,[0.432, 0.718]$; with the upstream channel $0.860\,[0.733, 0.942]$): the activation used in practice sits at the open end of the channel.

E Post-hoc aggregation of C3

C3 was pre-registered as estimation-only, with no kill condition, and the frozen analysis script contains no aggregate for it (Section 6.4). The paragraph below was produced after the confirmatory data were seen, at the request of an external reviewer, and is exploratory throughout: nothing in it tests a pre-registered hypothesis, and with 13 episodes over 10 seeds the rank intervals are wide by construction.

In 10 of the 13 episodes with an open area the gate is open at the moment of death, $t_{\mathrm{area}} = t_{\mathrm{death}}$ exactly, so in those episodes the lead time $t_{\mathrm{res}} - t_{\mathrm{area}}$ is identical to the duration of the episode and carries no information of its own. Controlling for when the episode began, the partial Spearman correlation

between t_{area} and t_{res} given t_{death} is $+0.10$, with a 95% seed-clustered bootstrap interval $[-0.43, +0.78]$ that crosses zero: there is no evidence that the zero of the area predicts the return time beyond the onset of the episode. Under this clock the gate does not warm up; in most episodes it is already open at death, and what varies is how long the rise takes. The mean drive of the episode is not recoverable from the instrument, which persists only the triple of times, z_{park} and the channel value, so the question cannot be pushed further with these runs. Persisting the drive series per episode is inherited as a requirement for the next stage.

Generated by `agrega_c3_posthoc.py` in the confirmatory's directory.

Algorithm 3 Hysteresis counter with sustained return (transformer block).

Require: arm, attention factor TRAIN/FROZEN, injection depth $\kappa \in \{0.3, 2.0\}$, band δ_h, horizon H, seed

Ensure: per unit: $(\text{revived}, \text{sustained}, \text{hover}, t_{\text{res}}, t_{\text{area}}, \mathcal{A}_{\text{drive}}, \mathcal{A}_{\text{mlp}}, \text{own-move}, \text{channel})$

Calibration of δ_h (offline, once). Run the frozen flow with no injection; on units alive throughout, take the half peak-to-peak amplitude of $z^{\max}$ in windows of 200 steps and set δ_h to the $q_{0.99}$ of that amplitude. This gives $\delta_h = 0.0476$, and it is frozen in the config before the grid runs.

Injection. Compute $z_k^{\max,0}$ and pick the unit k of lowest initial margin; shift $b_1[k]$ by $-\kappa - z_k^{\max,0}$, so that $z_k^{\max} = -\kappa$ exactly. The depth is a two-level factor across the grid, shallow -0.3 against deep -2.0, straddling the GELU sign-flip barrier $z^* = -0.752$.

$\text{alive}_k \leftarrow$ TRUE for all k

for $t = 0$ **to** $H - 1$ **do**

  forward pass; $z_k^{\max} \leftarrow \max_j z_{jk}$ over all $N \cdot S$ tokens; $j_k^* \leftarrow \arg\max_j z_{jk}$

  for all k with $\text{alive}_k \wedge (z_k^{\max} < -\delta_h)$ **do**

    confirmed death: $\text{alive}_k \leftarrow$ FALSE; open an episode and snapshot $(W_1[k], b_1[k])$

  end for

  for all k with an open episode **do**

    $c_k \leftarrow (w_{\text{out}} W_2)_k$; $\mathcal{A}_k^{\text{mlp}} \mathrel{+}= c_k\big[(y - \hat{y})_{j_k^*} + c_k a_{j_k^* k}\big]\eta$; $\mathcal{A}_k^{\text{drive}} \mathrel{+}= c_k (y - \hat{y})_{j_k^*} \, g'_\beta(z_k^{\max})\, \eta$

    record the first t at which each area turns positive

  end for

  for all k with $\neg\text{alive}_k \wedge (z_k^{\max} > +\delta_h)$ **do**

    confirmed return: $\text{alive}_k \leftarrow$ TRUE, $t_{\text{res}} \leftarrow t$; close the episode

    own-move $\leftarrow \|(W_1[k], b_1[k]) - (W_1, b_1)_{\text{death}}\|_2$; channel $\leftarrow$ UPSTREAM if own-move $< 10^{-9}$, else OWN

  end for

  gradient step on the trainable parameters only (under FROZEN, attention and the readout are held fixed)

end for

close every open episode as censored, and per unit set sustained $\leftarrow$ revived $\wedge$ alive_k, hover $\leftarrow$ revived $\wedge \neg\text{alive}_k$

Algorithm 4 Pre-registered decision procedure (part 1 of 2): classification, eligibility and gate.

Require: runs; horizon H; return-time quantile $\tau = q_{0.9}$ of the pilot; gate n^*; kill threshold ϑ; floor π_0; $\alpha = 0.05$

Ensure: per estimand, one of REPRODUCES/SUSTAINS/KILLS/INCONCLUSIVE (for the barrier estimand the three-way label SUSTAINS/CONSISTENT-CONFOUNDED/KILLS of the depth-matched rule)

1: **Classification.** For each dynamic death:

- *return* if the unit revived on the ground-truth clock; it is *ordered* if moreover $t_{\text{death}} \leq t_{\text{area}} < t_{\text{res}}$ with $t_{\text{area}} \geq 0$;
- *censored* if it did not revive but its area had opened;
- *terminal* if it did not revive and its area never opened.

2: **Eligibility.** The death is eligible iff $t_{\text{death}} \in [0, H - \tau]$, the only window in which a typical return still fits inside the horizon. In the transformer the injected deaths all occur at $t = 0$, so there the window is applied to t_{res} instead and yields a secondary robustness count, not the primary estimand.

3: **Gate.** $n^* \leftarrow$ the α-quantile of Poisson(μ), with μ the expected number of deaths implied by the pilot's bootstrap lower bound on the death rate; here $n^* = 10$ for the MLP and 6 for the transformer. It is fixed before the run.

Algorithm 5 Pre-registered decision procedure (part 2 of 2): intervals, estimands and the validity control.

Interval. $(\hat{p}, \ell, u) \leftarrow$ ClopperPearson(k, N, α), where ℓ solves $\Pr[X \geq k \mid \mathrm{Bin}(N, p)] = \alpha/2$ and u solves $\Pr[X \leq k \mid \mathrm{Bin}(N, p)] = \alpha/2$, each by 200 bisection steps on the exact binomial tail; $\ell = 0$ when $k = 0$ and $u = 1$ when $k = N$.
Rate estimand (k eligible returns out of N eligible deaths):
if $N < n^*$ **then**
  return INCONCLUSIVE {never negative on insufficient N}
else if $u < \vartheta$ **then**
  return KILLS
else
  return REPRODUCES
end if
Ordering estimand (k ordered out of N returns):
if $N < n^*$ **then**
  return INCONCLUSIVE
else if $\ell < \pi_0$ **then**
  return KILLS
else
  return SUSTAINS
end if
Validity control. If any run of the zero-gradient arm (ReLU in the MLP, relu-frozen in the transformer) records a return, declare the whole run **invalid**; no verdict on any substantive estimand may then be claimed.
Staircase estimands are reported with intervals and carry no kill condition, by pre-registration.